

XSLT and XPath for Visual Studio Code

User Guide

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Overview

DeltaXignia's XSLT/XPath extension for Visual Studio Code supports XSLT 3.0 and XPath 3.1, and the XSLT 4.0 and XPath 4.0 features implemented in Saxon 13.

The aim is to make writing, reviewing, testing and debugging XSLT and XPath easier.

The extension understands XPath expressions within the XSLT semantically, so it can detect problems immediately, whether code is written manually or generated by an AI assistant, navigate directly to declarations across imported modules, safely rename variables, functions and templates, and step through XPath expressions in the debugger. It distinguishes variables, functions and node tests, tracks where each is declared, and uses this information throughout editing, navigation, refactoring and debugging.

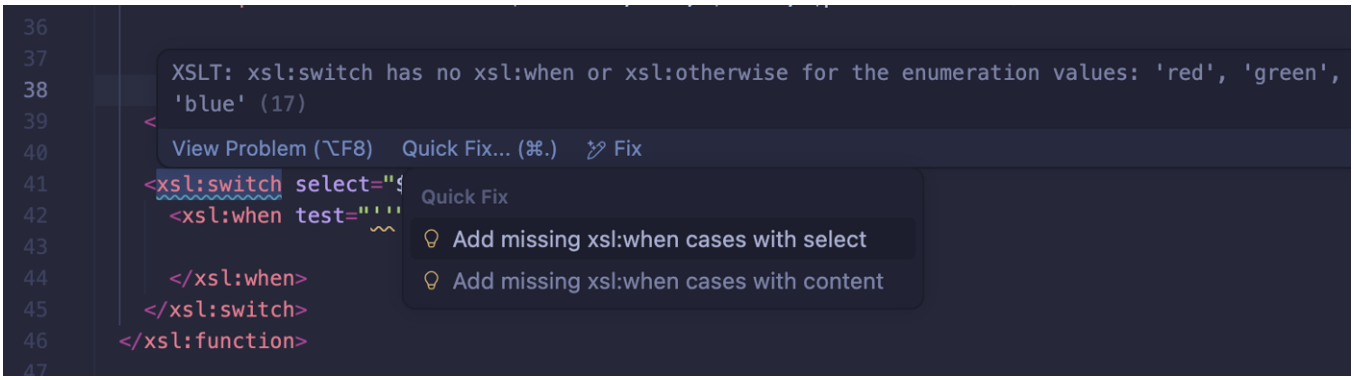
> *For an overview of the extension's capabilities, see Features. For what's new in the latest release, see What's New.*

Getting started

1. Install the extension and open an XSLT file - editing, highlighting and checking need no setup.
2. To run or debug stylesheets, set `XSLT.tasks.saxonJar` to the path of your Saxon jar (Saxon-HE is fine), with a Java runtime on the PATH - see The Saxon jar.
3. Press ⌘⌥R (Ctrl+Alt+R on Windows/Linux) to Quick Run the stylesheet with the XML file you last had open - shown in the status bar, where you can pick another.

Writing XSLT

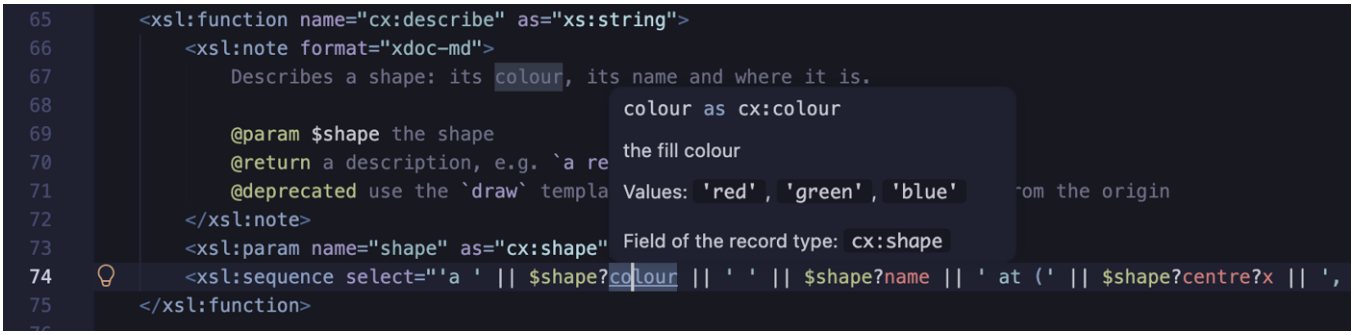
Problems are reported as you type - in XPath syntax, XSLT instructions, types and function calls, and references to declarations in other modules - many with a quick fix. Auto-completion offers the variables, functions, templates and modes in scope, and node names from the XML file you're working with. See Editing XSLT/XPath.



An `xsl:switch` that doesn't cover an enumeration's values, with quick fixes to add the missing cases

Reviewing XSLT

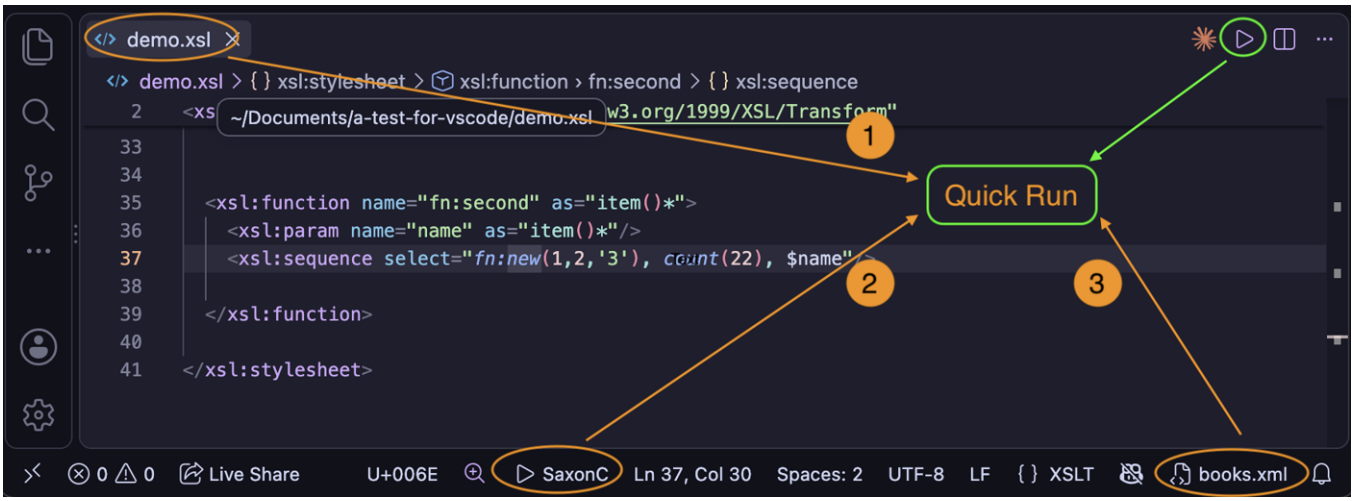
Hovering over a name shows its declaration - its type, and its documentation from an `xsl:note` - wherever it's declared. Go to Definition, Find References, the Outline and the XSLT Imports view help you find your way around a large stylesheet and its modules. See Code Navigation and Code Documentation.



Hovering over a record field in a lookup: its type, documentation and allowed values

Running XSLT

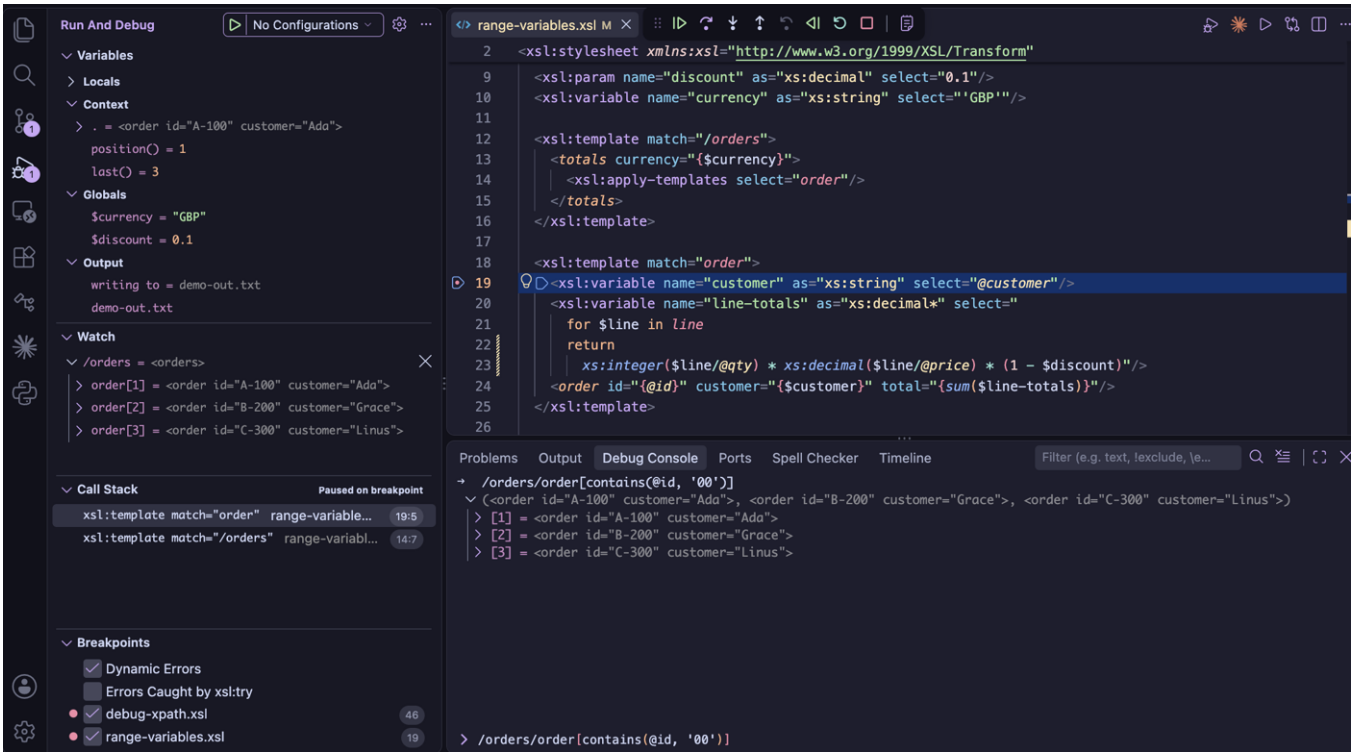
Quick Run runs the stylesheet in the active editor with SaxonJ, SaxonJS or SaxonC. The first run creates a task in `tasks.json`, which you can edit - and you're asked for the values of any required stylesheet parameters. Saxon's errors are listed in the Problems view, linked to the code. See Quick Run.



Quick Run uses the stylesheet in the active editor (1), the processor (2) and the XML context file (3), shown in the status bar

Debugging XSLT

The debugger sets breakpoints and steps through templates, functions and multi-line XPath expressions, showing variables, the context item and the output written so far. Where Saxon's standard tracing stops at XSLT instructions, this debugger uses Saxon's own APIs to inject extra trace points and re-parse XPath to locate sub-expressions, so it can stop deep inside multi-line expressions and inline functions. See Debugger.



Paused within an XPath expression, with its range variable, the context, global variables, inline values and the call stack

Scope

The extension supports the XSLT 4.0 and XPath 4.0 features that Saxon 13 implements. Its checks are made as you type, without compiling the stylesheet - Saxon reports anything else when the stylesheet runs. Schema-aware features aren't supported. The debugger needs SaxonJ 12 or later (any edition); SaxonJS and SaxonC stylesheets can be run, but not debugged.

Edit XML, XSLT or XPath

If you're new to VS Code, see Using VS Code for a brief introduction. For a comprehensive guide to Visual Studio Code's editor features, see the Microsoft Visual Studio Code User Guide.

Language Specific Features

XSLT and XPath specific editing features are described in the Editing XSLT/XPath page of the documentation that you are reading now. More general XML editing features are described in the Editing XML section.

The features described at Editing XSLT/XPath include:

- IntelliSense (Auto Complete)
- Syntax Highlighting
- Code Formatting
- Code Refactoring
- Symbol Renaming
- Hover Help and Signature Help
- XSLT/XPath Linter

Run XSLT - Quick Start

The quickest way to run the stylesheet in the active editor is Quick Run: press ⌘⌥R (Ctrl+Alt+R on Windows/Linux) or click the play button in the editor title bar. The first run creates an XSLT task for you, which later runs reuse.

To step through the stylesheet instead, set a breakpoint and click the debug button in the editor title bar, or press F5 when there's no `launch.json` - see the Debugger.

XSLT transforms are run using Visual Studio Code Tasks. Before configuring an XSLT task you should ensure that you open a Workspace folder (⌘O) from Visual Studio Code. The configuration is saved in a `tasks.json` file in this workspace folder.

To create your first task, use the `Run Build Task` command (⇧⌘B). As there is no currently configured task, you should be prompted to configure one (if another extension inhibits this, run **Configure Task** instead). You can then select the **Saxon Transform (New)** (SaxonJ), **Saxon-JS Transform (New)** (SaxonJS) or **SaxonC Transform (New)** (SaxonC) task.

A task will be added to the `tasks.json` task with default properties. To further configure the task, these properties can be changed or other properties added. Auto-completion and hints are provided to assist in the task configuration. Next time you press ⇧⌘B your task will be run in a **Task** tab in the Terminal window (toggled with ⌘J).

More on running XSLT

XSLT tasks are explained in full on the Tasks page. More general information can be found in the Visual Studio Code Tasks documentation.

What's New

Highlights of the features added in releases 2.0 and 2.1:

Release 2.1

- **Step-through debugging** - debug stylesheets with SaxonJ (any edition, including Saxon-HE): breakpoints with conditions, hit counts and logpoints, stepping through templates, functions and multi-line XPath expressions - including inline functions - and the variables, context item and output so far, as you go
- **XSpec tests** - run and debug XSpec tests from the Test Explorer, for stylesheets and Schematron, with each failure's expected and actual results side by side, and breakpoints in the stylesheet or the XSpec file. XSpec files are checked as you edit them, against XSpec's schema and the stylesheet they test, with auto-completion, hover help and navigation between the tests and the stylesheet. XSpec 4.1.2 is included
- **XSLT packages** - Workspace packages are automatically discovered from their `xsl:package` name and version and supplied to Saxon when running or debugging stylesheets. Component visibility, `xsl:expose`, and `xsl:accept` are handled exactly as Saxon does. Issues with `xsl:accept`, `xsl:override`, and package modes are flagged while you edit, with quick fixes available.
- **Find Nodes with XPath** - run XPath expressions on an XML file from a query file below it, with the XPath editor's checks and auto-completion: the nodes found are listed to step through, and values are shown after each expression. Replaces **Goto XPath**
- **MCP server for AI assistants** (experimental) - tools an assistant such as Claude Code or GitHub Copilot can call to check stylesheets that aren't open, find declarations and references, show a module's imports, and describe item types and built-in functions
- **Tracing values with xdm:trace()** - trace a value anywhere within an XPath expression: like `trace()`, it returns the value unchanged, but writes it formatted as `xdm:debug()` does. **Wrap with xdm:trace()** adds it around a selected expression, and **Remove xdm:trace() - keep the expression** removes the call and its other arguments, leaving just the expression
- **Converting a library to a package** - **Analyse as Package...** reports what would have to change for a library that stylesheets import to become a package, and **Convert to Package...** makes the changes across the workspace - the library's visibility, the stylesheets' overrides, even a shared unnamed mode - shown in the Refactor Preview before they're made

Release 2.0

- **Full language support for XSLT 4.0 in Saxon 13** - type-aware auto-completion, checks, hover help and refactoring for the XSLT 4.0 and XPath 4.0 features Saxon 13 implements: new syntax and item types, keyword arguments and optional parameters, `xsl:switch`, and documentation notes with `xsl:note`
- **Code Documentation** - documentation within `xsl:note` elements provides hover assistance using tags based on xqdoc along with Markdown descriptions. In an XSLT 3.0 stylesheet, `xsl:note` needs Saxon PE/EE 12.8 or later and the `XSLT.validation.xslt30ItemTypesAndNotes` setting.
- **Records and Enums** - auto-completion, checks, hover help and refactoring for XPath 4.0 record and enumeration types, from map constructors to `xsl:switch` cases
- **JSON and JNodes** - XPath 4.0 path expressions on JSON, with auto-completion, checks and hover help for steps on JNodes with a record type, and JNode paths shown by `xdm:debug()`

- **Quick Run** - run the active stylesheet with a shortcut or the toolbar button, without configuring a task first - with SaxonJ, SaxonJS or the new SaxonC support
- **Inferred imports** - the top-level stylesheet of a module is found from the workspace, without opening it first, and the new **XSLT Imports** view shows its tree of imports and includes
- **XML Catalogs** - resolve the `href` of `xsl:import` and `xsl:include`, e.g. the `http:` URI of a shared library, to a local file with an OASIS XML catalog - also passed to Saxon when a stylesheet is run
- **Editing** - signature help and richer hover help for functions and templates, more auto-completion, and the new Wrap with... refactoring
- **Print debugging** - one auto-completed `xsl:message` for all the in-scope variables, formatted and colored with `xdm:debug()`

Features

The extension's features at a glance, grouped by task. Each one links to the part of this guide that describes it.

XSLT and XPath languages

- XSLT 3.0 and XPath 3.1
- XSLT 4.0 and XPath 4.0, as implemented in Saxon 13
- Advanced XSLT 3.0 package features
- SaxonJS interactive extensions (IXSL)
- Fast semantic highlighting, from a full XPath parser
- XML editing, formatting and navigation

XSLT 4.0

- Record and enum types, with type-aware checks
- Named item types: `xsl:item-type`
- `xsl:switch`, with quick fixes for missing cases
- Keyword arguments and optional parameters
- Functions in no namespace
- Path expressions on JSON (JNodes)

Editing

- Context-aware auto-completion, including node names from the XML context file
- Signature help and hover help
- Snippets, including new stylesheets for 3.0 and 4.0
- Formatting of instructions and multi-line XPath
- Rename symbol, across imported modules
- Code actions for quick fixes and refactoring

Linter

- XPath syntax and XSLT instructions, checked as you type
- Types, records, enums and function arguments
- Unused XSLT and XPath variables or parameters
- References across imported modules
- Saxon errors from runs, in the Problems view
- Quick fixes for common problems
- MCP server for AI assistants (experimental)

Navigation

- Go to (or peek) definition, and find references
- Call hierarchy of functions and templates
- Go to the declarations of record fields
- Outline, breadcrumbs and Go to Symbol
- Links for `xsl:import` and `xsl:include`
- XSLT Imports view, with inferred top-level stylesheets
- Find nodes with XPath in XML files

Refactoring

- Safe symbol renaming
- Extract `xsl:function` or `xsl:template`
- Extract `xsl:variable` from an XPath expression
- Extract a record type from a map
- **Wrap with...** an instruction (`⌥⇧W` / `Alt+Shift+W`)
- Add a documentation note

Running XSLT

- **Quick Run** (`⌘⌥R` / `Ctrl+Alt+R`)
- Tasks for SaxonJ, SaxonJS and SaxonC
- Required `xsl:param` values asked for when a task runs
- `xsl:param` values controlled by tasks
- XML catalogs for `xsl:import` URIs
- XSLT Packages - no configuration needed
- File pickers for task inputs
- Java extension functions

Debugging

- Step-through XSLT debugger (SaxonJ, any edition)
- Stepping through multi-line XPath
- Conditional breakpoints and hit counts
- Logpoints, with formatted and colored values
- Views of the locals, globals, context and output
- Watch expressions, hovers and the Debug Console
- The output so far, mapped back to the XSLT that wrote it
- Formatted and colored print debugging

Testing with XSpec

- Test Explorer for running tests and comparing output
- Debug tests in the XSpec file or from the XSLT source
- Run Schematron tests, compiled with SchXslt2
- Generate tests for a function or template
- Edit XSpec with checks, quick-fixes, auto-completion
- Test coverage percentages and line margin highlights
- Navigate between tests and the underlying code
- Checks and assistance for component calls

Code documentation

- `xsl:note` with Markdown and xqDoc-style tags
- Shown in hover and signature help
- `@see` references, with navigation and rename
- Checks of `@param` and `@field` tags
- Module notes for stylesheets
- Highlighting and auto-completion for tags and references

Using VS Code

These pages complement the extensive Visual Studio Code documentation that begins:

> *Visual Studio Code is a lightweight but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support for JavaScript, TypeScript and Node.js and has a rich ecosystem of extensions for other languages...*

The User Interface

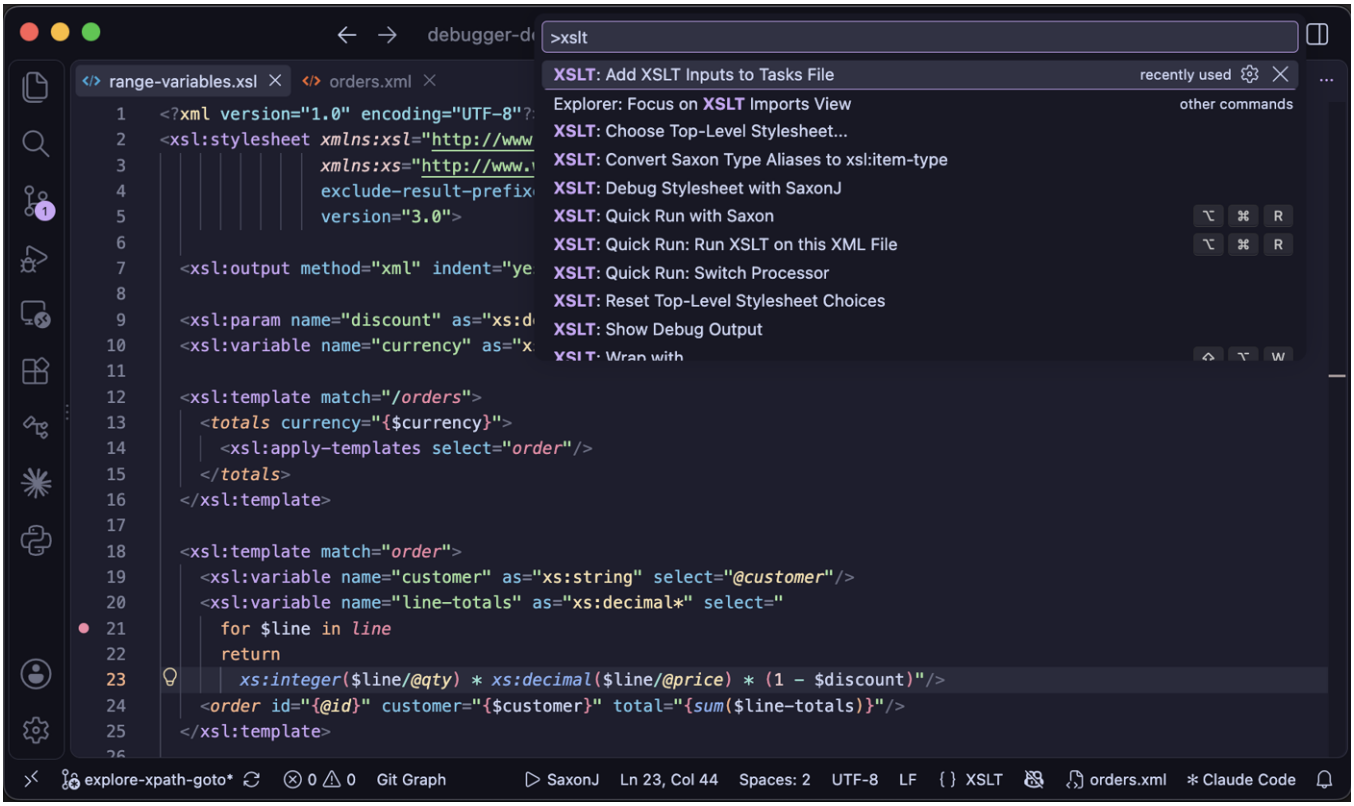
Visual Studio Code's user interface has a different look and feel to that of legacy XSLT editors. The Command Palette, along with keybindings to most common commands, play a significant role in providing a clearer environment to work in. User interface clutter is significantly reduced by eschewing previous designs which depended on a set of toolbars, each with an abundance of buttons.

The application **menu bar** and associated dropdown menus usefully group the more essential actions for managing your code. The **View** menu is particularly useful as it logically sets out the large set of view-configuration commands.

The sections below introduce you to the fundamental parts of the user interface from the perspective of an XSLT/XPath developer.

The Command Palette

A principal part of Visual Studio Code's user interface is the Command Palette, triggered by the keyboard shortcut ⇧⌘P. The Command Palette shows all available commands in a dropdown list. The list of commands is filtered as you type. The most recently used commands are shown at the top of the list.



VS Code's Command Palette - built-in commands are supplemented by specialist XSLT/XML commands.

Opening Workspace Files

The **File: Open Folder** command is used to select the starting 'Workspace' folder to be used by VS Code. More folders can be added to the Workspace subsequently using the **Workspaces: Add folder to workspace** command. The keyboard shortcut `⌘P` provides quick-access to all Workspace files from a pick-list. 'Fuzzy-matching' allows you to type in just a part of the filename to filter the pick-list files.

Keyboard Shortcuts

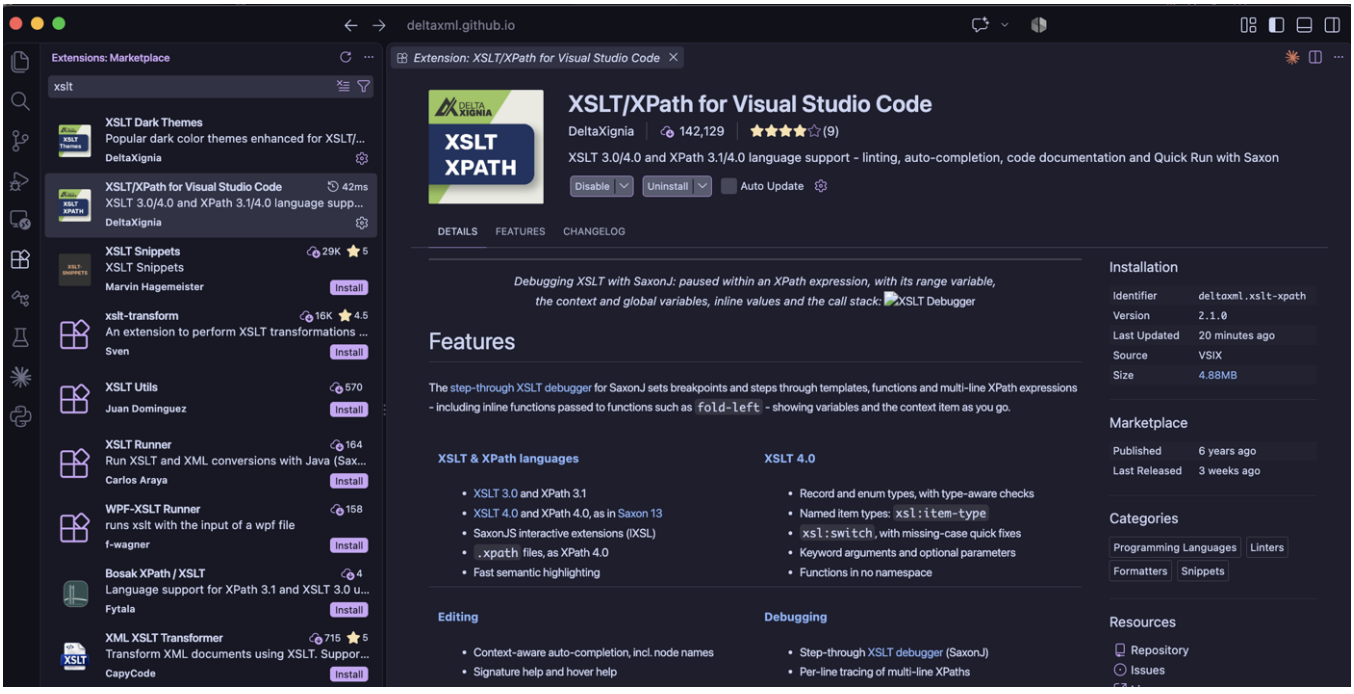
While the Command Palette provides convenient access to features you should find that keyboard shortcuts are more effective for commands you use most frequently. The keyboard shortcut for commands is shown alongside each listed command in the Command Palette. Not all commands have key bindings, but these can be configured using the command `Preferences: Open Keyboard Shortcuts` (`⌘K⌘S`). See: Key Bindings for Visual Studio Code

Settings

By adding or modifying Visual Studio Settings, using the `Preferences: Open User Settings` command, you can customize a range of editor and extension features to meet your needs. See the Settings section for details on some of the more important settings for developing in XSLT and XPath.

Install the XSLT/XPath Extension

Extensions are managed in the Extensions view (`⇧⌘X`) of the Side Bar. Type `XSLT XPath` in the Extensions view search bar and, on the **XSLT/XPath for Visual Studio Code** entry, press the `install` button.



Color Theme Extensions

Visual Studio Code includes a small set of light and dark color themes. Alternative color themes are installed as extensions. The number of available themes can be overwhelming, but VS Code has a built-in Theme Preview feature and there are websites like vscodethemes that provide a preview before installing the theme.

The XSLT Dark Themes extension provides a set of popular dark color themes specially enhanced for XSLT and XPath

> *The XSLT/XPath extension syntax highlighter uses Semantic Highlighting that should be enabled specifically in Settings to work with external color themes. Highlighter tokens are defined to provide effective highlighting for all the common color themes.*

Viewing HTML Output

When your XSLT generates HTML, you can view the result in Visual Studio Code's integrated browser, with no extension needed. When a transform writes an HTML result file, the notification that the task has completed has an **Open in Browser** button, as well as **Open** for the file's source. You can also choose **Open in Integrated Browser** from the editor title bar of an HTML file, or from its context menu in the Explorer. With the `workbench.browser.autoReloadOnFileChange` setting, the page reloads each time the transform writes the result file again - so you can edit the stylesheet, run it with Quick Run, and see the updated output alongside the XSLT.

Settings

Visual Studio Code's settings let you tailor your working environment to fit requirements for specific languages or extensions, or to suit your own preferences.

A general guide to settings is provided by the Visual Studio Code documentation: User and Workspace Settings. This page lists the XSLT/XPath extension's own settings, and the editor settings of most interest for XSLT and XPath.

Creating User and Workspace Settings

From the Command Palette (⇧⌘P) you can either invoke `Preferences: Open User Settings` or `Preferences: Open Workspace Settings`. Editor settings can also be applied in a language-specific way: for this, use the command `Preferences: Configure Language Specific Settings`.

Settings are either edited in the Settings form or in a JSON editor. Open the JSON editor by clicking the **Open Settings (JSON)** button at the top right of the Editor Group. This opens the corresponding `settings.json` file directly.

User settings apply wherever you use VS Code, so they suit paths on your own computer, such as the Saxon jar. Workspace settings are saved in the project's `.vscode/settings.json` and apply to everyone who opens the project, so they suit settings a team shares, such as an XML catalog or XSLT packages.

XSLT/XPath Extension Settings

To list the extension's settings in the Settings editor, type `XSLT` in its search box. All of them are listed below, with their defaults. The sections after the tables describe the most important ones.

Running and debugging

Setting	Default	What it does
`XSLT.tasks.saxonJar`	-	The path of your Saxon jar file, needed to run or debug stylesheets with SaxonJ - see The Saxon jar.
`XSLT.tasks.quickRunProcessor`	`"xslt"`	The processor Quick Run uses: `"xslt"` (SaxonJ), `"xslt-js"` (SaxonJS) or `"xslt-c"` (SaxonC). Switching processor in Quick Run saves the choice here.
`XSLT.tasks.java.enabled`	`true`	Offer SaxonJ (`xslt`) tasks - see Task types.
`XSLT.tasks.js.enabled`	`true`	Offer SaxonJS (`xslt-js`) tasks.
`XSLT.tasks.saxonc.enabled`	`true`	Offer SaxonC (`xslt-c`) tasks.
`XSLT.tasks.saxonCPath`	-	The folder containing the SaxonC command-line executable (`Transform` or `Transform.exe`) - see SaxonC tasks.
`XSLT.tasks.saxonCLicenseFileLocation`	-	The path of SaxonC's `saxon-license.lic` file, passed to SaxonC as `--licenseFileLocation:` - see Licensing.
`XSLT.tasks.saxonCLibraryPaths`	-	More folders to search for SaxonC's native libraries. The `lib` folder beside the executable is always included - see Native library path.

Setting	Default	What it does
`XSLT.tasks.htmlParserJar`	-	The path of the nu.validator htmlparser jar, added to SaxonJ's classpath for XSLT 4.0 stylesheets, for Saxon's `parse-html()` function.
`XSLT.debug.openOutput`	`false`	When debugging, open the output written so far beside the stylesheet, at the first pause where there is some - see Output while debugging.

Editing and checking

Setting	Default	What it does
`XSLT.resources.catalog`	`""`	An OASIS XML catalog file, for resolving the `href` of `xsl:import` and `xsl:include` - also passed to Saxon - see XML catalog.
`XSLT.resources.xsltPackages`	-	The files for `xsl:use-package` names - see XSLT packages.
`XSLT.resources.inferParentFromWorkspace`	`true`	For a module that another stylesheet imports or includes, find that stylesheet from the workspace's files, so references to its declarations are resolved without opening it first - see Inferred imports.
`XSLT.resources.useSourceFile`	`"latest"`	Completion of element and attribute names in XPath: `"latest"` suggests the names that can occur at that point in the path, from the XML context file; `"none"` suggests the names already used in the stylesheet - see IntelliSense for XPath locations.
`XSLT.validation.externalFunctionPrefixes`	`[]`	Prefixes of functions the processor provides, such as Java extension functions - calls of functions with these prefixes aren't checked. The quick fix for a function that isn't found adds its prefix - see Java extension functions.
`XSLT.validation.xslt30ItemTypesAndNotes`	`false`	Allow XSLT 4.0 named item types and `xsl:note` in XSLT 3.0 stylesheets, for Saxon PE and EE 12.8 or later - see below.
`XSLT.validation.xslt40UnionPriorities`	`false`	Warn in every stylesheet before XSLT 4.0 about union patterns whose branches have different default priorities - not only where XSLT 4.0's rule applies already - see Union patterns and XSLT 4.0.

XSpec tests

Setting	Default	What it does
`XSLT.xspec.home`	-	The folder of another XSpec installation, e.g. a newer release, to compile and run tests with - by default the XSpec 4.1.2 included. See XSpec tests.
`XSLT.xspec.allowSyntaxExtensions`	`"auto"`	Run tests with Saxon's syntax extensions (`--allowSyntaxExtensions:on`), so the stylesheet tested can use XPath 4.0 syntax, or e.g. record types in an XSLT 3.0 stylesheet: `"auto"` - on only for tests of a stylesheet that's XSLT 4.0, or uses item types or notes,

Setting	Default	What it does
		and not with Saxon-HE, which fails every transform with them on - `"on"` or `"off"`. See XSpec tests.
`XSLT.xspec.legacyCompatibility`	`false`	Run tests written for XSpec before 1.0 (e.g. 0.5) as that ran them, where XSpec 4 stops the run - see Tests written for older XSpec.
`XSLT.xspec.emphasiseLabels`	`true`	In XSpec files, show scenario labels in bold and expectation labels in italic, so the scenarios read like headings.
`XSLT.xspec.fadePending`	`true`	In XSpec files, fade the pending scenarios and expectations, other than the reason they're pending.

AI assistants

Setting	Default	What it does
`XSLT.mcp.enabled`	`false`	Run the extension's MCP server, with tools for AI assistants such as GitHub Copilot and Claude Code - see The MCP server.
`XSLT.mcp.port`	`6745`	The port the MCP server listens at, on `127.0.0.1`. With more than one window open, the first one opened has the server.

The Saxon jar

To run or debug stylesheets with SaxonJ - Saxon for Java - set `XSLT.tasks.saxonJar` to the full path of your Saxon jar file, usually as a user setting, since the path is specific to your computer. Any Saxon edition works, including Saxon-HE. The Debugger needs Saxon 12 or later, and a Java runtime must be on the PATH:

```
"XSLT.tasks.saxonJar": "/Users/me/saxon/SaxonHE12-9J/saxon-he-12.9.jar"
```

On Windows, write each `\` in the path as `\\`, as JSON requires - e.g. `"C:\\saxon\\saxon-he-12.9.jar"`. For Saxon-PE or Saxon-EE, keep the licence file, `saxon-license.lic`, in the same folder as the jar.

XSLT tasks refer to the setting as `${config:XSLT.tasks.saxonJar}`. For downloading Saxon and Java, see SaxonJ tasks.

XSLT packages

Packages in the workspace are found without a setting. For a package outside the workspace, add its path to `XSLT.resources.xsltPackages` - its name and version are read from its `xsl:package` element. See XSLT Packages.

```
"XSLT.resources.xsltPackages": [
  { "path": "../shared-xslt/strings.xsl" }
]
```

XML catalog

To resolve the `href` of `xsl:import` and `xsl:include` - and the URIs of `fixed-namespaces` - with an OASIS XML catalog, and pass the catalog to Saxon when a stylesheet is run, set `XSLT.resources.catalog` to the catalog file's path, relative to the workspace folder. See XML Catalogs.

```
"XSLT.resources.catalog": "catalog.xml"
```

Task types

The extension provides a task type for each Saxon processor - `xslt` for SaxonJ, `xslt-js` for SaxonJS and `xslt-c` for SaxonC - and all three are enabled by default. To stop a processor's tasks being offered, for example in **Run Build Task**, turn off its setting:

```
"XSLT.tasks.java.enabled": true,
"XSLT.tasks.js.enabled": true,
"XSLT.tasks.saxonc.enabled": true
```

See Tasks for configuring each task type.

XSLT 4.0 Item Types and Notes in XSLT 3.0

Saxon PE and EE 12.8 or later, run with syntax extensions, accept XSLT 4.0's named item types and `xsl:note` in a stylesheet with `version="3.0"`. To edit such stylesheets with the same support as for XSLT 4.0, enable the setting `XSLT.validation.xslt30ItemTypesAndNotes` - it's off by default. For a team, add it to the workspace settings, in the project's `.vscode/settings.json`:

```
"XSLT.validation.xslt30ItemTypesAndNotes": true
```

With the setting, in an XSLT 3.0 stylesheet:

- `xsl:item-type` declarations are accepted, with `record(...)` and `enum(...)` types - which can also be used directly in `as` attributes
- the record and enumeration type support applies: completions of record fields, map keys and enumeration values, checks of map constructors, lookups and values, hover help, and renaming a record field
- `xsl:note` needs no `use-when` attribute - see Code Documentation

Other XSLT 4.0 and XPath 4.0 features - such as `xsl:switch`, keyword arguments and `fn(...)` types - still need `version="4.0"`, and are reported as errors in an XSLT 3.0 stylesheet.

Without the setting, `xsl:item-type` and record and enumeration types are reported as errors in an XSLT 3.0 stylesheet, and an `xsl:note` gets a warning unless it's excluded with `use-when="false()"`, as an XSLT 3.0 processor reports `XTSE0010` for it.

To run the stylesheets, Saxon needs syntax extensions: the `--allowSyntaxExtensions:on` option on the command line, or `allowSyntaxExtensions="true"` in a Saxon configuration file. The extension's XSLT tasks and Quick Run enable it with the default `"allowSyntaxExtensions40": "auto"` task property, for Saxon PE and EE, when the stylesheet - or a module it imports, includes or uses - has item types or notes.

Converting Saxon type aliases

Saxon's earlier syntax for named types - `saxon:type-alias`, `tuple(...)` types and `~name` for a reference to a type alias - is ignored by Saxon 12.8 and later, which only report the references. The linter warns about a `saxon:type-alias`, and the command **XSLT: Convert Saxon Type Aliases to xsl:item-type** - also the warning's quick fix - converts them in all the workspace's files, shown for review in the refactor preview first:

- `<saxon:type-alias name="dfx:bounds" type="..."/>` becomes `<xsl:item-type name="dfx:bounds" as="..."/>`

- `tuple(t: xs:integer, label?: xs:string)` becomes `record(t as xs:integer, label? as xs:string)`
- `as="~dfx:bounds"` becomes `as="dfx:bounds"`

A type alias with an extensible tuple type, e.g. `tuple(a: xs:string, *)`, isn't converted - nor is one that references it - as XSLT 4.0 has no extensible record types. These are listed after the conversion, to convert by hand.

General Settings for XSLT

Default Editor Settings for XSLT, XPath and XML

The extension sets these VS Code editor settings for the `xslt`, `xpath` and `xml` languages - for `.xsl`, `.xslt`, `.xpath` and XML files. As they're set for each language, they don't affect files in other languages. Any of them can be changed in a language-specific setting, e.g. `"[xslt]": { "editor.tabSize": 4 }` - see Creating User and Workspace Settings.

Setting	XSLT and XPath	XML	What it does
`editor.defaultFormatter`	`"deltaxml.xslt-xpath"`		This extension formats the file - see Code Formatting.
`editor.formatOnType`	`true`		Lines are indented as you type, e.g. a new line, or a closing tag.
`editor.formatOnPaste`	`true`		Pasted text is indented to fit where it's pasted.
`editor.formatOnSave`	`false`		The file isn't formatted when it's saved.
`editor.formatOnSaveMode`	`"file"`		If `formatOnSave` is turned on, the whole file is formatted - or only the lines changed, with `"modifications"`.
`editor.tabSize`	`2`	`4`	The number of spaces for each level of indentation.
`editor.insertSpaces`	`true`		Indentation uses spaces, not tabs.
`editor.detectIndentation`	`true`		For an existing file, the tab size and spaces or tabs are detected from its content, instead of the two settings above.
`editor.wordSeparators`	`"`~!@#%^&*()=+[{]}\\\|;'\",:<>/?$"`		Unlike VS Code's default, `-` and `.` aren't word separators, so double-clicking selects a whole name, such as `current-group` - see Word Separators.
`editor.semanticHighlighting.enabled`	`true`	-	Syntax highlighting for XSLT and XPath, which uses semantic tokens - see Syntax Highlighting. It's also set for the `xsl` language.
`outline.showArrays`	`false`		The Outline view doesn't show arrays - see Code Navigation.

The default settings, as JSON

```
"[xslt]": {
  "editor.defaultFormatter": "deltaxml.xslt-xpath",
  "editor.formatOnSaveMode": "file",
  "editor.formatOnSave": false,
  "editor.formatOnPaste": true,
  "editor.formatOnType": true,
  "editor.semanticHighlighting.enabled": true,
  "editor.detectIndentation": true,
  "editor.wordSeparators": "`~!@#%^&*()=+[{]}\\|;'\",:<>/?$",
  "editor.insertSpaces": true,
  "editor.tabSize": 2,
  "outline.showArrays": false
},
"[xpath]": {
  "editor.defaultFormatter": "deltaxml.xslt-xpath",
  "editor.formatOnSaveMode": "file",
  "editor.formatOnSave": false,
  "editor.formatOnPaste": true,
  "editor.formatOnType": true,
  "editor.semanticHighlighting.enabled": true,
  "editor.detectIndentation": true,
  "editor.wordSeparators": "`~!@#%^&*()=+[{]}\\|;'\",:<>/?$",
  "editor.insertSpaces": true,
  "editor.tabSize": 2,
  "outline.showArrays": false
},
"[xml]": {
  "editor.defaultFormatter": "deltaxml.xslt-xpath",
  "editor.formatOnSaveMode": "file",
  "editor.formatOnSave": false,
  "editor.formatOnPaste": true,
  "editor.formatOnType": true,
  "editor.wordSeparators": "`~!@#%^&*()=+[{]}\\|;'\",:<>/?$",
  "editor.detectIndentation": true,
  "editor.insertSpaces": true,
  "editor.tabSize": 4,
  "outline.showArrays": false
},
"[xsl]": {
  "editor.semanticHighlighting.enabled": true
}
```

Built-in Editing Behaviour for XSLT

The extension also defines how the editor treats XSLT files in ways that aren't settings, so they can't be changed:

- **Comments** - **Toggle Block Comment** (⇧⌥A) wraps the selection in `<!--` and `-->`. In `.xpath` files it uses XPath's `(:` and `:)`.
- **Auto-closing** - typing `{`, `[`, `(`, a quote or a backtick adds the closing character. So does `&`, which adds `;` for an entity reference, and `(:`, which adds `:` to start an XPath comment.
- **Surrounding** - typing one of these characters with text selected surrounds the selection with the pair, e.g. selecting `$total` and typing `(` gives `($total)`.
- **Words** - a prefixed name such as `xsl:template`, a variable reference such as `$total-price`, and an attribute name test such as `@select` each count as one word - for example, when VS Code finds the word at the cursor for word-based suggestions.

- **Folding regions** - `<?region?>` and `<?endregion?>` processing instructions mark a region that can be folded - see Folding.

Code Formatting

The following settings directly affect code formatting in the XSLT/XPath extension:

```
"editor.defaultFormatter": "deltaxml.xslt-xpath",
"editor.formatOnSaveMode": "modifications",
"editor.formatOnSave": true,
"editor.formatOnPaste": true,
"editor.formatOnType": true
```

The `editor.formatOnSaveMode` setting can be used to only format XSLT lines changed since the last commit on Git.

Syntax Highlighting

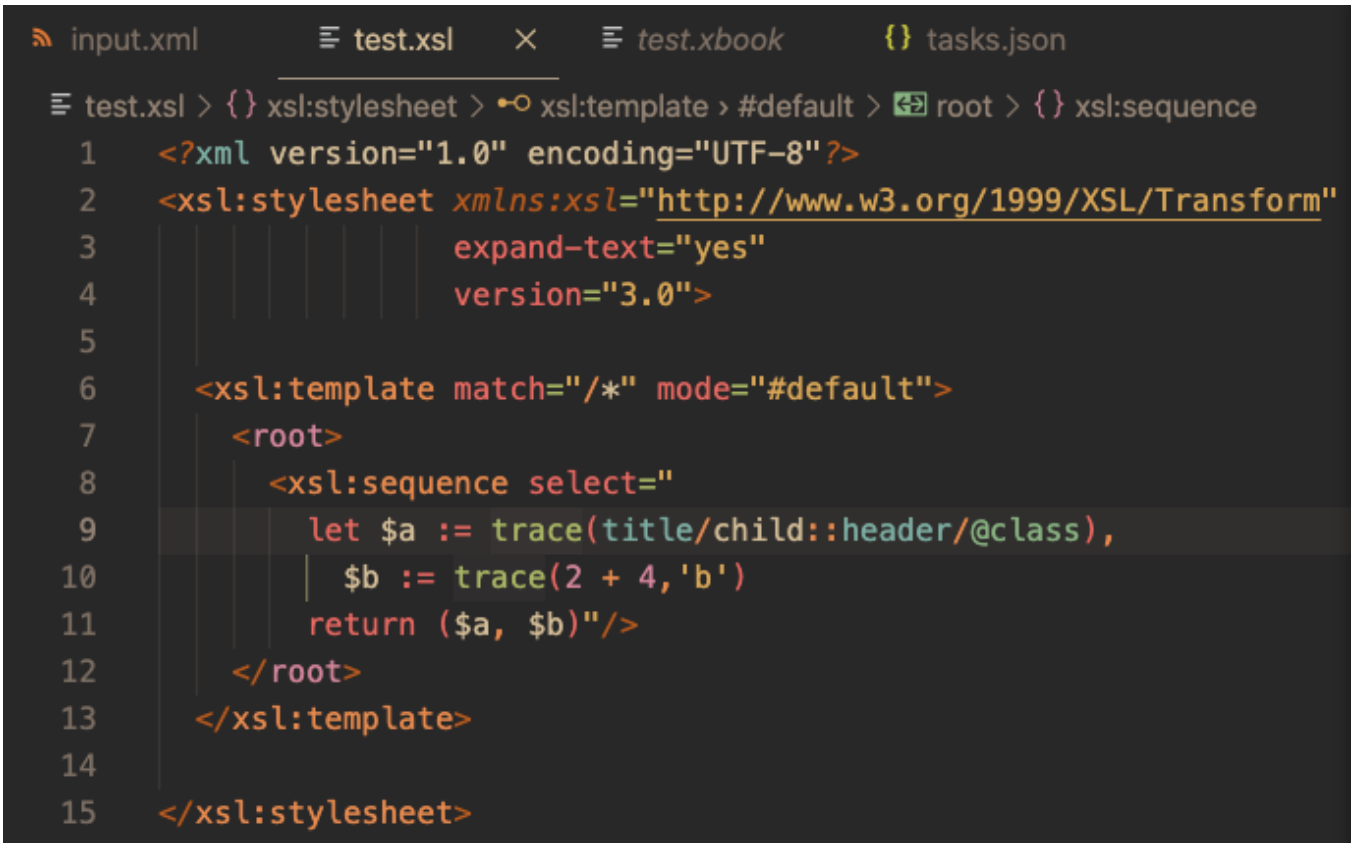
Syntax highlighting for XSLT and XPath is achieved using Visual Studio Code's Semantic Highlighting tokens alone. Almost all other languages in Visual Studio Code use a TextMate Grammar for tokens, which may be supplemented by Semantic Highlighting.

If XSLT and XPath are not syntax highlighted in a Color Theme, you can explicitly enable Semantic Highlighting with the setting:

```
"editor.semanticHighlighting.enabled": true
```

The XSLT Dark Themes extension provides a selection of popular dark color themes, enhanced for XSLT/XPath.

Screenshot of Gruvbox Theme extended for XSLT



X Gruvbox Material Dark theme

Word Separators

The `editor.wordSeparators` setting affects how the full 'word' is selected when you double-click on a part of the word. The characters listed are used as word separators. Typically in XSLT, you do not want characters like `-` and `.` treated as word separators, as these characters are valid within a QName like an attribute name or a variable name.

The Agents Window

Visual Studio Code's Agents window only runs the extensions on a list chosen by Microsoft, and the XSLT/XPath extension isn't on it yet - so, there, XSLT files have plain XML highlighting, with none of the extension's features. To run the extension in the Agents window, add this to your user settings:

```
"extensions.supportAgentsWindow": { "deltaxml.xslt-xpath": true }
```

The Agents window uses its own profile of extensions, so an extension you've disabled elsewhere - such as Red Hat's XML extension - may be enabled there. See AI Assistants.

Emmet Snippets

Emmet Snippets provide a convenient shorthand for inserting a number of literal result elements and attributes into your XSLT in one go. For example, entering:

```
ul>li.item$*2
```

provides the following output:

```
<ul>
    <li class="item1"></li>
    <li class="item2"></li>
</ul>
```

To enable Emmet snippets for XSLT, add the following setting:

```
"emmet.includeLanguages": {
  "xslt": "xml"
}
```

Editing XML

XSLT developers benefit from a set of XML editing features that are also available for any XML file. These general purpose features are described here.

XML Elements

Rename Tag

As you edit the start tag name of an element, the corresponding end tag name is changed at the same time to keep the XML well-nested.

Self-Close Element

To convert an element with a start-tag and end-tag to a self-closed element: With the cursor immediately to the left of the `>` at the end of the start-tag, type `/`. Any element contents and the end tag will be deleted automatically:

```
<d><para/>test</para</d>
--> <d><para/></d>
```

Rename all elements with selected name

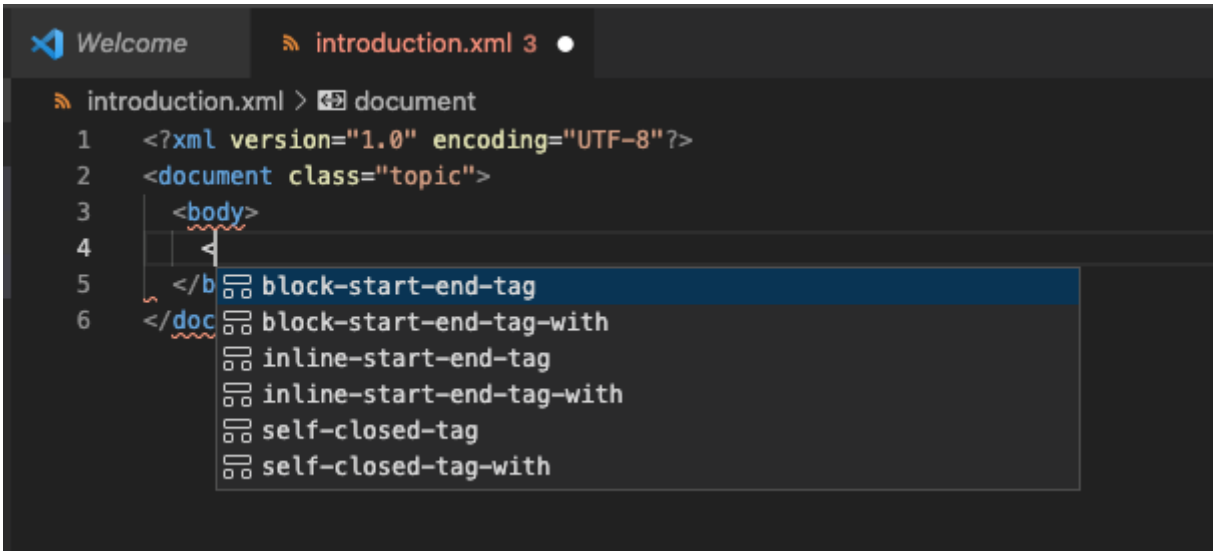
Select the element tag name you want to rename by double-clicking it, then press (⇧⌘L). This selects all matching tag names so you're in a multi-cursor edit mode. As you edit the tag name you selected all matching tag names are edited also. When done, press `Escape` to exit multi-cursor edit.

Auto tag-close

Element tags are auto-closed when typing `</` when an element is unclosed. Note that this feature is only enabled when the VS Code user-setting *Format on Type* is set (indentation is fixed at the same time).

Auto-Completion

Outside of the auto-complete (IntelliSense) features for XSLT and XPath, a small set of snippets for general purpose XML elements is accessible from the auto-completion list:



After a `<` within an element, the tag snippets insert an element with a placeholder for its name, which is used for both the start and end tags:

- `block-start-end-tag` - start and end tags on separate lines, with the cursor on the indented line between them
- `inline-start-end-tag` - start and end tags on the same line, with a placeholder for the text between them
- `self-closed-tag` - a self-closed element

Each has a `-with` variant, such as `block-start-end-tag-with`, that also inserts an attribute. Before the root element of a file, the `XML Root Element Snippet` inserts an XML declaration and a root element with an attribute and a child element.

With the cursor in the element start tag where you wish to add the attribute, press `^Space` and select `attribute` to insert the code snippet:

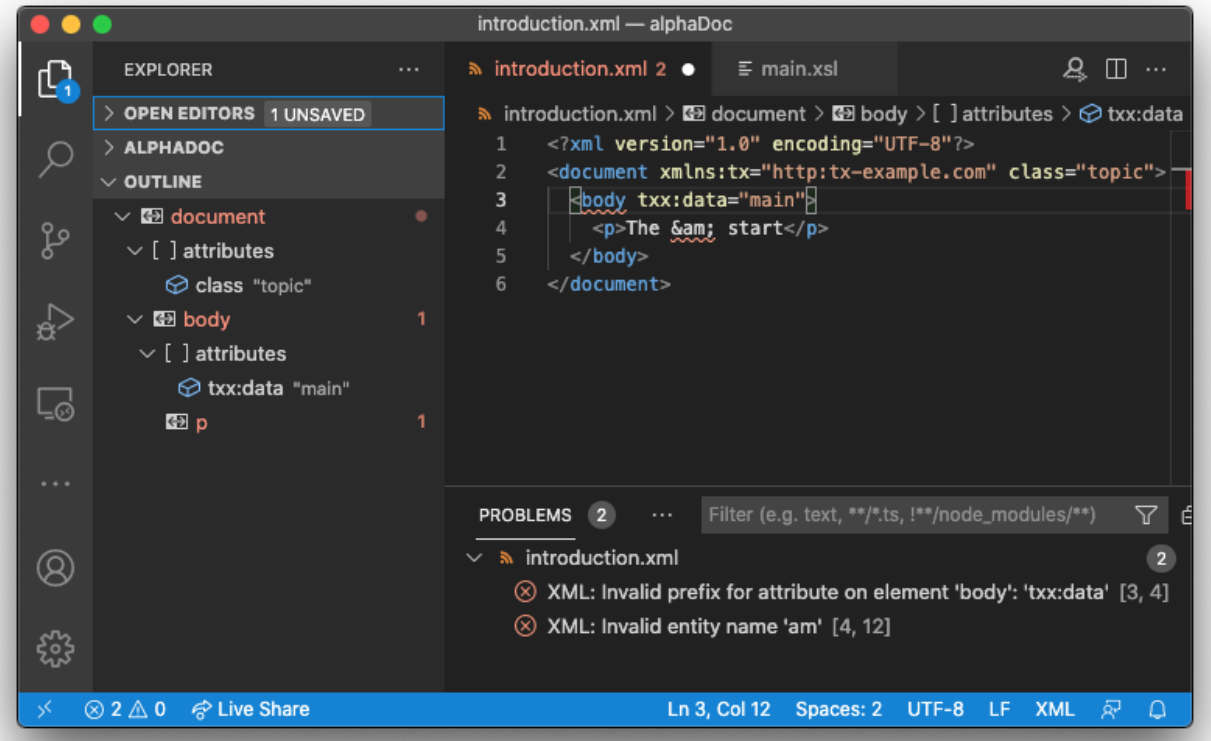
```
name="value"
```

Edit the `name` and then press `Tab` to edit the `value`.

Emmet

Emmet can be used to insert XML snippets using a shorthand notation. Emmet should be enabled in user settings. See the Settings section for an example.

Well-Formed XML Checks



As you edit an XML file, the syntax is checked for any well-formedness errors. Any problem tokens are highlighted with a squiggly underline in the editor. To see a description of the problem, hover over a highlighted token. To help navigate between problems, the Problems view (press `⇧⌘M` to toggle) shows a list of XML syntax problems. Problems are also highlighted in the Outline View. Selecting a Problems view list item, or Outline View node will navigate you to the corresponding token in the XML editor.

Entities

An XML or XSLT file can declare entities in its DOCTYPE's internal subset, to use as references such as `&company;`:

```
<!DOCTYPE xsl:stylesheet [
  <!ENTITY company "DeltaXignia Ltd">
]>
```

Each reference is checked: it must be one of the predefined entities - `<`, `>`, `&`, `"` and `'` - a character reference such as ` `, or an entity the internal subset declares. When the DOCTYPE names an external DTD, or the subset uses a parameter entity reference such as `%chars;`, any entity name is allowed, as its declarations can't all be seen.

- Type `&` to auto-complete an entity name: the declared entities, with their values, and the predefined entities
- Hover over a reference to see its entity's value, or the file of an external entity - or, for a character reference, the character's code point, such as U+00A0
- Go to Definition (`F12`) on a reference goes to its `<!ENTITY>` declaration

In the DOCTYPE, the names of declared entities are colored as references are, and quoted values as strings.

XML Formatting

When you have an XML file open, normal XML formatting features are available. With typical hand-edited XML, only the indentation is changed by the formatting. However, if no newline characters are found at the start of the XML file, it is assumed the contents are system generated and newline characters are added for elements that are not mixed content.

The formatter observes the normal rules for whitespace preservation in XML. So `xml:space` attributes affect whether indentation is preserved. Whitespace in `CDATA` sections and 'significant' whitespace in content inferred to be *mixed content* is also preserved.

XML attributes on newlines and multi-line attribute value are indented by the XML formatting provider to reflect the level of nesting.

The **XML: Format XML with newlines inside mixed-content** command, from the Command Palette (`⇧⌘P`), formats the whole file with a newline added before each start tag and end tag that doesn't already start a line, and indents the result - including inside mixed content, where whitespace is otherwise preserved.

Tab on an Empty Line

In an XML or XSLT file, pressing `Tab` on an empty or whitespace-only line, such as one you've clicked on, indents the line for its context, as if you had just pressed `Enter`, with the cursor at the end of the indentation. Elsewhere, or where the line is already indented that far, `Tab` works as normal. The key binding is for the **Tab: Indent an Empty Line for Its Context** command, and can be changed in the Keyboard Shortcuts editor.

XML Element Selection

The XSLT/XPath extension provides a set of commands for selecting an element. These commands can be invoked from the Command Palette (`⇧⌘P`), or from the **XML Selection** submenu of the editor's context menu (right-click), which also shows their keyboard shortcuts. From the keyboard, **XML: Select Element...** (`⇧⌘2`) shows the same commands in a quick pick. On Windows and Linux, the shortcuts use `Ctrl+Shift` instead of `⇧⌘`, for example `Ctrl+Shift+0`.

- **Select current element** - `⇧⌘0` extend selection to include containing element's start/end tags
- **Select parent element** - `⇧⌘9` extend selection to include parent element's start/end tags
- **Select first child element** - `⇧⌘8` select the first child element's start/end tags
- **Select following element** - `⇧⌘7` select next sibling's start/end tags
- **Select preceding element** - `⇧⌘6` select previous sibling's start/end tags

The submenu and quick pick also have **Find Nodes with XPath...** and **Add XPath Location to Queries** (`⌥G` / `Alt+G`) - see Find Nodes with XPath.

Editing XSLT/XPath

With DeltaXignia's XSLT/XPath extension, Visual Studio Code is a fully featured XSLT 3.0 editor, with support for the XSLT 4.0 and XPath 4.0 features implemented in Saxon 13. Standalone XPath 3.1 and XPath 4.0 files are also fully supported. Visual Studio Code's rich ecosystem is now available to XSLT and XPath developers.

Comprehensive, but language-neutral, descriptions of Visual Studio Code can be found in Microsoft's Visual Studio Code User Guide. Links to key sections are below:

- Keyboard shortcuts
- Multiple selections (multi-cursor)
- Find and Replace
- File encoding support
- Emmet (snippet abbreviations)

The documentation here focuses on XSLT and XPath language features. More general XML features are covered on the Editing XML page. Help on navigating code in the editor is provided in the Code Navigation page.

Syntax Highlighting

Eva Dark Theme



For improved performance and precision, this extension avoids using the default but problematic TM Grammar. Instead we use Visual Studio Code's Semantic Highlighting introduced in 2020. For color theme extensions you should explicitly enable Semantic Highlighting. See the Settings page for more information.

Github Light Theme

```
<?process 8798?>
<xsl:template match="item[@class eq 'random']" mode="#all">
  <!-- Gather together -->
  <xsl:variable name="list" as="xs:string*"
    select="
      for $num in 1 to 20 return
        substring($prefixes[$num], 5) =>
        string-join('end' || $num)"/>

  <body id="{$list[2]}">
    <text>All prefixes '{$list}' &lt;included.</text>
  </body>
</xsl:template>
```

The extension provides syntax highlighting for XSLT instructions, XML Literal Result Elements, XML attributes, XML character references, CDATA sections and all tokens within XPath expressions. XPath expressions within Attribute Value Templates (AVTs) and Text Value Templates are syntax highlighted also.

XSLT/XPath Dark Color Themes

The plethora of VS Code themes will highlight XSLT/XPath very well using Semantic Highlighting fallbacks for language-specific tokens. There is however a XSLT Dark Themes extension that enhances a set of popular dark themes specifically for XSLT and XPath:

1. Darcula From IntelliJ
2. Gruvbox Material Dark
3. Iceberg
4. Nord
5. Tokyo Night Storm
6. Zen

Formatting

Two commands are provided for formatting XML, XSLT or XPath expressions:

- **Format Document** (⇧⌥F) - Format the entire active file.
- **Format Selection** (⌘K ⌘F) - Format the selected content.

As well as explicitly invoking commands, formatting can be triggered as you type, when you save a file, or when you paste content from the clipboard. The following settings control this:

- `editor.formatOnType` - indents a new line as you press `Enter`, and indents and completes an end tag as you type `</`.
- `editor.formatOnSave` - formats the file when it is saved.
- `editor.formatOnPaste` - formats the pasted content.

- `editor.formatOnSaveMode` - with the *modifications* setting, only the changed code is formatted on save.

The XPath formatter indents code blocks within `{}`, `[]` and `()` bracket-pairs. Indentation is also adjusted for `if/else` blocks and range-variable blocks.

Folding

Folding with this extension uses indentation to determine fold regions. To set a region code-folding block, surround it with `<?region?>` and `<?endregion?>` processing instructions. You may optionally include a label for the processing instructions, for example:

```
<?region modeA?>
...
<?endregion modeA?>
```

For more detail and a full list of Folding-related actions see the Folding section in Visual Studio Code's User Guide.

Imports and Top-Level Stylesheets

The functions, templates, variables and other global declarations of a stylesheet's imported and included modules are used throughout the editor: by auto-completion, hover and signature help, go to definition, find references, symbol renaming and the linter.

A 'non-standalone' module - one that's imported by a parent stylesheet - often relies on declarations that only the parent imports. Without knowing the parent, those references can't be resolved: they're missing from auto-completion and hover help, and the linter reports spurious problems that may obscure real ones.

With **inferred imports**, the top-level ('parent') stylesheet that imports or includes the active module, directly or indirectly, is found and imported along with the module's own imports, so those references can be resolved.

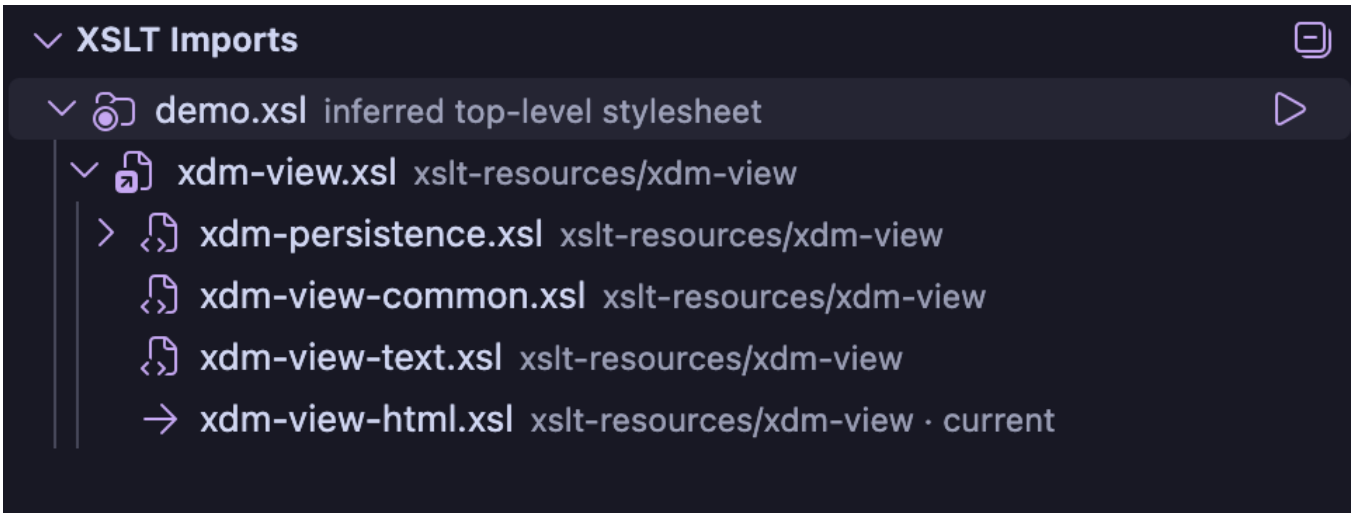
A parent stylesheet opened recently in Visual Studio Code is used first. Otherwise, the parent is found from the workspace's XSLT files, which are scanned in the background the first time they're needed, so the parent doesn't have to be opened first. The scan only reads the `href` attributes of `xsl:import` and `xsl:include` instructions, and is kept up to date as files change. Where a module is imported by more than one stylesheet, the one in the nearest folder is used. The workspace scan can be turned off with the `XSLT.resources.inferParentFromWorkspace` setting.

To use a different top-level stylesheet - for example, one of two that import the same modules - run **XSLT: Choose Top-Level Stylesheet...** from the Command Palette, or the button in the title bar of the XSLT Imports view. The list shows the top-level stylesheets that import or include the active module, directly or indirectly: the one chosen is used for all the modules of its tree. **Automatic** goes back to the nearest folder, and **Browse...** picks any stylesheet, for the active module alone. The choices are saved for the workspace, and **XSLT: Reset Top-Level Stylesheet Choices** clears them all. The XSLT Imports view marks a chosen stylesheet *chosen*, rather than *inferred*.

The XSLT Imports view

The **XSLT Imports** view in the Explorer shows, for the XSLT module in the active editor, its top-level stylesheet - marked *inferred* when it was found from the workspace - with its tree of imports and includes. The tree is expanded down to the active module, which is marked *current*, and clicking a module opens it. The tooltip for a module lists any other stylesheets that import or include it.

The ▶ button on the top-level stylesheet runs it with Quick Run, using the current XML context file - whichever module of its tree is being edited.



The XSLT Imports view in the Explorer, with the ▶ Quick Run button for the top-level stylesheet.

IntelliSense (Auto Complete)

IntelliSense suggestions include all in-scope XSLT and XPath symbols from `xsl:accumulator` names to `xsl:variable` names, to anonymous XPath function parameters. There is IntelliSense also for XSLT and XPath functions - help for built-in functions is shown alongside the suggestions list. Symbols from included/imported files are included in the suggestions list.

Hover Help and Signature Help

Hovering over a function call shows the function's signature and description. This works for built-in functions, with a link to the function's definition in the W3C specification, and for user-defined `xsl:function`s, including those in included and imported stylesheets. Hovering over the name of an `xsl:call-template` shows the named template's parameters. Hovering over a mode name, in an `xsl:mode`, `xsl:template` or `xsl:apply-templates`, shows where the mode is applied from across the stylesheet's modules: the template rules of other modes, named templates and functions with an `xsl:apply-templates` for it - see also Checks on modes.

While typing the arguments of a function call, signature help (parameter hints) shows the function's parameters, with the current parameter highlighted. Signature help is also shown for the `xsl:with-param` children of an `xsl:call-template`. Signature help can be triggered manually with `⇧⌘Space`.

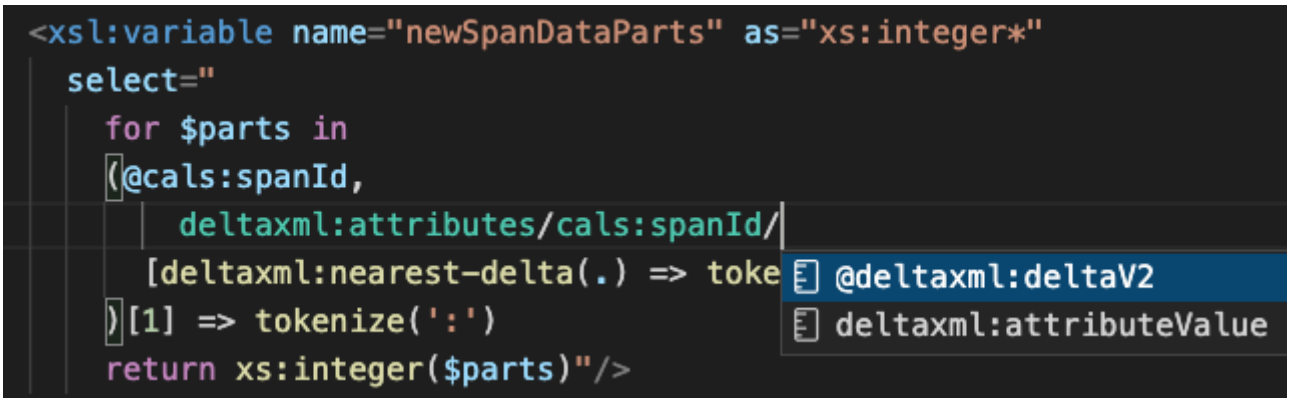
In an XSLT 4.0 stylesheet, a documentation note on a function or template is included in its hover help and signature help.

IntelliSense for XPath Locations

An XML source file is used for context to provide XML node-names when editing XPath location steps. This is initially the first non-XSLT file opened in Visual Studio Code. When the context file is set, the status-bar for the XSLT file editor will show the context filename. The label **[XML context file]** is shown if the context file has not been set, and **[no XML context]** if 'none' was picked. The same file is the XML source for Quick Run.

Locking the context file

To lock the auto-completion context file to a specific file, click on the status-bar item for the context file. You can then select from the list of recently opened XML files or pick a different file using the system's file explorer dialog. The file you pick is for the stylesheet you're editing (and the modules it imports or includes) - each stylesheet keeps its own, see Each stylesheet has its own XML context file.



Variables and XPath Locations

XPath is evaluated as you type so node-names in the auto-completion list are filtered to be just those that are possible given previous XPath steps such as axes and node-name tests. If an XPath location expression starts with a variable that also contains an XPath location, that context from that variable is also used to limit the node-names shown in the auto-completion list.

Triggering the auto-complete list

For XSLT IntelliSense, the `<` character triggers suggestions. XSLT instruction suggestions fit the context of the cursor location. On accepting a suggestion a Code Snippet is inserted which will include common attributes for the selected instruction.

IntelliSense is manually triggered with `^Space`, with `Tab` or `Enter` used to accept suggestions. These key bindings are fully customizable.

XPath Expressions and Operators

At the start of an XPath expression, the suggestions include snippets for the expressions with several parts: `for`, `let`, `some`, `every`, `if`, `map` and `array` - for example, `let $x := … return …`, with placeholders for each part.

After an operand, such as a variable or a closing bracket, only the keyword operators that can follow are suggested, such as `cast as`, `instance of` and `union` - and `return` or `satisfies` within a `for`, `let`, `some` or `every` expression. Short operators such as `and`, `eq` and `div` are quicker to type, so aren't included.

Parameters for xsl:call-template

Typing `<` within an `xsl:call-template` suggests an `xsl:with-param` for each of the called template's parameters that isn't passed yet, for example `xsl:with-param colour`, which inserts the parameter name. The same applies to the parameters of the enclosing `xsl:iterate` within an `xsl:next-iteration`. The `name` attribute of an `xsl:with-param` also offers the parameter names.

XSLT Instruction Snippets

In an empty file, the `xsl:stylesheet` and `xsl:package` suggestions insert a boilerplate stylesheet complete with namespace declarations, including one for your own functions - an 'identity transform', or one starting from `xsl:initial-template`. Each is available for XSLT 3.0 and XSLT 4.0, shown on the right of the suggestion.

The `xsl:message` suggestions include snippets that use the `xdm:debug()` function to show the names and values of in-scope variables and parameters - see Print Debugging.

More Snippets

You can define your own snippets in Visual Studio Code in a declarative way, without writing an extension. See the Visual Studio Code documentation: Create your own snippets.

Emmet Snippets provide a useful shorthand for inserting literal result elements. They can be enabled for XSLT in Visual Studio Code Settings.

Bracket Matching

Matching brackets in XPath expressions are highlighted when the cursor is near one of them. You can jump to the matching bracket with ⇧⌘\

Symbol Renaming

The **Rename Symbol** command F2 (available in the context-menu) updates all in-scope usages of the symbol - across all imported stylesheet modules. Press ⇧Enter to preview the updates or Enter to perform updates immediately.

Code Refactoring

Code refactoring features are provided via VS Code's Quick Fix (⌘ .) feature and the Refactor command:

- **xsl:function - full refactor** and **xsl:function - partial refactor** - extract an `xsl:function` from selected XSLT instructions
- **xsl:function (XPath) - full refactor** and **xsl:function (XPath) - partial refactor** - extract an `xsl:function` from a selected XPath expression
- **xsl:template** - extract an `xsl:template` from selected XSLT instructions
- **xsl:variable** - extract an `xsl:variable` from a selected XPath expression
- **Wrap with...** (⌥⇧W) - wraps the selected instructions, or the element whose start tag is at the cursor, in an instruction such as `xsl:if`, `xsl:choose` or `xsl:for-each`, or a literal result element - chosen from a list of the instructions allowed at that point, with the recently used ones first. The selected lines are indented within the new instruction, and the cursor is placed in its first attribute, such as `test` or `select`

- **Extract record type** and **Use record type** - XSLT 4.0, for a map constructor or `xsl:map`, see Records and Enums
- **Add documentation note** - XSLT 4.0, with the cursor on the start tag of an `xsl:function` or `xsl:template`, see Code Documentation
- **Override Package Components...** - with the cursor in an `xsl:use-package`, adds declarations to its `xsl:override` for the package components you pick, see Overriding package components

An available refactoring for a selection is indicated by a light-bulb 💡 adjacent to the selection. Clicking on the 💡 will display the available refactorings. The light-bulb is only shown when the code selection is a valid expression or a set of XSLT instructions that would be valid inside the new component. If you don't see the light-bulb when you expect to, check that the code selection **completely encloses** the XSLT instruction or XPath expression.

The name of an extracted new component should be changed with the `Rename Symbol` feature, this is automatically triggered when the new component is added.

After refactoring, the XSLT is updated to fix any context problems introduced when code is moved. The new component has `xsl:param` instructions added to provide context-properties required. For extract to xsl:function two modes are available (fully documented here):

- **full refactoring** - fixes 'missing-context' problems by updating XSLT to reference the context-property parameters.
- **partial refactoring** - highlights 'missing context' problems in the XSLT.

```
<xsl:template match="fn:map" mode="indent" expand-text="yes">
    <xsl:value-of>
        <xsl:variable name="depth" select="count(ancestor::*) + 1"/>
        <xsl:for-each select="*">
            <xsl:sequence select="dx:extractFunction(., position(), last(), $depth)"/>
        </xsl:for-each>
    </xsl:value-of>
</xsl:template>

<xsl:function name="dx:extractFunction" expand-text="yes" as="item()*">
    <xsl:param name="c.x" as="item()*"/>
    <xsl:param name="c.p" as="xs:integer"/>
    <xsl:param name="c.l" as="xs:integer"/>
    <xsl:param name="depth" as="item()*"/>
    <xsl:if test="$c.p gt 1">
        <xsl:text>{$depth} of {$c.l} on {name($c.x)}</xsl:text>
    </xsl:if>
    <xsl:apply-templates select="snapshot($c.x/@key)" mode="key-attribute"/>
    <xsl:apply-templates select="$c.x" mode="#current"/>
</xsl:function>
```

dx:extractFunction
Enter to Rename, ⇧Enter to Preview

Refactoring with 'Extract to xsl:function' in full mode, referencing context-parameters $c.x, $c.p and $c.l

Quick Fixes

Some problems reported by the linter have a Quick Fix, shown with the 💡 light-bulb when the cursor is on the problem:

- **include XSLT module for xdm:debug and xdm:trace** and **copy xdm-view library into workspace** - for the `xdm:debug()` or `xdm:trace()` function when the xdm-view library isn't included, see Print Debugging
- **Don't report unknown '*prefix*:' functions in this workspace** - for a call of a function with a prefix, such as an extension function, that isn't found; the prefix is added to the `XSLT.validation.externalFunctionPrefixes` setting (in user settings when no folder is open), see Java extension functions
- **Set 'htmlParserJar' in User Settings...** - for the `parse-html()` function, which needs the `XSLT.tasks.htmlParserJar` setting
- **Convert all Saxon type aliases in the workspace to xsl:item-type...** - for a `saxon:type-alias`, see Converting Saxon type aliases
- **Add missing record fields** - for a map constructor or `xsl:map` without all the required fields of its record type, see Records and Enums
- **Change to '*name*'** - for an unknown field of a record type, one for each field name that could replace it
- **Add missing xsl:when cases with select** and **Add missing xsl:when cases with content** - for an `xsl:switch` on an enumeration type that doesn't test all its values, see xsl:switch on an enumeration
- **Remove duplicate enum value** - for an enumeration type with a duplicate value
- **Add missing @param**, **Add missing @field** and **Add missing @variable** - for a documentation note without an `@param` for each parameter, an `@field` for each record field, or (in a module note) an `@variable` for each global variable, see Code Documentation
- **Exclude with use-when="false()"** - for an `xsl:note` in an XSLT 3.0 stylesheet, see Notes in XSLT 3.0
- **Add mode="#current"** (preferred) and **Add mode="#default"** - for an `xsl:apply-templates` without a `mode` in a template rule with a mode, see Checks on modes
- **XSLT packages** - for an `xsl:use-package` whose package isn't found, abstract components that aren't overridden, problems with `xsl:accept` and `xsl:override`, and modes a package doesn't declare, see a package that isn't found, abstract components, problems with xsl:accept and xsl:override and declared modes

XSLT Linter

The XSLT/XPath linter performs checks on your code as you type, performing symbol reference tests across all included modules. Any problems are shown in the Problems view and also highlighted inline with a squiggly underline for the specific token at issue.

A module that relies on its parent stylesheet's imports is checked with those imports too - see Imports and Top-Level Stylesheets.

Checks made by the linter

1. XML Syntax
2. XSLT/XPath Variable name references

3. XSLT/XPath Parameter name references
4. **All** other symbol name references (`xsl:accumulator` `xsl:key` etc.)
5. Function name and arity (number of params)
6. `xsl:with-param` names
7. File locations in `xsl:import` `xsl:include` `xsl:use-package`
8. Order of operators/operands in XPath
9. Matching of brackets in XPath
10. Presence of the context-item for expressions where it is required
11. XSLT instructions and their attributes, and their structure - for example an `xsl:choose` without an `xsl:when` , or two `xsl:with-param` elements with the same name
12. Duplicate global symbol names
13. Attribute Value Template Syntax
14. Text Value Template Syntax
15. Item types on `as` attributes of XSLT instructions
16. Accumulator names in `use-accumulators` attributes, and accumulators not used by any `use-accumulators` attribute
17. Operators in patterns - `or` , `and` , `eq` or `,` outside a predicate or function call, for example `match="a or b"` instead of `match="a | b"`
18. Duplicate keys in map constructors and in the `xsl:map-entry` children of an `xsl:map`
19. Braced URI literals, for example `Q{http://example.com}name`
20. The order of `xsl:param` , `xsl:on-completion` and `xsl:next-iteration` within `xsl:iterate`
21. Modes - see Checks on modes
22. XSLT 4.0: record and enumeration types, see Records and Enums
23. XSLT 4.0: the `@param` tags of documentation notes, see Code Documentation

The linter follows the XSLT rules that affect which code is valid:

- Names of functions, templates, modes and variables are matched by namespace across modules, so modules may use different prefixes for the same namespace.
- A module may use a prefix other than `xsl` for XSLT instructions, for example `xslt:template` .
- Forwards compatible mode: for an element with a higher `version` , an unknown XSLT element with an `xsl:fallback` child, and an unknown attribute, aren't reported.
- Simplified stylesheets: a literal result element with `xsl:version` as the outermost element.
- A call of a function that isn't known is not reported when it's guarded by `use-when="function-available('name')"` .

Checks on modes

A large transform may run in many modes, and a template rule applied in the wrong mode fails silently. The linter warns about:

- **An unknown mode** - an `xsl:apply-templates` whose `mode` isn't declared by an `xsl:mode` or used by any template rule, in all the stylesheet's modules - often a typo in the mode name.
- **An `xsl:apply-templates` without a mode in a template rule with a mode** - it applies the default mode, not the template rule's mode. This commonly happens when a template rule is added for a new mode and `mode="#current"` is forgotten. The quick fixes add `mode="#current"`, or `mode="#default"` to show that the default mode is intended. There's no warning when the template rule's modes include the default mode - the unnamed mode, or the mode named by the innermost `[xsl:]default-mode` attribute - or `#all`, nor for a template rule within an XSLT 4.0 `xsl:mode`, where the enclosing mode is the default.
- XSLT 4.0: an `xsl:mode` with enclosed template rules but no `name`, and an enclosed template rule with a `mode` or `name` attribute, or without a `match`.

Union patterns and XSLT 4.0

The linter warns about a template rule without a `priority` attribute whose `match` is a union with branches of different default priorities, e.g. `match="@d:* | @other"` (-0.25 and 0). XSLT 3.0 makes a rule for each branch, with its own priority, but XSLT 4.0 makes one rule, with the highest - as Saxon reports with its warning SXWN9052. So another template rule may be chosen: here, `match="@d:keep"` would no longer win over the union. The warning lists each branch's priority. Give the template a `priority`, or split it into a template for each branch, so it runs the same either way.

It's shown where XSLT 4.0's rule applies:

- in an XSLT 4.0 stylesheet
- in a stylesheet before 4.0 that's run with Saxon's syntax extensions, as it - or a module of its stylesheet - uses item types or notes: Saxon 13 then uses 4.0's rule
- in any stylesheet before 4.0, with the setting `XSLT.validation.xslt40UnionPriorities` - e.g. to prepare for XSLT 4.0

Other stylesheets before 4.0 are run without the syntax extensions by default, so they keep XSLT 3.0's rule.

Unused Variables

The linter detects unused variables and parameters declared in either XSLT or XPath expressions. These are highlighted by being 'grayed out' in the editor. Global variables are shown as unused if they are not referenced in the current file, the 'unused' state is not affected by imported stylesheets they may reference the same variable.

Compile-time and Run-time problems

The linter is designed to find most basic code problems due to typos or changes to referenced symbols. When XSLT is run, an attempt is made to parse error messages from the Saxon XSLT processor (shown in the Task tab of the Terminal View) so that the problem token is highlighted, and the error message is shown in the Problems view.

> *Advantages of using the linter to supplement compile-time error detection:*
>
> - *Improved response times (lower CPU usage) as you type*
> - *Improved precision on error location*
> - *Improved potential to report multiple errors*
> - *Provides warnings on unused variable references etc.*

You can quickly create a new XSLT file when adding a `xsl:import`, `xsl:use-package` or `xsl:include` XSLT instruction. Simply enter the file path in the href, press `⌘`+click and then, in the 'Unable to Open...' dialog press the **Create File** button.

> *For `xsl:use-package`, a package in the workspace is found by its package URI - the `name` in its `xsl:package` element - one outside the workspace needs its path in the setting `XSLT.resources.xsltPackages` - see XSLT Packages.*

> *An `href` can also be a URI that an XML catalog maps to a file, e.g. `http://example.com/xslt/strings.xsl` - see XML Catalogs.*

Code Navigation

Visual Studio Code includes a compelling set of features for code navigation, the language-neutral details are outlined here.

This page outlines code navigation from an XSLT and XPath development perspective.

File Navigation

While the Explorer View is used for workspace browsing, pressing `⌘P`, for the **Quick Open** command, provides a quick-pick list of available files that is filtered/highlighted as you type.

File tabs are arranged in one or more Editor Groups. Editor Groups are arranged either adjacent or above/below other Editor groups. The same file may be open in more than one Editor Group. There are settings to control whether file tabs are scrollable or wrapped within an Editor Group.

Hold `^` and press `Tab` for a list of files in an Editor Group. Keep pressing `Tab` until you reach the file you want and then release `^` to open it.

Previous/Next Location

Many actions can be used to navigate you to a new location. Press `^-` for **Previous Location** and `^⇧-` for **Next Location**. Locations may be in the same file or different files.

File Links

In XSLT, the `href` attribute values of `xsl:import`, `xsl:include` and `xsl:use-package` have a solid underline if the file path is resolved. Press `⌘+Click` on the file path to navigate to that file.

XSLT/XML Breadcrumbs

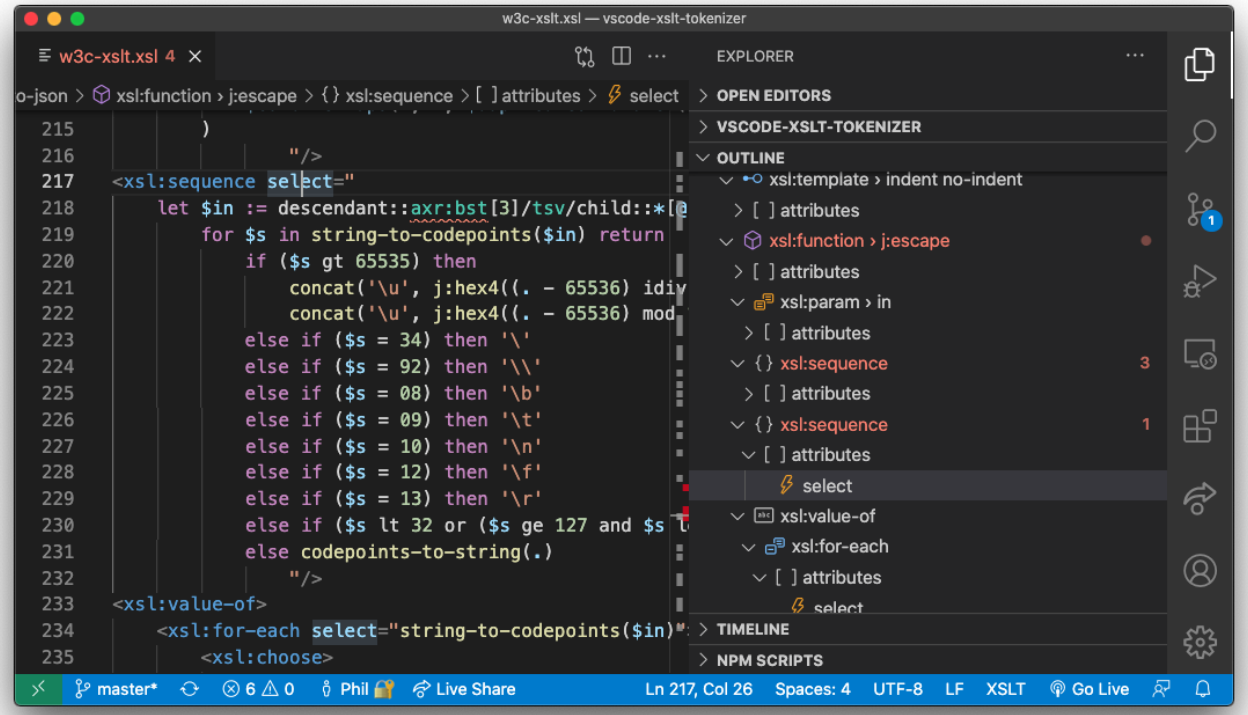
The Breadcrumb View, shown at the top of the editor, shows the current element or attribute location within an XML or XSLT file with respect to ancestor elements. In XSLT Breadcrumbs, an identifier is often shown as well as the element name. This may be for example the `name` of an `xsl:variable` or the `mode` of an `xsl:template`. The symbol names are the same as those shown in the Outline view.

Clicking on a Breadcrumb item will show a dropdown Outline view which you can then navigate through with the mouse, touch or cursor keys.

Keyboard Navigation

Press `⇧⌘.` (note the 'dot' character) to select the last Breadcrumb item and open **with** a dropdown outline. You can then use cursor keys to navigate the dropdown outline. Alternatively, press `⇧⌘;` (note the 'semi-colon' character) to select the last Breadcrumb item **without** a dropdown outline. Use the `←` and `→` keys to select a Breadcrumb item or the `↓` for the dropdown outline.

Outline View



The Outline View (see right-hand panel) is a panel that, by default, is positioned below the File Explorer in the Primary Side Bar. This can be moved however into other panels such as the Secondary Side Bar (see Sidebars). The Outline view renders the XML or XSLT document element hierarchy as a tree, symbol names in the tree include element identifiers such as the `mode` for an `xsl:template`.

Symbols in the Outline View can be sorted by position, name or category. There's also a '**Filter on Type**' mode that filters the Outline View while you type in part of a symbol name.

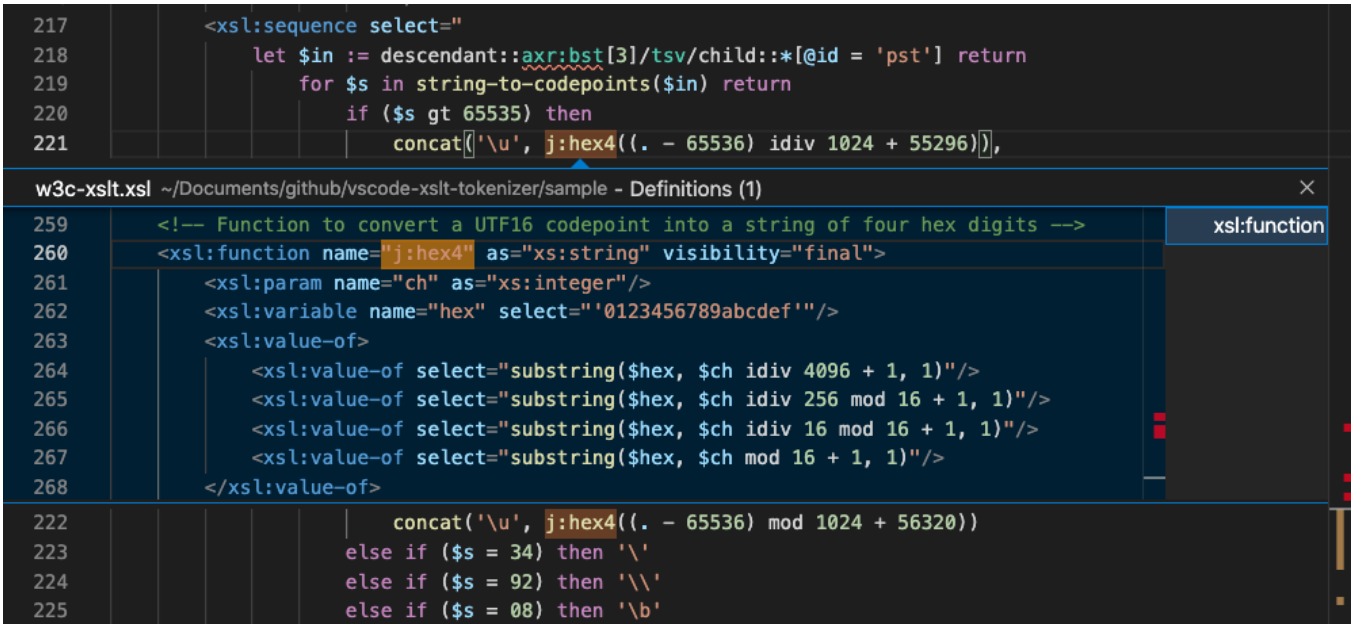
> *Note: XML Attributes are shown in the outline view by default, they are grouped inside an element within an attributes 'array' symbol. If you find attributes clutter the XML element outline you can hide them in Settings with the property:* `outline.showArrays` .

Peek/Go to Definition

A symbol, such as a variable or function, is defined in one location in a stylesheet file, but used in other locations. The **Peek** and **Go to** commands let you either view inline or navigate to the definition of a selected symbol.

Peek Definition

Invoke the **Peek Definition** command by pressing (`⌥F12`) on any symbol name such as a variable or function name. This command embeds a view of the definition inline avoiding the context-switch arising from the **Go to Definition** command. The inline definition view permits quick edits of the definition also. Press `Escape` to close the definition view.



Go to Definition

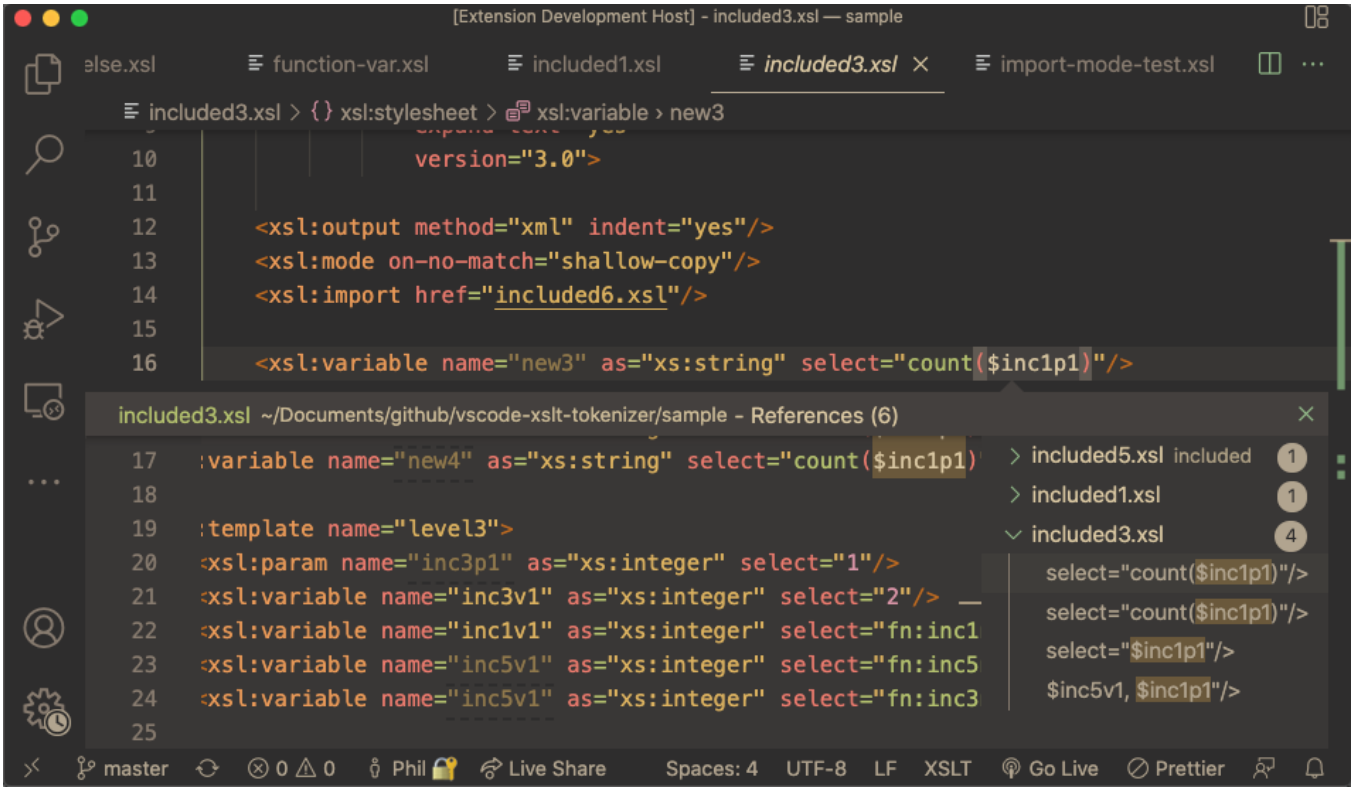
Invoke the **Go to Definition** command by pressing (F12) on any symbol name such as a variable or function name. This command navigates you to the location of the symbol definition (for example, the corresponding `xsl:variable` instruction), opening the containing stylesheet file in the editor group if not already open.

Go to (Peek) References/Find All References

This feature is about navigating to **references** to symbols, in contrast to the previous section that covers navigating to symbol **definitions**. The **Peek** and **Find All** commands let you either view inline or navigate through all the references (including the definition) of a selected symbol.

Go to (Peek) References

Invoke the **Go to (Peek) References** command by pressing (⇧ F12) on any symbol name such as a variable or function name. This command embeds a view of the resolved references inline, use the cursor keys to navigate between each reference listed. The inline view permits quick edits of the reference also. Press Escape to close the references view.



Find All References

Invoke the **Find All References** command by pressing (⌥⇧F12) on any symbol. A References Panel, shown in the application Side Bar, lists all references, grouped by their stylesheet module. Use the cursor keys to move through the references and press `Enter` to open the editor in the reference location.

For XSLT packages, Find All References includes the declarations that override a package component, and **Go to Implementations** and **Go to Declaration** navigate between them - see Navigating between overrides.

For a function or named template, Find All References also lists its uses in the XSpec files that test it, and from an XSpec file, Go to Definition opens the function or template a scenario calls - see Navigating between tests and the stylesheet.

After navigating to a reference or definition, you can quickly navigate to the location you were in using `^-`

Call Hierarchy

Use the **Show Call Hierarchy** command (⇧⌥H , or from the editor's context menu) on a call of a stylesheet function or named template, or anywhere within a template or function, to see what calls it and what it calls. The **Call Hierarchy** view, in the Side Bar, shows either:

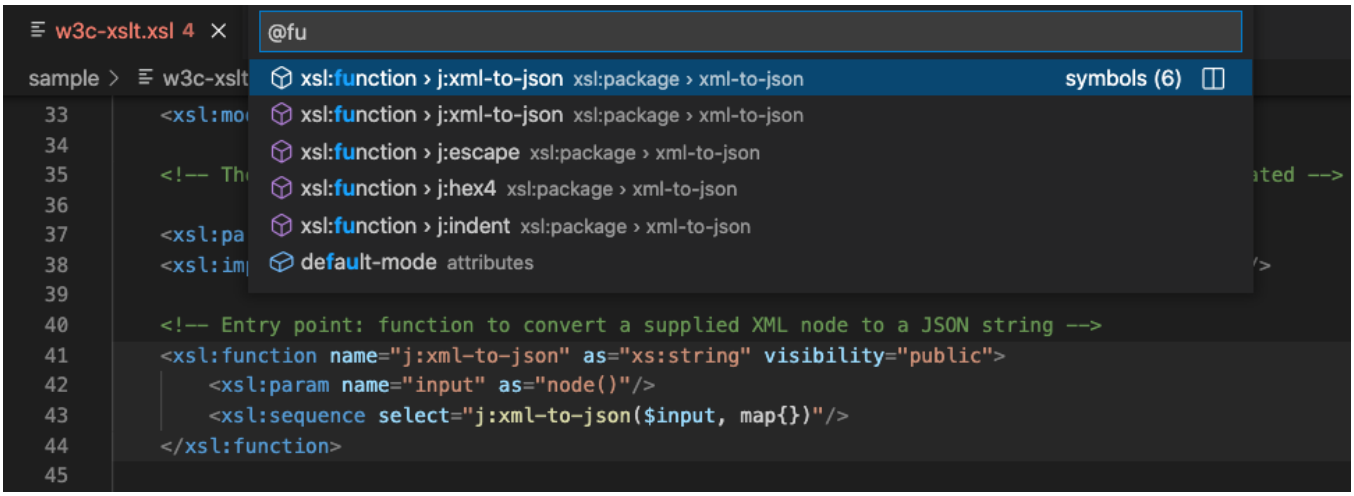
- **Incoming calls** - the template rules, named templates, functions and other declarations (e.g. global variables) that call the function or named template, in all the stylesheet's modules - each with its calls, to go to
- **Outgoing calls** - the stylesheet functions and named templates that a template or function calls, with `xsl:call-template` , a function call, or a named function reference such as `f:double#1`

Expand an entry to see its own calls, and so on. Use the button at the top of the view to switch between incoming and outgoing calls. Where functions or templates with the same name are declared in more than one module, calls go to the one with the highest import precedence - the one Saxon uses.

A template rule is called by `xsl:apply-templates`, which depends on the node being processed when the stylesheet runs, so template rules are shown as callers, but not as being called.

Go to Symbol

Use the **Go to Symbol in Editor** command ⇧⌘O for a filtered quick-pick list of symbols used in the current editor. The text you enter filters the list, the ↑ and ↓ keys navigate you through the symbol list, with the current symbol item highlighted in the editor. Press Enter to navigate to the symbol in the current editor or ⌘Enter to open an adjacent editor and navigate to the symbol there.



Find Nodes with XPath

In an XML file, to go to the nodes an XPath expression finds - such as `//section[@id]` or `/root/section[2]/para[3]` - write it in the query file and run it: the nodes are listed as references are, to step through. ⌥G / Alt+G adds the location of the node at the cursor to the query file. See Find Nodes with XPath.

XSLT Imports View

For the XSLT module in the active editor, the **XSLT Imports** view in the Explorer shows its top-level stylesheet with its tree of imports and includes, expanded down to the active module. Click a module in the tree to open it. The view is collapsed initially - click its title in the Explorer to expand it. See The XSLT Imports view for details.

Refactoring

The Refactoring section of the Editing XSLT/XPath part of this guide describes XSLT refactoring in general. This section provides additional detail on the **'extract to xsl:function refactor.**.

Introduction

There are two available modes for extracting selected code to an xsl:function: **full** and **partial**. In this refactor, required context properties must be passed as parameters when the new xsl:function is invoked, this is because context properties are cleared inside an xsl:function.

Partial Mode

In the **partial** mode, required parameters are passed to the new function, but code in not refactored to reference the new parameters. Instead, the 'missing-context' properties are highlighted as problems.

Full Mode

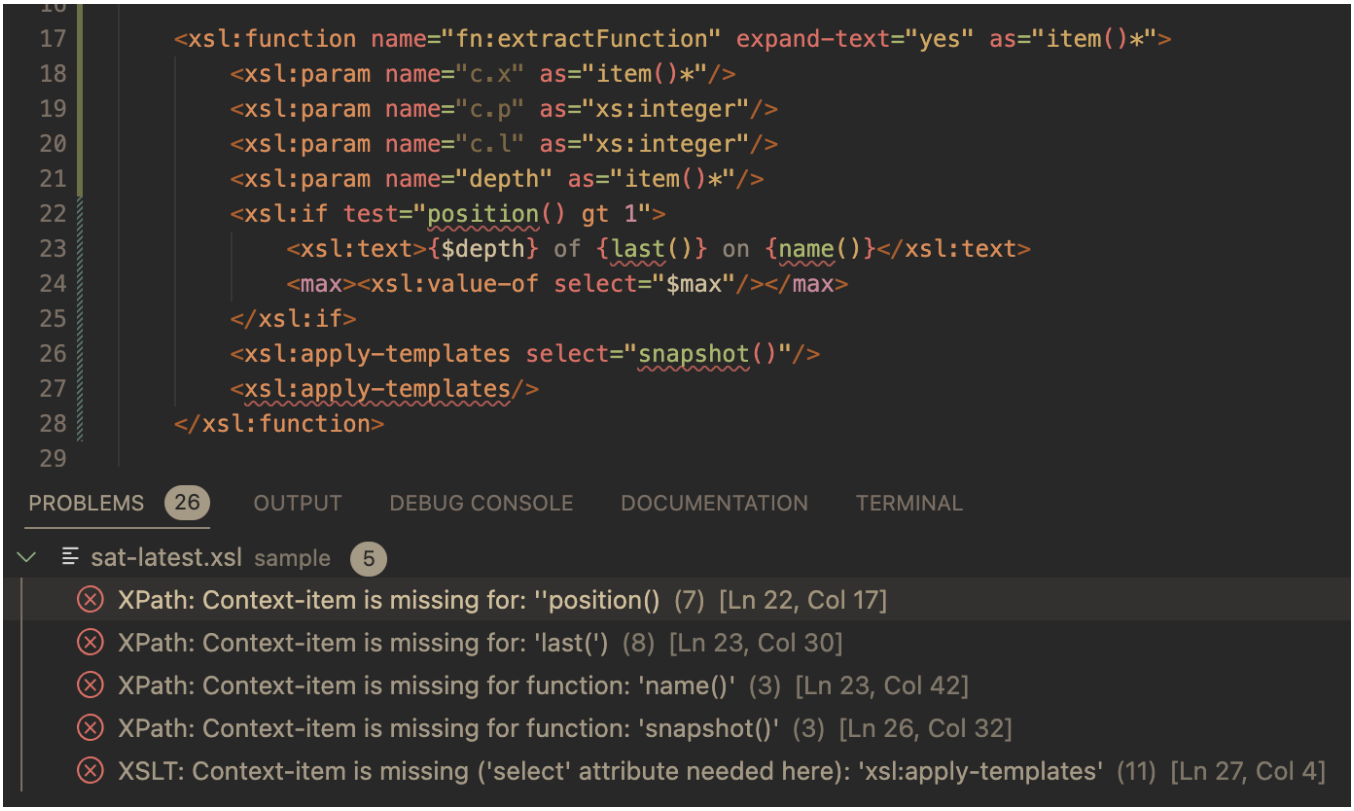
The **full** refactor fixes 'missing-context' properties by refactoring the code to reference the relevant parameters.

A Sample Refactor

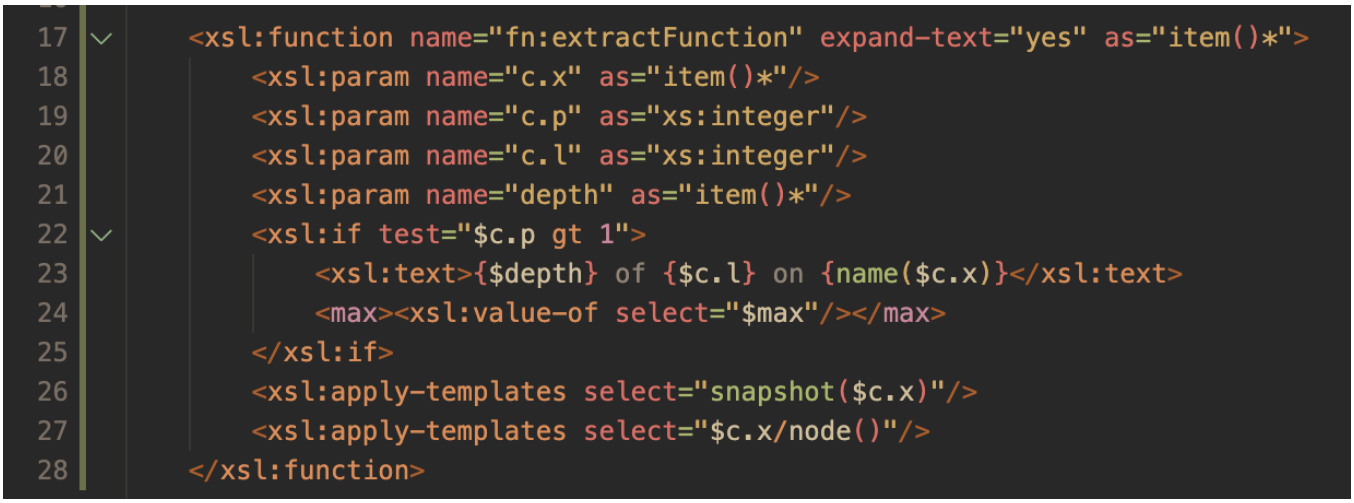
The screenshots below show sample code refactored to extract to an xsl:function.

```
8     <xsl:template match="book" mode="indent" expand-text="yes">
9         <xsl:value-of>
10            <xsl:variable name="depth" select="count(ancestor::*) + 2"/>
11            <xsl:for-each select="*">
12                <xsl:if test="position() gt 1">
13                    <xsl:text>{$depth} of {last()} on {name()}</xsl:text>
14                    <max><xsl:value-of select="$max"/></max>
15                </xsl:if>
16                <xsl:apply-templates select="snapshot()"/>
17                <xsl:apply-templates/>
18            </xsl:for-each>
19        </xsl:value-of>
20    </xsl:template>
```

1. The original code selected for refactoring



2. After a **partial** refactor: 'missing-context' problems are highlighted



2. After a **full** refactor: there are no 'missing-context' problems as code is updated in the refactor to let parameters provide the context

Known Restrictions

The code selected for a refactor is important. For example, in many cases it may be more effective to extract an entire xsl:merge instruction instead of a set of instructions within the child xsl:merge-action instruction.

There are known limitations in what can be achieved when passing context to an xsl:function, these are outlined below. Please note also that in general there will be cases where, because of context, it would be advisable to extract to xsl:template instead of xsl:function.

- Potential ‘missing context’ problems for the xsl:call-template instruction are highlighted but not fixed. In such cases, it’s advisable to undo the refactor and use the **extract to xsl:template** refactoring instead.

- The regex-group($n) expression (where $n is not a literal integer) with ‘missing context’ is a special refactoring case. It is resolved to `$r.group($n)` where `$r.group` references a context xsl:param of type `map(xs:integer, xs:string)`, The number of items in the generated code for `$r.group` is set to `99` this number should be corrected manually to the expected number of regex groups given the regex used.

Context Properties

The table below lists the context properties, along with the corresponding xsl:function parameter names that will, if required, be created for an xsl:function refactor.

Context property	**Parameter**
current() or '.'	c.x
position()	c.p
last()	c.l
current-group()	g.current
current-grouping-key()	g.key
regex-group($n)	r.group($n)
current-merge-group()	m.current
current-merge-key()	m.key
current-merge-group($name)	m.groups($name)

XML Catalogs

An **XML catalog** maps the URIs used in a stylesheet to the files they're resolved to. With a catalog, a stylesheet can import a shared library by a fixed name, such as `http://example.com/xslt/strings.xsl`, wherever the library's files are kept - only the catalog changes when they move. Catalogs are an OASIS standard, XML Catalogs 1.1, and Saxon supports them with its `-catalog` option.

The XSLT/XPath extension uses a catalog to resolve the `href` of `xsl:import` and `xsl:include` - so the declarations of a module imported by its catalog name are known to the linter, auto-completion, hover help and go to definition - and passes the same catalog to Saxon when a stylesheet is run.

Catalogs are useful when:

- a team shares XSLT libraries across projects, and each project keeps its copy of a library in a different place
- stylesheets import libraries by `http:` URIs, and the files should be read locally - for speed, or to work offline
- libraries are reorganised, or a new version is tried, without editing the stylesheets that import them

To try it out, see the catalog demo.

Setting up a catalog

Set `XSLT.resources.catalog` to the path of the catalog file - relative to the workspace folder, or with `${workspaceFolder}`. For a team, add it to the workspace settings, in the project's `.vscode/settings.json`:

```
"XSLT.resources.catalog": "catalog.xml"
```

The setting names one master catalog. It can hand lookups on to any number of other catalogs, with `nextCatalog` entries - for example, a catalog for each library.

When there's no setting, and the root of the workspace folder has a `catalog.xml` that's an XML catalog, you're asked whether to use it: **Use** adds the setting to the workspace settings, and **Don't Ask Again** is remembered for the workspace.

> *To use no catalog, and not be asked, set the setting to* `""`.

Changes to the setting, and to the catalog files, are picked up straight away.

Catalog files

A catalog is an XML file whose root element is `catalog`, in the namespace `urn:oasis:names:tc:entity:xmlns:xml:catalog`. Each entry maps a URI - the name used in an `href` - to the URI of a file. A relative path in a catalog is relative to the catalog file itself.

```
<catalog xmlns="urn:oasis:names:tc:entity:xmlns:xml:catalog">
  <uri name="http://example.com/xslt/strings.xsl" uri="lib/strings/strings.xsl"/>
</catalog>
```

With this catalog, `<xsl:import href="http://example.com/xslt/strings.xsl"/>` imports `lib/strings/strings.xsl`. The name needn't be a real web address: it's an identifier for the library.

The extension uses these entries:

- `uri` - maps one URI, its `name`, to the URI of a file, its `uri`
- `rewriteURI` - maps every URI that starts with its `uriStartString`, replacing that part with its `rewritePrefix`, and keeping the rest - e.g. a whole folder of libraries:

  ```
  <rewriteURI uriStartString="http://example.com/lib/" rewritePrefix="lib/"/>
  ```

  maps `http://example.com/lib/strings/strings.xsl` to `lib/strings/strings.xsl`
- `uriSuffix` - maps every URI that ends with its `uriSuffix` to the URI of a file
- `nextCatalog` - another catalog file, its `catalog`, to try when this catalog has no entry for a URI
- `group` - groups entries, e.g. to give them an `xml:base` - which can also be on the `catalog` element or an entry, to set the base URI that its relative paths are resolved against

In each catalog, an exact `uri` entry is used first, then the `rewriteURI` entry with the longest `uriStartString` that matches, then the `uriSuffix` entry with the longest `uriSuffix` - and only if none matches, the `nextCatalog` entries, in order. An `href` is looked up as it's written, and then, if it's relative, as the absolute URI it resolves to.

In the editor

For an `xsl:import` or `xsl:include` whose `href` the catalog resolves:

- the module's declarations are known, as for a module imported by its file path
- Cmd/Ctrl+click on the `href` opens the module's file
- hover help on the `href` shows how the catalog resolved it: the entry that matched, and the catalog files it was found through - e.g. *Resolved by the XML catalog: the `uri` entry for `http://example.com/xslt/strings.xsl` in catalogs/libraries.xml, via catalog.xml*



Hover on a catalog-resolved href, showing the entry and the catalog chain.

An `href` that's a URI the catalog doesn't resolve to a file is reported, naming the catalog:

- an `http:` or `https:` URI is a warning - its declarations aren't known, but Saxon will try to fetch it when the stylesheet is run
- another URI, e.g. `urn:example:strings`, is an error - Saxon can't resolve it either

In the Problems view, the problem has a link to the catalog - and Cmd/Ctrl+click on the `href` opens the catalog, to add an entry for it. With no catalog, the problem suggests using one.

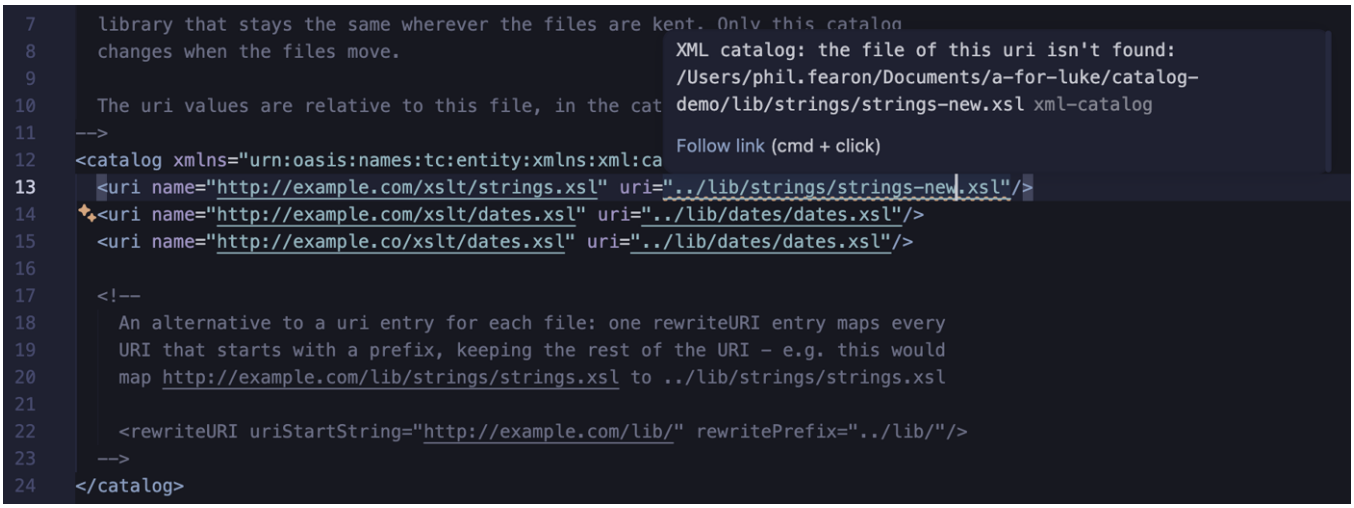


The warning for an http: href that isn't in the catalog.

Catalog files in the editor

In a catalog file - one that the catalog of the setting reads, or any XML file with a `catalog` root element in the catalog namespace:

- the `uri` of each entry, and the `catalog` of each `nextCatalog` entry, is a document link, so you can Cmd/Ctrl+click from the master catalog, through the catalogs, to a library file
- an entry whose file - or, for a `rewritePrefix`, folder - isn't found has a warning, e.g. after a library has moved



A catalog file, with a warning for an entry whose file isn't found

Running stylesheets with the catalog

The XSLT tasks for SaxonJ (`xslt`) and SaxonC (`xslt-c`) - including those Quick Run creates - pass the catalog of the setting to Saxon, with the `-catalog` option, so the stylesheet is run with the same imports as it's edited with. A task's own `catalogFilenames` property is used instead, when it has one:

```
"catalogFilenames": "${workspaceFolder}/other-catalog.xml"
```

> *Saxon uses the catalog for all the URIs it resolves - including `doc()`, `document()` and `unparsed-text()`, and DTDs - not only imports. So the catalog is always passed, even when the stylesheet's imports don't need it.*

Limitations

- The setting names one master catalog - use `nextCatalog` entries for others.
- The editor uses the `uri`, `rewriteURI`, `uriSuffix` and `nextCatalog` entries - not `system`, `public` or `delegate` entries, which are for DTDs and other external entities. Saxon uses them all.
- Saxon-JS tasks aren't passed the catalog.

Catalog demo

The catalog-demo folder is a small, complete example: a master catalog that hands lookups on to a secondary catalog, which maps the URIs of two XSLT libraries, in different folders, to their files - and a top-level stylesheet that imports both libraries by those URIs. Clone the repository, or copy the files, to try it.

```
catalog-demo/
├── catalog.xml              master catalog: a nextCatalog entry for catalogs/libraries.xml
├── catalogs/
│   └── libraries.xml        secondary catalog: a uri entry for each library
├── lib/
│   ├── strings/strings.xsl  imported as http://example.com/xslt/strings.xsl
│   └── dates/dates.xsl      imported as http://example.com/xslt/dates.xsl
├── main.xsl                 imports both libraries by their catalog names
└── .vscode/settings.json    "XSLT.resources.catalog": "catalog.xml"
```

How `http://example.com/xslt/strings.xsl` is resolved:

1. `catalog.xml` has no entry for it, so its `nextCatalog` entry is tried: `catalogs/libraries.xml`
2. `libraries.xml` has a `uri` entry with that `name`, whose `uri` is `../lib/strings/strings.xsl` - relative to `libraries.xml`
3. so the import is `lib/strings/strings.xsl`

Try it

1. Open the `catalog-demo` folder in VS Code (**File > Open Folder...**), so that its `.vscode/settings.json` applies
2. Open `main.xsl`: there are no problems, and hover help on `str:shout` shows the function's signature, from `strings.xsl`
3. Hover on an `href`, to see the entry that resolved it - and Cmd/Ctrl+click it, to open the library file
4. Open `catalog.xml`, and follow its links to `catalogs/libraries.xml` and the library files
5. Run `main.xsl` with Quick Run: the result is e.g. `CATALOGS WORK! | Wednesday 30 September 2026`

Then:

- Change an `href` to a URI that isn't in the catalog, e.g. `http://example.com/xslt/missing.xsl`, to see the warning - and Cmd/Ctrl+click it to open the catalog
- Move `lib/dates` to another folder: its entry in `catalogs/libraries.xml` has a warning. Update the entry's `uri` - `main.xsl` doesn't change
- Replace the two `uri` entries with the `rewriteURI` entry in the comment in `libraries.xml`, and change the `href`s in `main.xsl` to match
- Delete `.vscode/settings.json` and reopen the folder, to be offered `catalog.xml`

From the command line, the same catalog is passed to Saxon with `-catalog`:

```
java -cp saxon-he-13.0.jar net.sf.saxon.Transform -xsl:main.xsl -it -catalog:catalog.xml
```

XSLT Packages

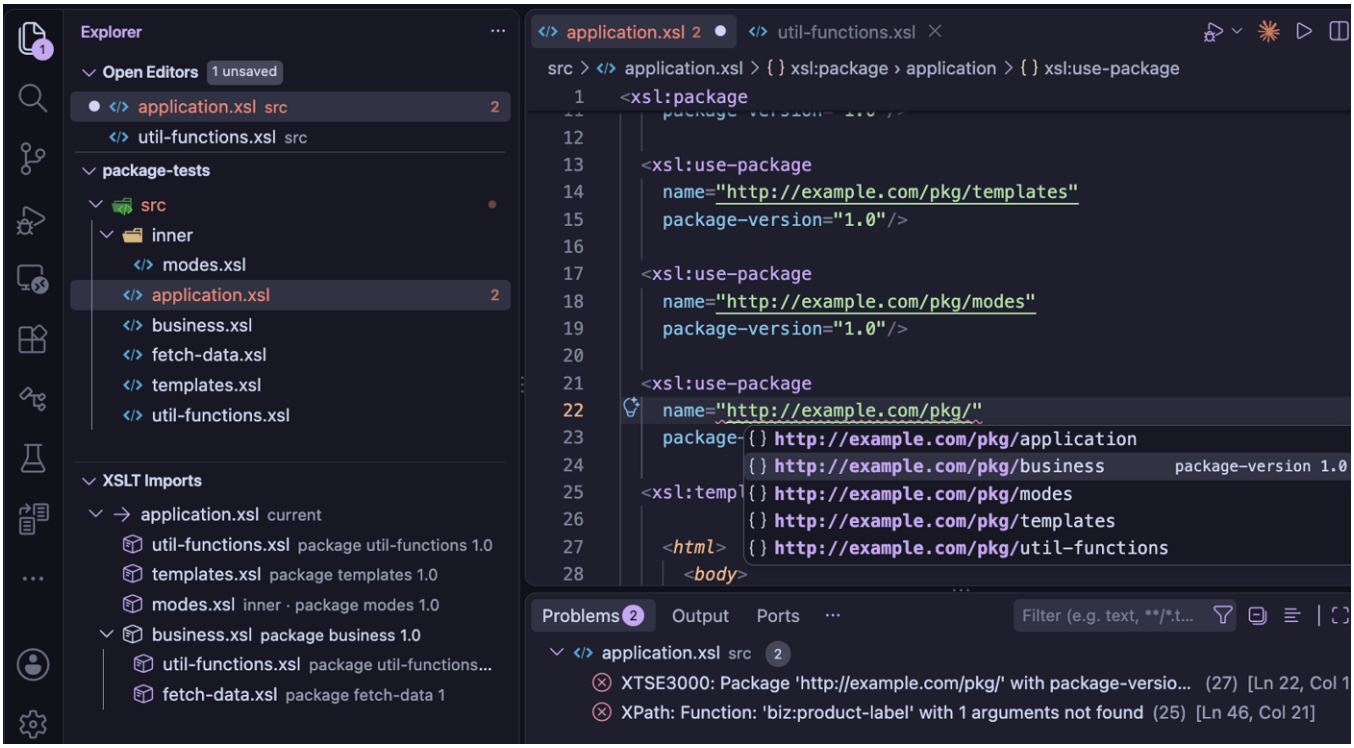
An **XSLT package** is a stylesheet module whose root element is `xsl:package`, with a `name` - usually a URI - and optionally a `package-version`. Another stylesheet uses it by name and version, with `xsl:use-package`, rather than by its file path:

```
<xsl:package name="http://example.com/strings" package-version="2.1"
             version="3.0" xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
    ...
</xsl:package>

<!-- in a stylesheet that uses it -->
<xsl:use-package name="http://example.com/strings" package-version="2+"/>
```

The package URI is an identifier, not a location: `xsl:use-package` matches it exactly against the `name` of each `xsl:package`, and it isn't resolved like the `href` of `xsl:import`. Neither the XSLT specification nor Saxon fixes where a package's file is.

The XSLT/XPath extension finds it, so that the declarations the package makes public are known to the linter, auto-completion, hover help and Go to Definition - and when a stylesheet is run, passes the packages it uses to Saxon.



The XSLT Imports view shows packages used by the current stylesheet, discovered during workspace scanning or declared in workspace settings. Use auto-completion on the `name` attribute of `xsl:use-package` to select and import from additional packages available.

Packages in the workspace

Every `.xsl` and `.xslt` file in the workspace whose root element is `xsl:package` is found automatically, by the same background scan of the workspace that finds the top-level stylesheet importing a module. Its name and version are read from its `xsl:package` element - a package with no `package-version` is version `1`. These packages need no setting, and they're read again when saved: change a package's name or version, and the stylesheets using it see the change.

> *The workspace scan is turned off with the **XSLT.resources.inferParentFromWorkspace** setting. Packages are then found only from the **XSLT.resources.xsltPackages** setting.*

Packages outside the workspace

For a package kept outside the workspace, for example a library shared between projects, add its path to the `XSLT.resources.xsltPackages` setting - relative to the workspace folder, or as a `file:` URI. The name and version are read from the file, so the path is all that's needed:

```
"XSLT.resources.xsltPackages": [
    { "path": "../shared-xslt/strings.xsl" },
    { "path": "file:///opt/xslt/dates.xsl" }
]
```

An entry can also give a `name` and `version`, which are then used instead of those in the file - for example, to try a copy of a package under a different version. Packages from the setting come before those found in the workspace when the same version of a package is in both.

Which version is used

When more than one file declares a package with the same name, the `package-version` of `xsl:use-package` chooses between them: the highest version in its range is used. With no `package-version`, the range is `*`, so the highest version of all is used.

package-version	Versions it matches
`*`	any version
`2.1`	2.1 only - also written 2.1.0
`2.*`	2, 2.0, 2.1, 2.1.5 ...
`2+`	2 or higher
`1.5 to 2.9`	from 1.5 to 2.9, inclusive
`to 2`	2 or lower
`1.0, 3.0`	any of the ranges separated by commas

Versions are compared part by part as numbers, so `1.10` is higher than `1.9`, and a missing part counts as 0, so `1` and `1.0.0` are the same version. A version with a name part, such as `3.0-beta`, is lower than the same version without one, `3.0`.

What a stylesheet can use from a package

Unlike an imported module, a package decides which of its declarations - its *components* - other stylesheets can use. A stylesheet using the package sees its public and final components only, as Saxon does, so the linter reports a reference to anything else, and auto-completion doesn't offer it.

Component	**Visibility, with no `visibility` attribute**
Function, named template, global variable, attribute set, mode	private - unless an `xsl:expose` in the package matches it
Global `xsl:param`	public (it can't have a `visibility` attribute)
Key, accumulator, character map, template rules in the unnamed mode	not exposed as package components: a stylesheet using the package can't refer to them - e.g. name a key in `key()`, or apply the unnamed mode's template rules - though the package's own code uses them as usual

```
<xsl:package name="http://example.com/strings" package-version="2.1" ...>
    <!-- functions in the package's namespace are final, unless they say otherwise -->
    <xsl:expose component="function" names="str:*" visibility="final"/>

    <xsl:function name="str:pad">...</xsl:function>                        <!-- final -->
    <xsl:function name="str:helper" visibility="private">...</xsl:function>  <!-- private -->
    <xsl:function name="str:format" visibility="abstract" as="xs:string"/>   <!-- to be
overridden -->
</xsl:package>
```

A `visibility` attribute on a declaration takes precedence over `xsl:expose`. Where several `xsl:expose` elements match, the most specific name wins: a name, then a wildcard such as `str:*` or `*:pad`, then `*`.

In the stylesheet using the package, `xsl:accept` can narrow a component's visibility - for example to `hidden`, so it can't be used at all - and `xsl:override` replaces a public or abstract component with a declaration of its own. An abstract component can only be used once it's overridden. Within the overriding declaration, `xsl:original` refers to the package's version.

```
<xsl:use-package name="http://example.com/strings" package-version="2+">
    <xsl:accept component="function" names="str:pad#2" visibility="hidden"/>
    <xsl:override>
        <xsl:function name="str:format" as="xs:string">...</xsl:function>
    </xsl:override>
</xsl:use-package>
```

A package that uses another package doesn't pass its components on: they're private to it, unless its own `xsl:accept` or `xsl:expose` makes them public.

Overriding package components

An `xsl:override` declaration must have the same signature as the component it replaces, so the extension writes it for you - as a Java IDE does for overriding methods - from the package's own declaration, with the stylesheet's prefixes:

- **Override Package Components...** - with the cursor anywhere in an `xsl:use-package`, this refactoring (from the 💡 light-bulb, or `^ ⇧ R`) lists the package's public and abstract components that aren't

overridden yet, with the abstract ones already ticked. The ones you pick are added to its `xsl:override`, which is created if needed

- **Auto-completion** - after `<` in an `xsl:override`, the list starts with an *override ...* item for each of those components, abstract ones first, with the package's documentation for it
- **Quick fixes** - for abstract components that aren't overridden, see below

An abstract component is a placeholder, so its declaration has a body that stops the transformation until you implement it, and a documentation note with a TODO and an `@see` link to the package's declaration. In an XSLT 3.0 stylesheet the note is excluded with `use-when="false()"`:

```
<xsl:use-package name="http://example.com/shapes">
    <xsl:override>
        <xsl:function name="shp:area" as="xs:double">
            <xsl:note use-when="false()" format="xdoc-md">
                TODO: implement the abstract function of package http://example.com/shapes.

                @see shp:area#1
            </xsl:note>
            <xsl:param name="shape" as="map(*)"/>
            <xsl:message terminate="yes">shp:area#1 is not implemented</xsl:message>
        </xsl:function>
    </xsl:override>
</xsl:use-package>
```

A public component's declaration calls the package's version with `xsl:original` instead, as Java's `super` does, so it behaves as before until you change it - for a template, with an `xsl:with-param` for each parameter. A global parameter keeps the package's default.

Abstract components that aren't overridden

Saxon compiles a stylesheet that doesn't override an abstract component it doesn't call - but calling a component of the package that depends on it stops the transformation, with `XTDE3052`. So the `xsl:use-package` has a warning that lists them, with quick fixes to override them all, or each one. A call of one in the stylesheet is reported as an abstract component that isn't overridden, with a quick fix to override it. A package can pass an abstract component on to the stylesheets that use it, with `xsl:accept visibility="abstract"`: there's no warning then.

Problems with xsl:accept and xsl:override

These static errors, which Saxon reports when it compiles the stylesheet, are reported as you edit - most with quick fixes, shown with the 💡 light-bulb:

Error	Reported for	Quick fixes
`XTSE3020`	a function name in `xsl:accept` without its arity, such as `str:pad` for `str:pad#2`	Change to the name with each arity the package has
`XTSE3030`	a name in `xsl:accept` that the package doesn't have	Change to a similar name, or remove it
`XTSE3040`	accepting a private component by name, or accepting a component with a wider visibility - a final component as public	Change the visibility to the component's, or remove the name

Error	Reported for	Quick fixes
`XTSE3051`, `XTSE0020`	a component both accepted and overridden by the same `xsl:use-package` - or `#unnamed`, the unnamed mode, which can't be accepted	Remove the name
`XTSE3050`	a declaration with the name of a component the package gives, outside `xsl:override`	Move it into the `xsl:override`
`XTSE3058`	overriding a component the package doesn't have	Change to a similar name, or remove the declaration
`XTSE3060`	overriding a private or final component	Remove the declaration
`XTSE3008`	an `xsl:use-package` in a module that's imported - directly, or through `xsl:include` - it belongs in the top-level stylesheet	
`XTSE3070`	an overriding declaration whose types differ from the original's - its `as`, and its parameters' types. A template must also keep each parameter of the original, with the same `required` and `tunnel` values (it may leave out a tunnel parameter), and any parameter it adds must be optional	Change the signature to match the package's - keeping the names of a function's parameters, and the parameters a template adds, made optional

A name removed from `xsl:accept` takes the `xsl:accept` with it, if it's the only one.

> *Types are compared as written, with their prefixes' namespaces and ignoring whitespace, because Saxon requires the same type - a subtype, such as `xs:NCName` for `xs:string`, isn't allowed. A missing `as` is `item()*`.*

Declared modes

In an `xsl:package`, each mode used must be declared with `xsl:mode` - the `declared-modes` attribute is `yes` by default for a package, unlike a stylesheet. That includes the unnamed mode: a package with template rules that have no `mode` attribute needs `<xsl:mode/>`. An undeclared mode is reported as `XTSE3085`, once, where it's first used: in the `mode` of a template rule or `xsl:apply-templates`, or the `default-mode` of the package. A mode declared in a module the package includes or imports, or a public mode of a package it uses, is declared.

The quick fixes add the declaration - **Add <xsl:mode name="..."/>** or **Add <xsl:mode/>**, after the package's other `xsl:mode` elements or before its first template - or **Set declared-modes="no" on the xsl:package**.

Hover help

Hover over the `name` of an `xsl:use-package` to see the package's documentation - its module note, the `xsl:note` that's the first child of its `xsl:package` - the version and file used, and the components the stylesheet can use, by kind. These are after its `xsl:accept` elements, and without those it overrides, so the list is the package's public interface as this stylesheet sees it.

Hover help for a function or named template from a package says which package it's from, and its visibility - for example, *Final function of package `http://example.com/strings` 2.1, declared in strings.xsl* - with its documentation note.

For a declaration in `xsl:override`, and the calls of it, the hover help says what it overrides - for example, *Overrides the abstract function of package* `http://example.com/shapes`, *declared in shapes.xsl* - and if it has no documentation note of its own, it has the package declaration's, as an overriding method's documentation is inherited in Java. In the note of an overriding declaration, an `@see` of the component it overrides links to the package's declaration.

Converting a library to a package

A library that stylesheets use with `xsl:import` or `xsl:include` can become a package, but not just by changing its root element: a package's components are private by default, a stylesheet's declaration with the name of the library's must move into `xsl:override`, and some declarations aren't exposed as package components at all. **Analyse as Package...** shows what would have to change for the stylesheets to work as before - it changes nothing.

Run it from the Command Palette, with the library open, or from the context menu of a module in the XSLT Imports view. It checks each top-level stylesheet in the workspace that imports or includes the library, directly or indirectly, and opens a report in the Markdown preview, with links to each place:

- **To rework first** - what would fail in a package:
  - a global variable whose value uses the context item, e.g. `select="/*"`, which has none in a library package (`XPDY0002`)
  - a stylesheet's `xsl:global-context-item` that overrides the library's by import precedence, but with a different type, which an override can't have (`XTSE3070`)
  - a name the library uses that isn't declared in it or the stylesheets found to use it, e.g. in a module whose import isn't resolved, as a package must declare all it uses. The report lists imports that aren't found: the XML catalog setting may resolve them, e.g. DITA-OT's `plugin:` URIs
  - code that would move into `xsl:override` from a module outside the workspace, or that stylesheets which don't use the library use too, as they'd lose it
- **Made public** - the library's functions, templates, variables and attribute sets that the stylesheets use
- **Overrides** - the stylesheets' declarations that override the library's by import precedence, which move into `xsl:override`
- **Abstract components** - functions, templates and variables that the library uses but only the stylesheets declare, which the package declares as abstract
- **Modes and template rules** - the library's named modes that the stylesheets use, and their template rules in them, which move into `xsl:override` - and the unnamed mode, if the library shares it with the stylesheets, which becomes a named mode, as a package keeps its unnamed mode to itself (see below)
- **Not exposed by a package** - keys, accumulators, decimal formats, character maps, `xsl:strip-space` and `xsl:output`, which apply only to the package's own code, so the stylesheets need their own copies of those they rely on

The report starts with a verdict: whether the library can be converted, with how many changes, or which items need a decision or reworking first. Each item has a link to its place, then a list of the other modules involved - for a component made public, the stylesheets that use it.

> *The analysis is lexical, like the linter: it finds names as they're written, with their namespaces, so a name built at run time, e.g. by `function-lookup()`, isn't found. To find a global variable that uses the context item, the linter checks its value as the body of a function, where there's no context item.*

Convert to Package...

When nothing needs reworking first, **Convert to Package...** makes the changes, from the Command Palette or the context menu of a module in the XSLT Imports view. It asks for the package's name and version, then shows the changes to every module in the **Refactor Preview**, where you can check them before they're made:

- the library's root element becomes `xsl:package`, with the name and version, and `declared-modes="no"`
- its components that the stylesheets use or override get `visibility="public"`, its named modes they use a public `xsl:mode`, and the functions it calls that only the stylesheets declare an abstract declaration
- in each top-level stylesheet, the `xsl:import` or `xsl:include` of the library - in any of its modules - is replaced by an `xsl:use-package` in the top-level module, after its imports: `xsl:use-package` can't be in an imported module (`XTSE3008`)
- the stylesheet's overriding declarations and template rules move into its `xsl:override`, named with its prefixes, and `xsl:apply-imports` in them becomes `xsl:next-match`
- the stylesheet gets its own copies of the keys, decimal formats, character maps, `xsl:strip-space` and `xsl:output` it relied on

Any items the analysis listed as needing a decision are left as they are - after the conversion, the linter reports what still needs to change, mostly with quick fixes.

The unnamed mode

A library and the stylesheets that import it often share the unnamed mode: the stylesheets apply the library's template rules, and add rules of their own, which take precedence. A package's unnamed mode is private to it, so the conversion asks for the name of a public mode that it becomes - e.g. `lib:default`, with a prefix the library declares - and then:

- each of the library's modules gets `default-mode="lib:default"`, so its template rules and `xsl:apply-templates` without a `mode` are in the named mode - and its `xsl:mode` for the unnamed mode, if it has one, gets the name and `visibility="public"`
- the stylesheets' template rules in the unnamed mode move into `xsl:override`, with the mode, and their `xsl:apply-templates` without a mode get it
- each top-level stylesheet gets a template rule that applies templates in the named mode: the transformation starts in the stylesheet's unnamed mode, as a package's mode can't be the initial mode

```
<!-- the transformation starts in the unnamed mode: this applies templates in the mode of
package urn:lib that it became -->
<xsl:template match=".">
    <xsl:apply-templates select="." mode="lib:default"/>
</xsl:template>
```

A stylesheet's own `xsl:mode` for the unnamed mode, e.g. with `on-no-match`, has to move to the library first, as a mode can't be overridden.

A template rule that's also a named template, with both `match` and `name`, can't move into `xsl:override` as it is, where it would also override a named template the package doesn't have. It's split: the named template stays where it is, and a template rule that calls it, passing its parameters, moves into `xsl:override`. A rule that uses `xsl:next-match` or `xsl:apply-imports` can't be split like this, as they depend on the current template rule, so it has to be reworked first.

Running and debugging

When a stylesheet that uses packages is run - with Quick Run, an XSLT task for SaxonJ or SaxonC, or the debugger - the package files are passed to Saxon with its `-lib` option. These are the packages used by the stylesheet, by the modules it imports or includes, and by the packages themselves - each the version chosen as above. Saxon reads each package's name and version from the file, as the extension does, so packages are only ever declared in one place: the packages themselves.

Without this, Saxon reports `XTSE3000 Cannot find package` for each package used.

> *A task with a `configFilename` property - a Saxon configuration file - doesn't get the `-lib` option, as the configuration file may declare the packages itself, in its `xsltPackages` element. SaxonJS tasks don't get it either.*

A package that isn't found

An `xsl:use-package` whose package isn't found - no package with the name, or none with a version in its `package-version` range - is reported as an error, `XTSE3000`, as Saxon reports it. Two quick fixes are offered:

- **Create Package '...'** - choose where to save a new package file, by default beside the stylesheet and named from the package URI's last part. It has the package's name, and the version the `package-version` asks for (the lowest, for a range such as `2+`)
- **Add Existing Package File...** - pick a package file, for example outside the workspace, to add to the `XSLT.resources.xsltPackages` setting. Its `xsl:package` must have the name, and a version in the range

Auto-completion

In the `name` of an `xsl:use-package`, auto-completion lists the URIs of the packages found - in the workspace and the `XSLT.resources.xsltPackages` setting - with each package's versions. In its `package-version`, it lists `*` and the versions of the named package, highest first.

Navigating to a package

The `name` of an `xsl:use-package` is a link to the package's file - press `⌘+click` to open it. Go to Definition, hover help and auto-completion for the package's public functions, templates, variables and other components work as for an imported module.

Navigating between overrides

Between a package's components and the declarations that override them, VS Code's own commands work:

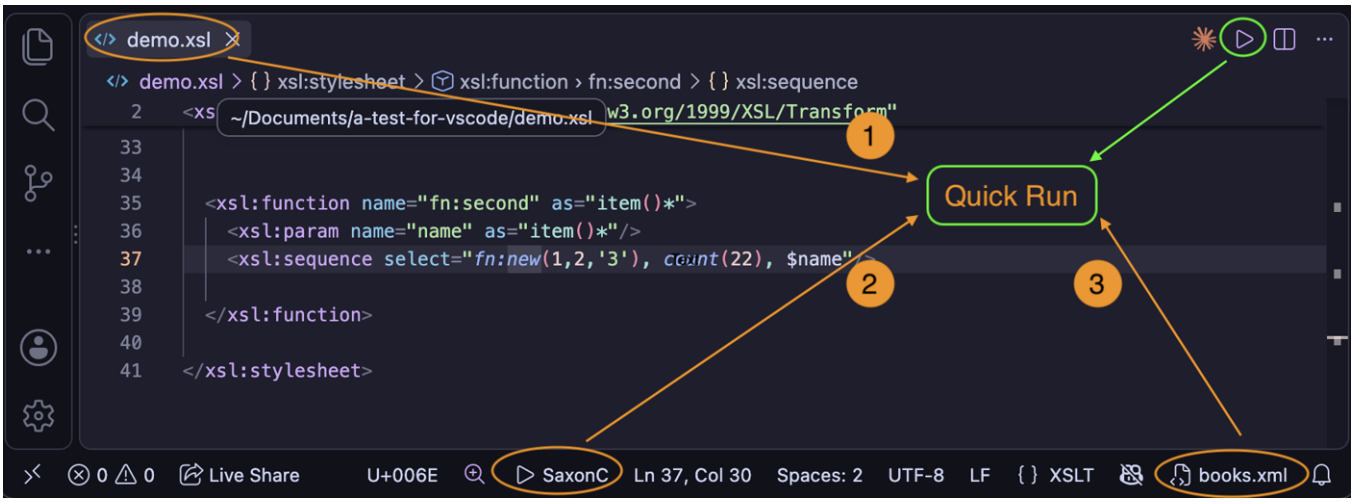
- **Go to Implementations** (⌘F12) - from a package component, or a call of it, to the declarations in the workspace that override it
- **Go to Declaration** - from an overriding declaration's name, or `xsl:original` within it, to the package's declaration
- **Find All References** (⌥⇧F12) - for a package component, includes the declarations that override it - and for an overriding declaration, the package's declaration

With code lenses on (the **editor.codeLens** setting), an overriding declaration shows *overrides public function str:pad#2 of package http://example.com/strings*, which goes to the package's declaration, and a package component shows *overridden in 2 modules*, which lists the declarations that override it.

> *Renaming a package component renames it in the package only, not the declarations overriding it in the stylesheets that use the package - nor their calls of it.*

Quick Run

Quick Run is the fastest way to run an XSLT stylesheet, without first creating a task by hand. It can run the stylesheet with any of the three Saxon processors: **SaxonJ** (Java), **SaxonJS** (JavaScript) or **SaxonC** (native C/C++). It reuses the same command-line arguments the corresponding XSLT task would use, but takes the stylesheet and XML source directly from the editor rather than requiring a pre-configured `tasks.json` entry.



(1) XSLT in active editor, (2) the Saxon processor, (3) the XML context file - Click the play button to run

Running a stylesheet

With an XSLT file open in the active editor, invoke Quick Run in one of three ways:

- Press `⌘⌥R` (`Ctrl+Alt+R` on Windows/Linux)
- Click the play button in the editor title bar - its tooltip names the processor currently in use, for example **Quick Run with SaxonC**
- Run `Quick Run with Saxon` from the Command Palette

The stylesheet is the active editor's file. The XML source is the current XML context file - the same file used to provide XML node-names when auto-completing XPath location steps, shown in the status bar. This means the context file now serves two purposes: XPath auto-completion, and as the Quick Run source document.

An XML context file must be set before Quick Run can be used - unless the stylesheet declares an `xsl:initial-template`, which Quick Run then starts from (see Starting from xsl:initial-template below). Each processor also needs its own setup:

- **SaxonJ** - the **XSLT.tasks.saxonJar** setting must point at your Saxon jar file. See the **Saxon (Java) setup** section on the XSLT Tasks page.
- **SaxonJS** - requires a NodeJS installation; the `xslt3` package is run using NPX. See the **Saxon-JS Tasks** section on the Tasks page.
- **SaxonC** - the **XSLT.tasks.saxonCPath** setting must point at the folder containing the SaxonC `Transform` executable (`Transform.exe` on Windows). See the **SaxonC Tasks** section on the Tasks page for setup details, including native library paths and licensing.

> *Tasks for each processor can be turned off with the **XSLT.tasks.java.enabled**, **XSLT.tasks.js.enabled** and **XSLT.tasks.saxonc.enabled** settings. Quick Run reports an error if the selected processor's tasks are disabled.*

Running a module that another stylesheet imports

A module that another stylesheet imports or includes may use functions, variables or templates declared in other modules of that stylesheet. In the editor, these references are resolved through the top-level stylesheet - see Inferred Imports - but the module can't be compiled, and so can't be run, on its own.

When you Quick Run such a module, it's checked first - with the modules it imports and includes - and if it depends on the top-level stylesheet, you're offered the choice:

- **Run *top-level stylesheet*** - runs the top-level stylesheet instead, with the same XML context file and processor
- **Run Anyway** - runs the module itself

The choice is remembered for the module until Visual Studio Code is restarted. A module that can be run on its own is run straight away, with no prompt. The top-level stylesheet can also be run from the ▶ button in the XSLT Imports view, whichever of its modules is being edited.

Choosing the processor: SaxonJ, SaxonJS or SaxonC

Quick Run uses SaxonJ by default. When an XSLT or XML file is active, the processor currently selected is shown in the status bar, for example ▶ **SaxonJ**. To switch processor, use any of the following:

- Click the processor name in the status bar
- Hold `⌥` (`Alt` on Windows/Linux) and click the play button in the editor title bar - while the key is held, the play button changes to a 'switch' button
- Run `XSLT: Quick Run: Switch Processor` from the Command Palette

Each of these shows a list of the three processors to pick from, with the current one marked. The play button's tooltip and the status bar update straight away.

The choice is stored in the **XSLT.tasks.quickRunProcessor** setting, whose values are the task types: `xslt` (SaxonJ), `xslt-js` (SaxonJS) and `xslt-c` (SaxonC). You can also edit this setting directly, either as a User setting or as a Workspace setting - for example, to always use SaxonC for one project. Switching processor updates the setting where it is already defined (the workspace folder, then the workspace), otherwise it updates your User settings.

> *SaxonC, a native executable, starts much faster than SaxonJ, which needs a few hundred milliseconds to start the Java VM and load Saxon on each run. So SaxonC usually gives the quickest turnaround when running a stylesheet on small or medium inputs. For longer transforms, SaxonJ's Java JIT compiler makes it faster: in our tests, SaxonJ was ahead once execution took more than about two seconds.*

First use: a task is created for you

The first time Quick Run is used for a given stylesheet/context-file pairing, a real task is added to your workspace's `tasks.json` file (creating the file if it doesn't exist yet) - not just an ephemeral, one-off run. The generated task:

- is a normal task for the selected processor: an `xslt` task for SaxonJ, an `xslt-js` task for SaxonJS or an `xslt-c` task for SaxonC
- is labelled from the stylesheet and context file names, for example **demo with books.xml**. SaxonJS and SaxonC task labels have the processor name added, for example **demo with books.xml (SaxonC)**
- uses the `xslt-xpath.pickResultFile` command for its result path, so you're prompted to choose (or reuse a recently used) output location - the same 'recent files' behaviour described in the Variable References in Tasks section of the XSLT Tasks page
- refers to the processor's setting (for example `${config:XSLT.tasks.saxonCPath}`) rather than copying its current value, so the task keeps working if you later change the setting
- includes an entry in its `parameters` array for each stylesheet parameter that must be set - including those declared in imported or included stylesheets. Parameters with a default are left out, so the stylesheet's own defaults apply

Required parameters

A stylesheet parameter must be set when it has `required="yes"`, or when it has no default (no `select` attribute or content) and an `as` type that doesn't allow an empty sequence - for example `as="xs:string"` or `as="element()+"`, but not `as="xs:string?"`. A parameter without an `as` attribute defaults to a zero-length string, so it isn't added.

The value of each one comes from an input that's added to the `inputs` of `tasks.json`, so you're asked for it each time the task runs:

```
"parameters": [
    { "name": "title", "value": "${input:xsltParam-title}" },
    { "name": "?config", "value": "${input:xsltParam-config}" }
]
...
"inputs": [
    {
        "id": "xsltParam-title",
        "type": "command",
        "command": "xslt-xpath.enterParameterValue",
        "args": { "name": "title", "type": "xs:string", "xpath": false }
    },
    ...
]
```

You can type a value, or pick one you've used before for a parameter of that name - the last one used is first, so Enter uses it again. Press Escape to cancel the run. Once you've settled on a value, you can replace the `${input:...}` reference with the value itself. Parameters with the same name, in different tasks, share an input.

> *For an atomic type, such as `xs:string`, `xs:integer` or `xs:anyURI`, the value is typed as it is, without quotes - Saxon converts it to the declared type. For other types, such as maps, arrays and nodes, the parameter name is prefixed with `?`, for example `?config`, which tells Saxon to evaluate the value as an XPath expression: `map{'a': 1}`, `doc('data.xml')`, or `'text'` for a string.*

Subsequent runs

Once a task exists for a stylesheet/context-file pairing, Quick Run re-runs that same task rather than creating another one. This means you can freely edit the generated task afterwards - change a parameter value, adjust the result path, add Saxon features or additional parameters - and Quick Run will keep using your edits on every subsequent run, just like running any other task.

Editing the task

To go straight to the task for the stylesheet and the current XML context file, use the command **XSLT: Edit Quick Run Task** in the Command Palette, or the **Edit Quick Run Task** link in the tooltip of the XML context file in the status bar. This opens `tasks.json` with the task highlighted - if there isn't a task yet, it's created first, as Quick Run would create it.

Debugging always uses the SaxonJ (`xslt`) task. The **Edit Debug/Run Task (SaxonJ)** button in the editor's title bar, beside the Run and Debug buttons, opens that task whichever processor is selected. With SaxonJS or SaxonC selected, the editor's **...** menu also has **Edit Quick Run Task (SaxonJS)** or **Edit Quick Run Task (SaxonC)**. When Quick Run or debugging first creates a task, a notification also offers **Edit Task**.

For different sets of parameters, copy the task in `tasks.json` and give the copy a new label. When more than one task is for the stylesheet/context-file pairing, Quick Run (and debugging) asks which task to run - the tasks you've used most recently are listed first.

Tasks are kept separately for each processor. If you switch processor, the next Quick Run for the same stylesheet/context-file pairing creates a new task for that processor; your existing tasks for the other processors are left unchanged, and are used again if you switch back.

If you want a fresh task instead, either edit the existing one directly, or remove it from `tasks.json` and Quick Run will create a new one next time it's used for that stylesheet/context-file pairing.

> *An alternative way to run a task created with Quick Run is to invoke the **Tasks: Run Task** command from the Command Palette, this will list all your tasks in a pick list. The tasks are sorted with the most recently run tasks at the top.*

Starting from xsl:initial-template

A stylesheet that declares an `xsl:initial-template` can be run with no source document at all. When no XML context file has been set yet, Quick Run does this automatically. To keep running without a source document once XML files are opened - which otherwise become the context file - click the XML context file in the status bar and choose **None**, the last entry in the list of files. The status bar then shows **[no XML context]**.

With no XML context file, Quick Run starts the stylesheet from `xsl:initial-template` (Saxon's `-it` command-line option) instead of applying templates to a source document. The `xsl:initial-template` can be declared in the stylesheet itself or in any module it imports or includes. If there isn't one, Quick Run reports an error and offers to pick an XML context file instead.

The task created for this has an empty `xmlSource` and an empty `initialTemplate` property, and is labelled with the stylesheet name, for example **report with xsl:initial-template**. It is kept separately from any task for the same stylesheet that has an XML source, so you can switch between the two just by changing the context file.

None stays selected until you pick a file again - opening or viewing other XML files does not replace it. While it is selected, XPath auto-completion has no source document to take element and attribute names from.

> *A stylesheet can declare `xsl:initial-template` and also contain match templates designed for a source document, so Quick Run doesn't guess which way to run it: while an XML context file is set, it always applies templates to that file. The first time it creates a task for such a stylesheet, a notification points out the alternatives - add `"initialTemplate": ""` to the task to start from `xsl:initial-template` with the context file as the global context item, or choose None to run with no source document at all.*

Running from an XML file

Quick Run can also be started from an XML file, to run a stylesheet on it without opening the stylesheet first. With an XML file open in the active editor, invoke it in one of three ways:

- Press `⌘⌥R` (`Ctrl+Alt+R` on Windows/Linux)
- Click the play button in the editor title bar
- Run `XSLT: Quick Run: Run XSLT on this XML File` from the Command Palette

This shows a list of the XSLT tasks that can run on the file, grouped as:

- **tasks for books.xml** (named after the XML file) - every task whose `xmlSource` is this file, for any processor. This includes each task Quick Run has created for a stylesheet with this file as its source, so any stylesheet you've already run on the file is one click away.
- **tasks for the current file** - tasks whose `xmlSource` is `${file}`, which Visual Studio Code replaces with whichever file is active when the task runs. Use this for general-purpose stylesheets you want to run on any XML file.
- **Pick Stylesheet...** - choose a stylesheet from the recently used list, from any `<?xml-stylesheet?>` processing instruction in the XML file, or from the file explorer. Quick Run then creates and runs a task for that stylesheet and this file, using the selected processor, exactly as it would from the stylesheet itself - so it appears under **tasks for books.xml** next time.

Each task is listed with its processor and stylesheet path, and you can type to filter on any of these. If there are no matching tasks yet, Quick Run goes straight to picking a stylesheet. As with the stylesheet's play button, holding `⌥` (`Alt` on Windows/Linux) while clicking the XML file's play button switches the processor instead.

A task that runs on the current file looks like this:

```
{
        "type": "xslt",
        "label": "Summarise",
        "saxonJar": "${config:XSLT.tasks.saxonJar}",
        "xsltFile": "${workspaceFolder}/tools/summary.xsl",
        "xmlSource": "${file}",
        "resultPath": "${command:xslt-xpath.pickResultFile}"
}
```

Each stylesheet has its own XML context file

The XML context file you pick from the status bar is for the stylesheet in the active editor, so you can switch between stylesheets that process different documents and each one keeps its own. A module imported or included by a top-level stylesheet shares that stylesheet's context file.

For a stylesheet you haven't picked a context file for, the source of its Quick Run task in `tasks.json` is used - so a stylesheet you've run before keeps the file it was run with. Otherwise, it's the context file you last picked for any stylesheet, or with none picked yet, the first XML file you viewed.

The XML context file is remembered

Your chosen XML context files are restored automatically when Visual Studio Code is reopened, rather than defaulting back to the last XML file you happened to view. They're stored per workspace, separately from the 'recently used' file list, so viewing other XML files - for example a transform's result - does not change what gets restored. Choosing **None** is remembered in the same way.

If the context file has since been deleted or moved, Quick Run - and debugging - say so instead of running, with a **Pick XML Context File** button to choose another.

> *For custom parameters and features beyond what's auto-populated, chaining tasks together, or any other task configuration in detail, see the Tasks page - a task Quick Run creates is a normal task, so everything described there still applies to it.*

Tasks

> *Just want to run a stylesheet without configuring a task first? See Quick Run.*

Visual Studio Code's Tasks include custom task types for configuring and running XSLT transforms with SaxonJ (Java), SaxonJS (JavaScript) or SaxonC (C/C++). The Saxon XSLT processor is called via its command-line interface from the Integrated Terminal in Visual Studio Code. Task configurations effectively control command-line arguments for the XSLT Processor. An attempt is made to parse any error messages returned from Saxon, so that the position of any Saxon-reported errors in the XSLT can be highlighted.

XSLT is run or debugged using custom `xslt`, `xslt-c` or `xslt-js` Visual Studio Code Tasks for SaxonJ (requires Java and Saxon installation), SaxonC or SaxonJS (requires NodeJS installation) respectively.

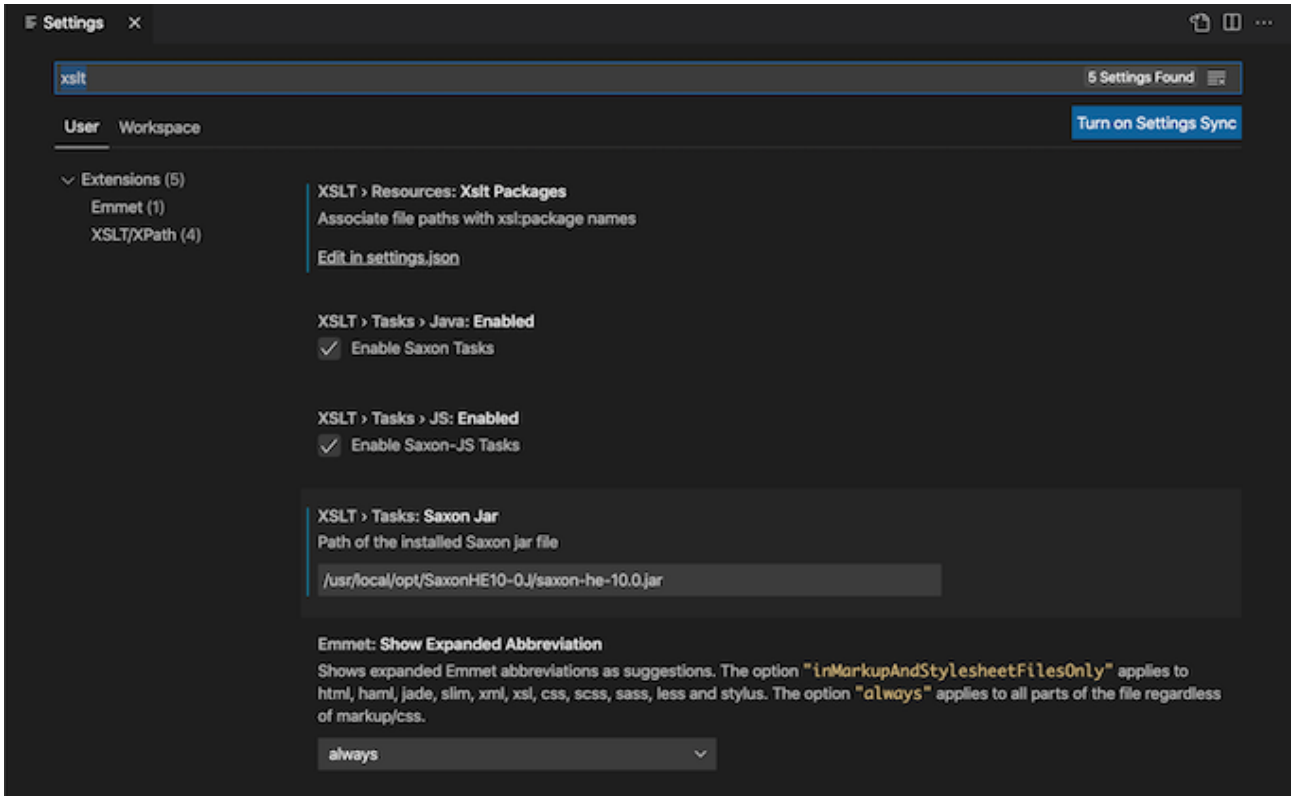
Saxon (Java) setup

1. Ensure that you have a workspace folder open in Visual Studio Code (use **File > open...** from the menu). The configuration will be saved in a `tasks.json` file in this workspace folder.

2. If Java is not already installed, Java 1.8 or later can be downloaded and installed from here

3. If Saxon is not already installed, Saxon-HE can be downloaded and installed from here

4. Saxon (Java) tasks need the path to the installed Saxon jar file to be included in a property in your Visual Studio Code settings. Invoke `Preferences: Open User Settings` from the Command Palette (⇧⌘P) and type `xslt` to show a short-list of settings:



5. Enter the Jar file path in the **Tasks: Saxon Jar** property (see above)

Saxon-JS Tasks

1. Ensure that you have a workspace folder open in Visual Studio Code (use **File > open...** from the menu). The configuration will be saved in a `tasks.json` file in this workspace folder.

2. If NodeJS is not already installed, it can be downloaded and installed from here

3. Initialize NPM in the current folder and then install the Saxon-JS package (xslt3) by running the following commands in the Visual Studio Code terminal (⌘J):

```
npm init -y
npm install --save-dev xslt3
```

> *The Saxon-JS task uses NPX to invoke Saxon-JS task. NPX will download the Saxon-JS package if you choose not to follow the previous step and it can't find the Saxon-JS package in your global space.*

SaxonC Tasks

The `xslt-c` task type runs XSLT transforms using SaxonC, Saxonica's native C/C++ port of Saxon. Its command-line arguments mirror the Saxon (Java) task closely, since SaxonC's `Transform` command-line tool follows the same conventions.

Setup

1. Install SaxonC and note the folder containing its `Transform` (or `Transform.exe` on Windows) command-line executable.

2. Set the **XSLT.tasks.saxonCPath** user setting to that folder.

3. Invoke `Tasks: Run Build Task` from the Command Palette (⇧⌘B) and configure a new `xslt-c` task, following the same steps as for a Saxon (Java) task described above.

> *SaxonC tasks can be disabled with the **XSLT.tasks.saxonc.enabled** setting, matching the equivalent settings for Saxon (Java) and Saxon-JS tasks.*

Native library path (macOS/Linux)

Some SaxonC distributions are built without an embedded library search path, so the native shared libraries alongside `Transform` - and any others it depends on - may not be found automatically. The `xslt-c` task always searches the `lib` folder next to the configured `saxonCPath`, and you can list additional folders (for example, a separately-packaged core library) in the **saxonCLibraryPaths** task property, or the equivalent **XSLT.tasks.saxonCLibraryPaths** user setting.

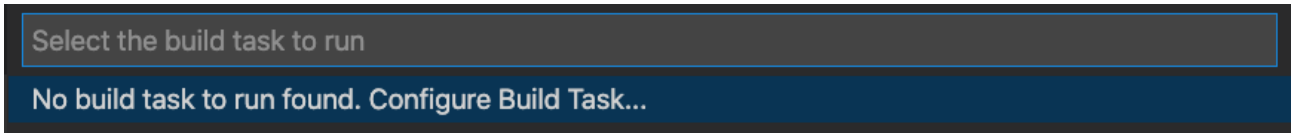
> *If a transform fails with an error like `Library not loaded: ...dylib ... no LC_RPATH's found`, add the folder containing that library to **saxonCLibraryPaths**.*

Licensing

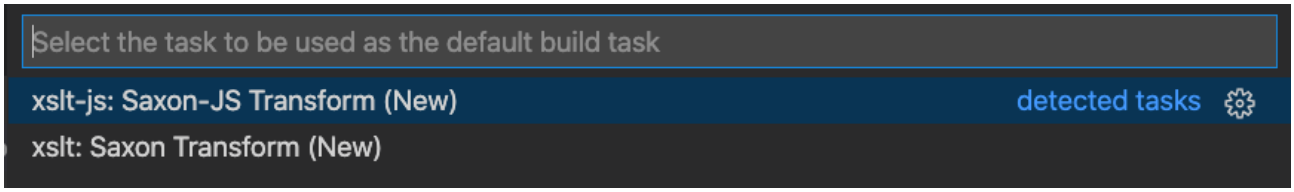
If SaxonC reports `No license file found – running with licensable features disabled`, set the **licenseFileLocation** task property (or the equivalent **XSLT.tasks.saxonCLicenseFileLocation** user setting) to the path of your `saxon-license.lic` file. This is passed to `Transform` as `--licenseFileLocation:`, the same option Saxon (Java) tasks use.

To configure the initial task:

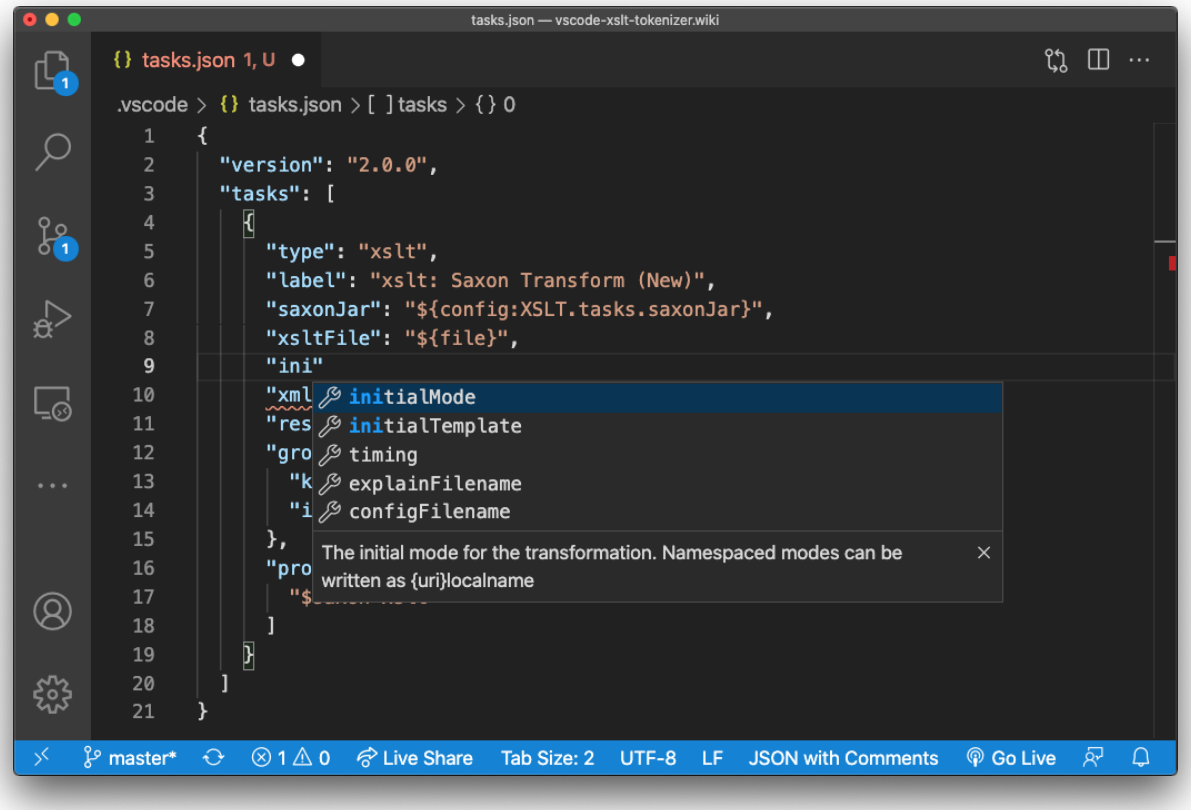
1. Invoke `Tasks: Run Build Task` from the Command Palette (⇧⌘B):



2. Press `Enter` to **Configure Build Task**:



3. Select from the list (shown above) the template task for the processor you require: **Saxon Transform (New)** for SaxonJ, **Saxon-JS Transform (New)** for SaxonJS or **SaxonC Transform (New)** for SaxonC (you may see more task types in the list).



4. The `tasks.json` file will then be opened with a template task added for the task type specified. The task shown above is for a SaxonJ (`xslt`) task. Edit this task to fit your XSLT transform requirement. It's a good idea to change the `label` property (the value shown in the tasks list) to describe your XSLT transform. Auto-completion with hints is available as you type. The actual file type is 'JSON with Comments' so you can add comments and leave trailing commas etc.

5. You can create any number of tasks for different XSLT transform scenarios. Note that the `group.isDefault` property controls which task is automatically launched when you next invoke `Run Build Task` from the Command Palette or when using `⇧⌘B` .

6. Save any changes to the `tasks.json` file `⌘S`

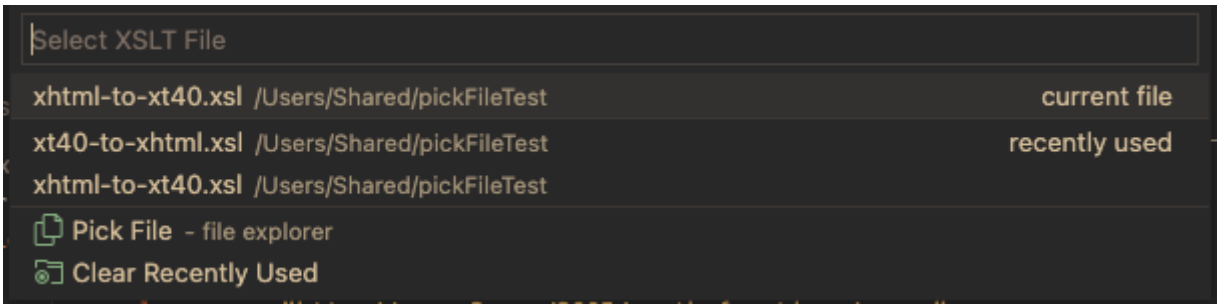
7. Run the newly configured XSLT transform by invoking `Run Build Task` from the Command Palette or when using `⇧⌘B` .

Variable References in Tasks

Visual Studio Code supports variable substitutions for Task configuration files. Variable substitution uses the `${variableName}` syntax. Commands that return a string can be invoked via variable substitution: the boilerplate XSLT task uses the `${command:xslt-xpath.pickXsltFile}` and `${command:xslt-xpath.pickXmlSourceFile}` variable references, and writes its result to `${workspaceFolder}/xsl-out/result1.xml` .

User-input for file-selection within a Task

A set of 'Pick File' commands suitable for XSLT tasks is available to allow user-selected file paths to be used within an XSLT task. An example is shown below:



Each command prompts the user to select a file. This is from: the current file, a 'recent files' list for that command (optionally from a specified 'previous stage' command), or from a 'Open File' or 'Save File' VS Code dialog as appropriate. The available commands are listed below:

Command	Description
xslt-xpath.pickFile	General file selection with arguments to provide custom behaviour. This must be invoked with arguments that are supplied in the 'inputs' property of the tasks.json file. Invoked by ${input:<input-id>}
xslt-xpath.pickXsltFile	Returns a user-selected XSLT file
xslt-xpath.pickXmlSourceFile	Returns a user-selected XML Source file
	Returns a user-selected Stage2 XML source file

Command	Description
xslt-xpath.pickStage2XmlSourceFile	
xslt-xpath.pickResultFile	Returns a user-selected result file
xslt-xpath.pickStage2ResultFile	Returns a user-selected result file for Stage2

A full list of variable references is given in Visual Studio Code's Variables Reference

The 'pickFile' command

The general purpose command 'xslt-xpath.pickFile' is the base command for specific 'pick file' commands like 'pickXsltFile'. After a task is invoked, the user is prompted to select a file for opening or saving. A Pick List is shown with options to select files in up to 5 ways:

- the current file (if its extension is compatible with the extensions list)
- from recently used (current stage) files - from list for the current label e.g. 'Select XSLT File'
- from recently used (previous stage result) files - shows result files from a previous task
- from the `<?xml-stylesheet href="...?>` href in the current file
- from an 'Open File' or 'Save File' dialog

The 'recently used' file data is maintained per workspace and is restored when VS Code is closed and reopened. A 'clear recently used' option is also listed when there are files already in the list.

The 'pickFile' command must be invoked from an 'input' property of a task, as it requires an 'args' argument. A sample 'inputs' property can be added to the 'tasks.json' file with the **XSLT: Add XSLT Inputs to Tasks File** command.

Command Options for 'pickFile'

The 'pickFile' command's 'args' object has the following properties:

- label - the prompt message for the user, also the identifier for the 'recently' used list
- extensions? - an array of filter file-extensions (when set, a '*' option can also be selected from the dialog)
- isResult? - causes the 'Save File' dialog to be shown when set true
- prevStageLabel? - shows the recently used files for the specified label
- prevStageGroup? - the group-name for the 'previous stage' files - default: 'recent files from previous stage'
- xmlStylesheetPI? - set 'true' for pick-list to include a `<?xml-stylesheet?>` href from current editor

Sample input property definitions for the 'pickFile' command are listed below:

```
"inputs": [
/*
1. the specialist 'pickFile' command equivalents for the inputs defined here are:
        "xsltFile": "${command:xslt-xpath.pickXsltFile}",              // can select from
current file - if xslt - or recently used xslt files
        "xmlSource": "${command:xslt-xpath.pickXmlSourceFile}",        // can select from
current file or recently used stage1 source files
        "xmlSource": "${command:xslt-xpath.pickStage2XmlSourceFile}",  // can select from
recently used stage2 source files or stage1 result files
        "resultPath": "${command:xslt-xpath.pickResultFile}",          // can save to recently
used stage1 result files
        "resultPath": "${command:xslt-xpath.pickStage2ResultFile}",    // can save to recently
used stage2 result files

2. these inputs invoke the 'pickFile' command with args for custom behaviour:
*/
{
        /* --- Usage: ---
        "xmlSource": "${input:xmlFile}",
        */
        "id": "xmlFile",
        "type": "command",
        "command": "xslt-xpath.pickFile",
        "args": {"label": "Select XML File", "extensions": ["xml", "xhtml", "svg"] }
},
{
        /* --- Usage: ---
        "xmlSource": "${input:xmlFile2}",
        */
        "id": "xmlFile2",
        "type": "command",
        "command": "xslt-xpath.pickFile",
        "args": {"label": "Select Stage1 XML File", "extensions": ["xml", "xhtml", "svg"],
"prevStageLabel": "Select Result File", "prevStageGroup": "recent files from previous stage" }
},
{
        /* --- Usage: ---
        "xsltFile": "${input:xsltFile}",
        */
        "id": "xsltFile",
        "type": "command",
        "command": "xslt-xpath.pickFile",
        "args": {"label": "Select XSLT Stylesheet", "extensions": ["xsl", "xslt"],
"xmlStylesheetPI": true }
},
{
        /* --- Usage: ---
        "resultPath": "${input:resultFile}",
        */
        "id": "resultFile",
        "type": "command",
        "command": "xslt-xpath.pickFile",
        "args": {"label": "Select Result File", "isResult": true }
}
]
```

Configuring more tasks

To create more tasks, just invoke `Tasks: Run Build Task` from the Command Palette and choose the task type you need. Provided you've changed the `label` property from the template task for that type, a new task will be created from the template task.

Visual Studio Code tasks are very flexible. You can combine XSLT tasks with other task types, for example, to run a shell command. The `dependsOn` task property is used to chain tasks together. The **Compound Tasks** section of the Visual Studio Code Tasks documentation covers this concept in more detail.

XML catalogs

The `xslt` and `xslt-c` tasks pass the catalog of the `XSLT.resources.catalog` setting to Saxon, with the `-catalog` option - unless the task has its own `catalogFilenames` property. See XML Catalogs.

Java extension functions

An `xslt` task can run a stylesheet that calls Java code. Add the folder or jar with your classes to the task's `classPathEntries` property. Saxon supports two kinds of Java extension function:

- **Reflexive** - a namespace with a `java:` URI names a Java class, and the functions in it call the class's methods, e.g. `math:sqrt(2)` with `xmlns:math="java:java.lang.Math"`. These need Saxon-PE or Saxon-EE.
- **Integrated** - a class that extends Saxon's `ExtensionFunctionDefinition`, registered by a Saxon configuration file. Set the task's `configFilename` property to the file. These work with any Saxon edition, including Saxon-HE.

```
"classPathEntries": ["${workspaceFolder}/lib/my-functions.jar"],
"configFilename": "${workspaceFolder}/saxon-config.xml"
```

The configuration file names each integrated function's class:

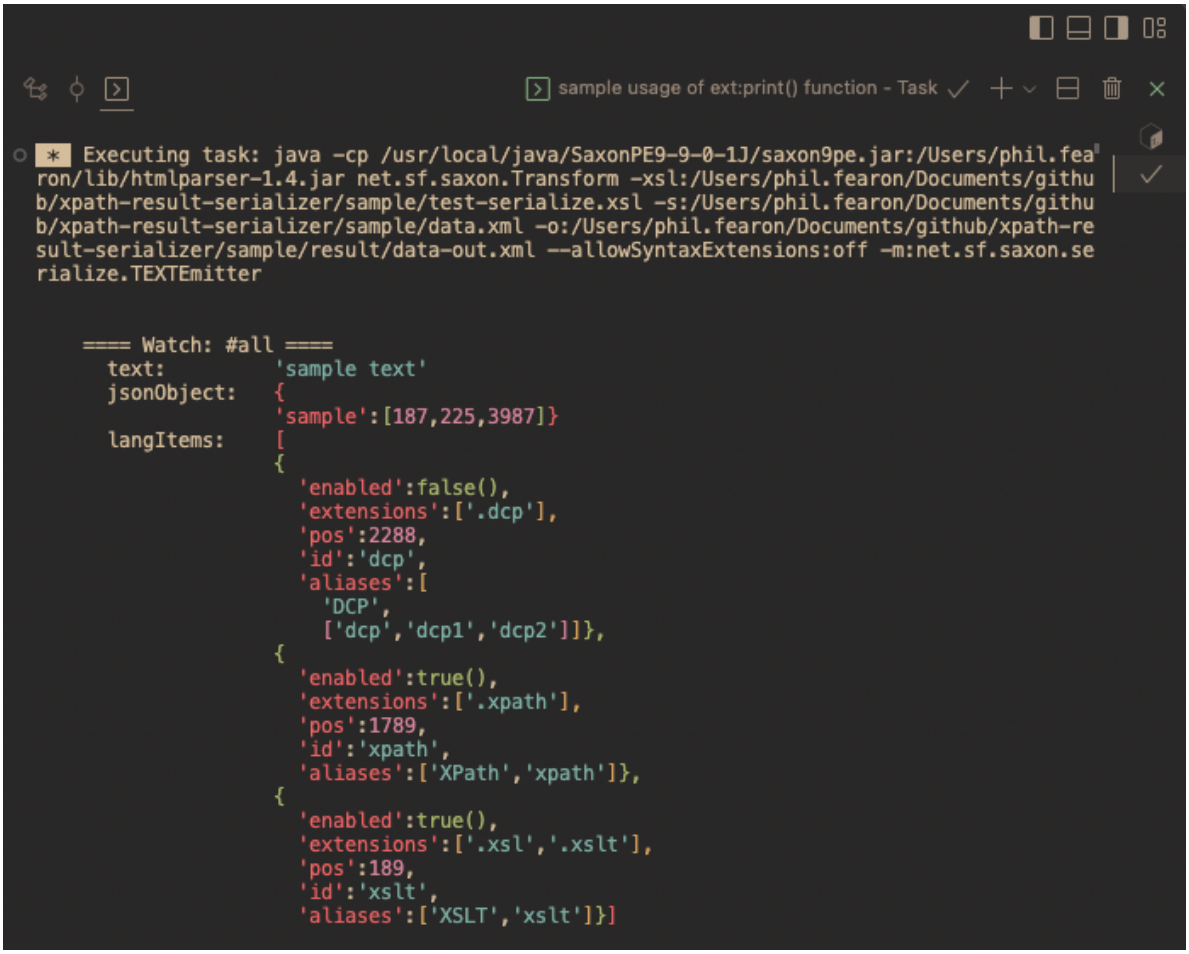
```
<configuration xmlns="http://saxon.sf.net/ns/configuration" edition="HE">
  <resources>
    <extensionFunction>com.example.Shout</extensionFunction>
  </resources>
</configuration>
```

The debugger runs the stylesheet with the task's properties, so Java extension functions work when debugging too - including in Watch expressions.

The linter doesn't check calls of a `java:` namespace's functions. For an integrated function, add its prefix to the `XSLT.validation.externalFunctionPrefixes` setting - the quick fix for a function that isn't found does this - and calls of functions with that prefix aren't checked.

The Visual Studio Code terminal

When you start a Task, it is run in a *Task* tab in Visual Studio Code's Integrated Terminal. The panel hosting the Terminal view is toggled with `⌘J`. The tab label reflects the Task configuration `label` property. When a Saxon XSLT task is run, `xsl:message` output and any Saxon errors will be shown in the terminal window.



> *The above screenshot shows formatted and colorized xsl:message output available through special auto-completion options. See Print Debugging for more detail.*

Debugger

XSLT stylesheets can be debugged with SaxonJ in VS Code's debugger: set breakpoints, step through templates, functions and multi-line XPath expressions, and inspect variables, the context item and the output written so far.

If you're new to debugging in VS Code, see Debugging in the VS Code documentation for the debug toolbar, views and keyboard shortcuts.

For debugging by adding `xsl:message` instructions to the stylesheet, see Print Debugging.

> ***Requirements:*** *SaxonJ 12 or later - any edition, including Saxon-HE - set with the* `XSLT.tasks.saxonJar` *setting, and a Java runtime on the PATH. Debugging with SaxonJS or SaxonC isn't supported. For XPath 4.0 syntax in an XSLT 4.0 stylesheet, Saxon-PE or Saxon-EE is needed.*

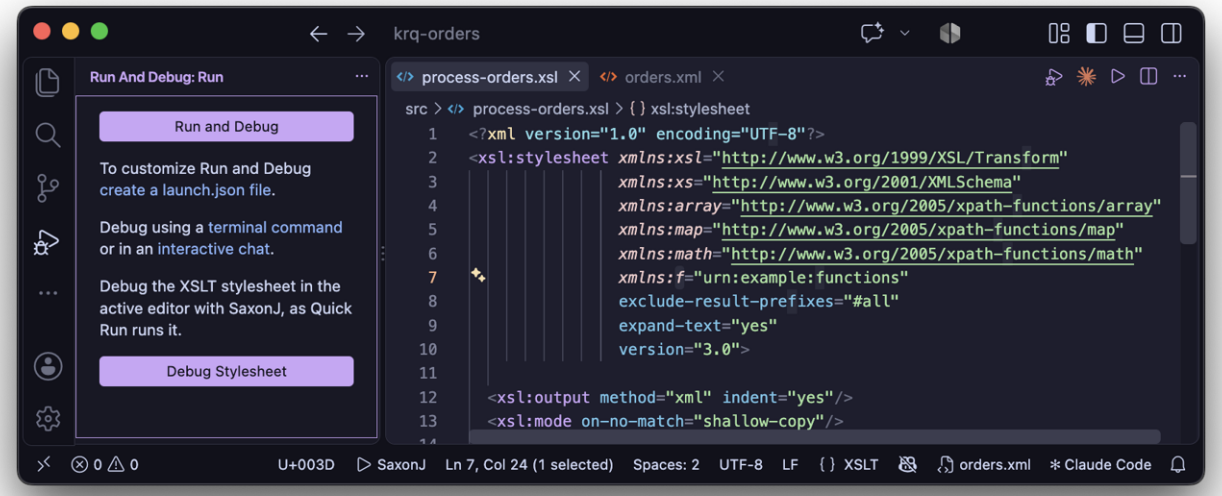
Starting a debug session

Open the stylesheet and set a breakpoint - click in the margin to the left of a line number, or press `F9`. Then start debugging in any of these ways:

- the **Debug Stylesheet with SaxonJ** button in the editor title bar
- **XSLT: Debug Stylesheet with SaxonJ** from the Command Palette
- `F5`, when there's no `launch.json` - or the **Debug Stylesheet** link in the **Run and Debug** view

The stylesheet runs as Quick Run runs it: with the XML context file, or from `xsl:initial-template` when there's no context file. It uses the same Quick Run task, so parameters added to that task also apply when debugging - if there's more than one task for the stylesheet and context file, you pick which. When the stylesheet is imported or included by another, you're offered the choice of running the parent stylesheet instead.

A Quick Run task includes the stylesheet's required parameters, with values from `tasks.json` inputs. You're asked for these values when the debug session starts, as when the task runs. If a task refers to an input that `tasks.json` doesn't define, the session doesn't start, and you're offered **Open tasks.json**.



The Run and Debug View provides options for how to launch the debugger.

Launch configurations

With no `launch.json`, the **Run and Debug** view offers **Debug Active Stylesheet**, and a **Debug *task label*** configuration for each `xslt` task in `tasks.json` (see Tasks). **create a launch.json file** adds these to a new `launch.json`.

To debug a task without a `launch.json`, choose **XSLT (Saxon)...** in the **Run and Debug** view's dropdown, or use **Debug: Select and Start Debugging** from the Command Palette, then the task's **Debug *task label***. The configuration stays selected in the dropdown, so F5 debugs that task again - whichever file is open.

A launch configuration uses the `xslt-saxon` debugger type, with one of:

- `"quickRun": true` - the stylesheet in the active editor, as Quick Run runs it
- `"task"` - the label of an `xslt` task in `tasks.json`, whose properties are used; properties set in the configuration override them
- the same properties as an `xslt` task, such as `xsltFile`, `xmlSource`, `initialTemplate`, `resultPath` and `parameters`

```
{
    "type": "xslt-saxon",
    "request": "launch",
    "name": "Debug report",
    "task": "report with books.xml",
    "stopOnEntry": false
}
```

`"stopOnEntry": true` pauses at the first instruction executed. In `launch.json`, **Add Configuration...** offers four snippets: **XSLT (Saxon): Debug Active Stylesheet**, **Debug a Task**, **Debug Stylesheet** (with a source file) and **Debug from Initial Template**. See Debug configuration in the VS Code documentation for more on `launch.json`.

> *An XSpec test can be debugged from the Test Explorer, with breakpoints in the stylesheet under test and in the XSpec file itself - see Debugging a test.*

Breakpoints

A breakpoint can be set on a line where an XSLT instruction or a literal result element starts. A breakpoint anywhere in a multi-line start tag is moved to the tag's first line.

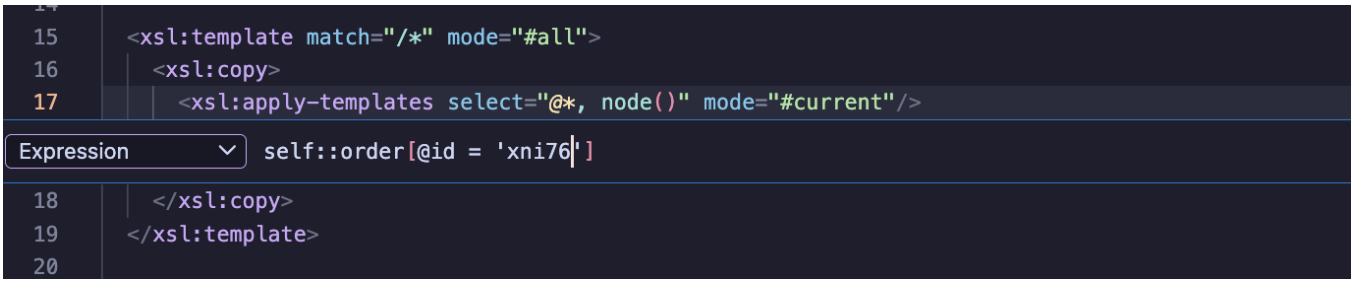
Breakpoints can also be set inside a multi-line XPath expression - see Stepping through XPath.

Conditional breakpoints and hit counts

Right-click in the margin and choose **Add Conditional Breakpoint...** - or right-click an existing breakpoint and choose **Edit Breakpoint**:

- **Expression** - an XPath expression, evaluated where the breakpoint is, with its variables and context item, e.g. `@customer = 'Grace'` or `$total gt 30`. If the expression raises an error, execution pauses and the error is shown.
- **Hit Count** - `5` or `==5` (the 5th hit), `>5`, `>=5`, `<5`, `<=5` or `%5` (every 5th hit). With an expression too, only the hits where the expression is true are counted.

An expression and a hit count can be given for a logpoint too, to choose when it writes its message.



Setting a conditional breakpoint with an XPath expression.

Logpoints

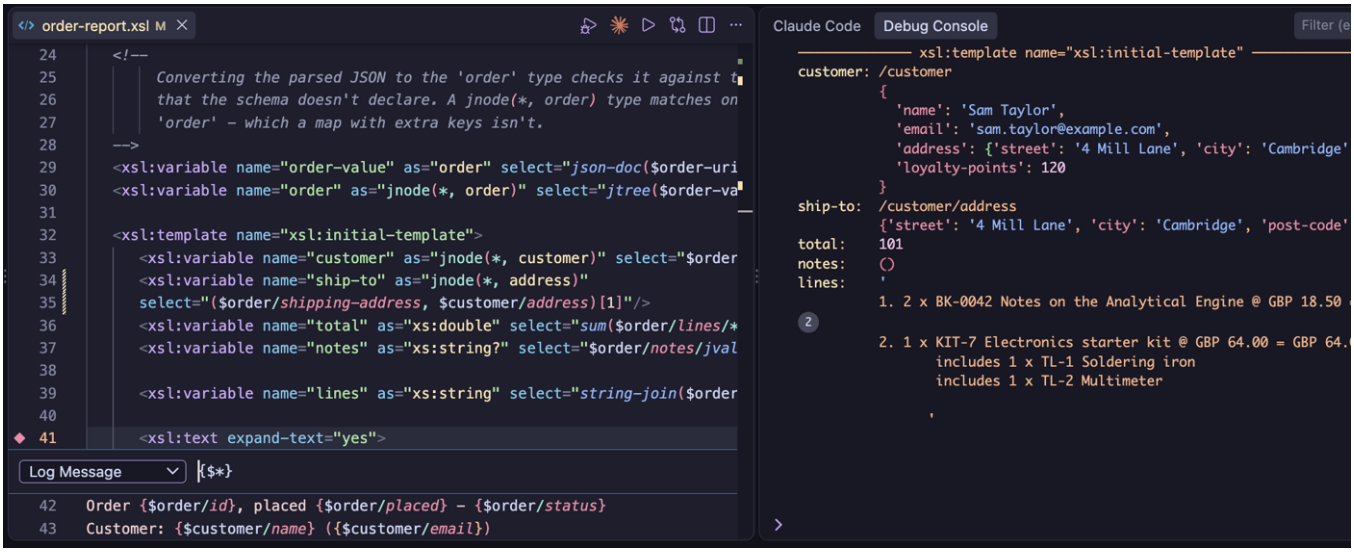
A logpoint writes a message to the Debug Console without pausing - print debugging without changing the stylesheet. Right-click in the margin and choose **Add Logpoint...**, or edit a breakpoint and choose **Log Message**.

Each `{xpath}` in the message is replaced by its value, evaluated where the logpoint is, e.g. `{$line/@product} costs {f:line-total($line)}`. Values are formatted and colored as `xdm:trace()` writes them, without the stylesheet including the xdm-view library:

- a node or JNode with its path above it, showing where it's from
- a string quoted, and a value's type where it isn't obvious, e.g. `2026-10-27 (xs:date)`
- a map or array over several lines, e.g. `order: {$order}`

An expression can't contain braces, as in a map constructor.

Logging all variables: `{$*}` is a special case - it's replaced by the local variables in scope at the logpoint, as `xdm:debug-color()` writes them: a banner with the name of the template, function or instruction, then each variable's name and value, e.g. `order {@id}: {$*}`. It's like an `xsl:message` with `xdm:debug-color()`, but without changing the stylesheet. With Saxon 12, whose maps don't keep their order, the variables may not be in the order they're declared.



A logpoint with the message `{$*}`, and what it writes to the Debug Console: a banner naming the template, then each local variable - JNodes with their paths, maps over one or several lines, and an empty sequence as `()`.

Function and named-template breakpoints

To stop whenever a stylesheet function or named template is called, without finding where it's declared, click **+** (**Add Function Breakpoint**) in the **Breakpoints** view and enter its name. The function or template can be in any module of the stylesheet, including imported and included modules. Execution stops at the start of the function or template, where the Variables view shows its parameters - for a named template, these are the values passed to it.

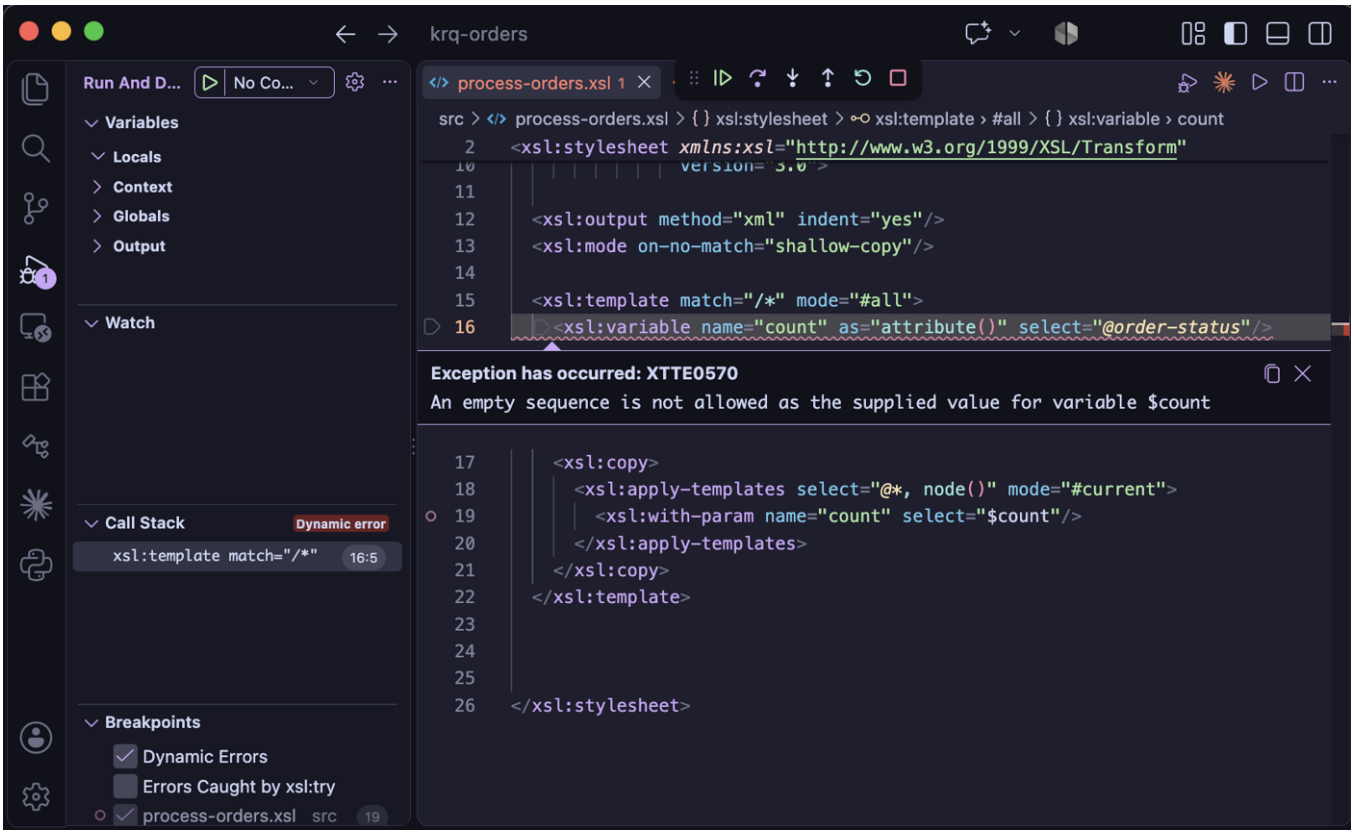
- **The name** is written as it is where the function or template is declared, with the same prefix, e.g. `f:total` or `cell` - or as an EQName, e.g. `Q{urn:f}total`. A `()` after the name is allowed.
- **An arity** selects one version of a function, e.g. `f:total#2`. With no arity, every version of the function stops.
- **Conditions and hit counts** work as for line breakpoints - right-click the breakpoint and choose **Edit Condition** or **Edit Hit Count**. The condition can use the parameters, e.g. `$n = 2`.

A name that isn't valid is shown as unverified straight away. A function or template that isn't in the stylesheet - or a function with no version of the arity given - is shown as unverified once the stylesheet has compiled; hover over the breakpoint for the reason. Template rules - templates with a `match` attribute - aren't stopped at this way, even when they also have a `name`: use a line breakpoint instead.

Pausing on errors

The **Breakpoints** view has two options for errors. Like the No Template Matched option, they're listed once a debug session has been started in the workspace:

- **Dynamic Errors** (on by default) - pauses where a dynamic error that ends the transform is raised, including `xsl:message` with `terminate="yes"`. The error's code and message are shown, and the variables at that point can be inspected.
- **Errors Caught by xsl:try** (off by default) - also pauses at errors that an `xsl:try` catches. This needs Saxon 12.5 or later.



The debugger paused on a dynamic error.

A *static* error - one Saxon finds when it compiles the stylesheet - can't be paused at, as the stylesheet never runs. This includes a type error that Saxon can tell will always happen: for example, an `xsl:variable` with `as="xs:string"` whose `select` is `count(...)`, which is always an `xs:integer`. The session then ends with a notification that gives the first error and its line, and the errors are listed in the **Problems** view.

Elements no template matched

When no template rule matches an element processed by `xsl:apply-templates`, XSLT's built-in template rules apply: the element's text is written to the output, and its children are processed. This is often intended - e.g. for a wrapper element - but it's also what happens when a template's `match` pattern is wrong, e.g. for an element in another namespace, so the expected template is never used.

With the **No Template Matched** option in the **Breakpoints** view (off by default), execution pauses at the first element of each name, in each mode, that no template rule matched. It pauses at the `xsl:apply-templates` instruction, in a call stack frame such as `no template matched: item (mode #default)`, whose context item is the element - so the **Context** view, Watch and the Debug Console show and evaluate that element. If no template has matched yet, from the start of the transform, the frame has no line.

Whether or not the option is on, when the transform ends the Debug Console lists the elements no template matched, e.g. `▸ Built-in template rules applied to 37 elements no template matched: item ×30 (mode #default), o:note ×7 (mode summary)`. Elements dropped by a mode with `on-no-match="deep-skip"` are listed on a line of their own. Modes whose `on-no-match` copies or skips elements otherwise - e.g. `shallow-copy` - aren't included, as there that's the intent; with `on-no-match="fail"`, it's a dynamic error.

Stepping

The debug toolbar has the usual **Continue** (`F5`), **Step Over** (`F10`), **Step Into** (`F11`), **Step Out** (`⇧F11`), **Pause** and **Stop** commands. In XSLT terms:

- **Step Over** goes to the next line in the current template or function, without stopping in the templates and functions it calls - for example, with `xsl:apply-templates` or `xsl:call-template`.
- **Step Into** stops in each template rule, named template or function called, and in a global variable when it's evaluated.
- **Step Out** continues until the current template or function returns.
- `xsl:for-each`, `xsl:for-each-group`, `xsl:iterate` and `xsl:analyze-string` have their own entry in the call stack, and are stepped like calls: **Step Over** runs all their iterations, **Step Into** stops in their body for each item, and **Step Out** from their body goes to the line after them. A breakpoint in the body still stops; after stepping over one by mistake, **Step Back** goes back to it.

Stepping goes line by line: execution stops once each time it reaches a line, not once for each instruction on it. A breakpoint in a loop stops once for each iteration.

The debug toolbar also has **Step Back** and **Reverse Continue** - see Going back - and **Show Debug Output** - see Output while debugging.

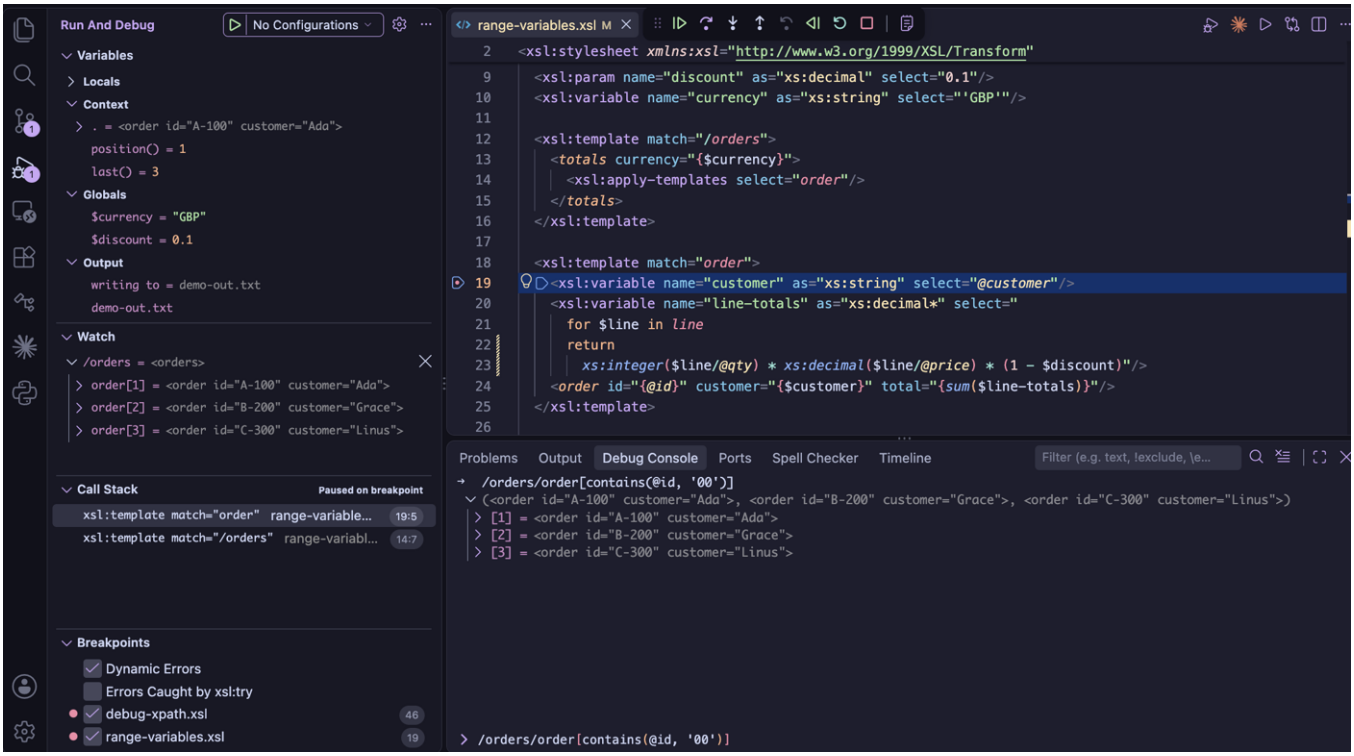
Stepping through XPath

A multi-line XPath expression in a `select`, `test` or `value` attribute - or in `xsl:map-entry`'s `key`, `xsl:evaluate`'s `xpath`, `context-item` or `with-params`, or the content of an XSLT 4.0 `xsl:select` - can be stepped through. An `xsl:when`'s `test` is stopped at where it's evaluated: only the tests reached, in order, until one is true. In XSLT 4.0, the `select` of an `xsl:when` or `xsl:otherwise`, and an `xsl:if`'s `then` and `else`, are stepped through only when that branch is taken; an `xsl:record`'s attributes - its fields' expressions - are stepped through as the record is built. Within `xsl:catch`, the `$err:` variables can be used, as in Watch. An `xsl:with-param`'s `select` is stepped through as part of the instruction it's in, e.g. `xsl:call-template`. An `xsl:sort`'s `select` is stopped at for each item, as its sort key is computed - before the first item is processed - with that item as the context item. Use **Step Into** at the instruction, or set a breakpoint on a line of the expression. Execution stops on each line where a part of the expression starts. A breakpoint on a line with only keywords such as `then`, `else` or `return`, and no expression starting on it, is unverified.

Step Into goes through an expression when that shows more than its result: when it binds variables (`for` , `let` , `some` or `every`), calls a stylesheet function or an inline function, or has an `if` whose branches are on lines of their own - execution then stops at the branch taken. Otherwise - for example, for comparisons joined by `or` - **Step Into** steps over it, and the variable's value is shown when it's in scope. A breakpoint on a line of the expression still stops there.

A simple map (`!`), a path step (`/`) or a predicate (`[...]`) is evaluated for each item of the expression before it. A breakpoint on a line of its right-hand side, or predicate, stops for each item, with that item as the context item and its `position()` and `last()` in the **Context** section. **Step Into** goes through such a loop only when what's evaluated for each item has something to show, as above - e.g. a function call - so that it doesn't stop for each item of a long path. A function argument that Saxon evaluates lazily - within the function, when it's first used - shows its context item, but `position()` and `last()` as `(not known)` .

The expression is evaluated separately, in the instruction's context, so that each part reports its own line. It appears in the call stack as, for example, `xsl:variable/@select (XPath)` , and its result is shown in the Debug Console, e.g. `▸ $line-totals (xsl:variable/@select, lines 25–28) returned (6.75, 10.8, 15.75)` - or the dynamic error it raised, e.g. one an `xsl:try` catches. The instruction itself then runs as usual, but without stopping again in the functions the expression calls, or writing their `xsl:message` output again - so each is stopped in, and its messages written, once. The body of an inline function is stepped through each time it's called - for example, by `fold-left()` .



Evaluation context, the call stack and the debug toolbar are shown when execution is paused.

Going back

The debug toolbar also has **Step Back** and **Reverse Continue**, for going back to an earlier point in the transform without starting a new debug session:

- **Step Back** goes to the line reached before, in the current template or function - or, at its first line, to the line that called it.

- **Reverse Continue** goes back to the last line with a breakpoint that was reached (breakpoint conditions aren't checked), or to the start of the transform if there's none.

To go back, the debugger runs the transform again, from the start, to that point - quietly, and stepping as before - then pauses there. Variables, the context item, Watch expressions and the call stack are then as they were at that point, and stepping or continuing goes on from there. `xsl:message` output and logpoint messages already shown in the Debug Console aren't shown again.

Call stack

The **Call Stack** view has a frame for each:

- template rule, with its match pattern and mode, e.g. `xsl:template match="order" mode="summary"`
- named template, e.g. `xsl:template name="header"`
- function, by name and arity, e.g. `f:line-total#1`
- global variable or parameter being evaluated, e.g. `xsl:variable $orders`
- `xsl:for-each`, `xsl:for-each-group`, `xsl:iterate` or `xsl:analyze-string` instruction, with its `select` expression
- XPath expression being stepped through, e.g. `xsl:variable/@select (XPath)`

Selecting a frame shows its location in the editor, and its variables in the Variables view.

Variables

The **Variables** view has four sections:

- **Locals** - the parameters of the template, function or `xsl:iterate`, then the variables in scope - from `xsl:variable`, and from XPath `for`, `let`, `some` and `every` expressions, including a positional `at $pos` variable. Tunnel parameters are shown as `$name (tunnel)`. A value that Saxon hasn't evaluated yet, and whose evaluation would run stylesheet code - for example, with Saxon 12, the accumulator passed to a `fold-left` function - is shown as `(not evaluated yet)`, with a button to evaluate it. Evaluating it runs that code straight away, without stopping at its breakpoints. Inside an inline function, only the function's parameters, the variables from outside it that it uses, and its own range variables are shown.
- **Context** - the context item (`.`), `position()` and `last()`, and where they apply, `current-group()`, `current-grouping-key()` and each `regex-group()`.
- **Globals** - the global variables and parameters, sorted by name. A global variable that hasn't been used yet is evaluated when it's shown. While a global variable is being evaluated, one that might use it - directly, or through a function - isn't evaluated, and is shown as `(not evaluated yet)`: evaluating it would be a circular reference. When an XSpec test is debugged, the test runner's own globals, such as `$x:xspec-uri`, aren't shown, but they can still be evaluated in the Debug Console.
- **Output** - which result is being written to, or that a temporary tree is being built (for example, an `xsl:variable`'s content), and the end of each output written so far.

Nodes, maps, arrays and sequences can be expanded. An element shows its attributes, then its child nodes, with the line number of the element in its source document. Long values are truncated, and an expanded value shows at most 1000 items.

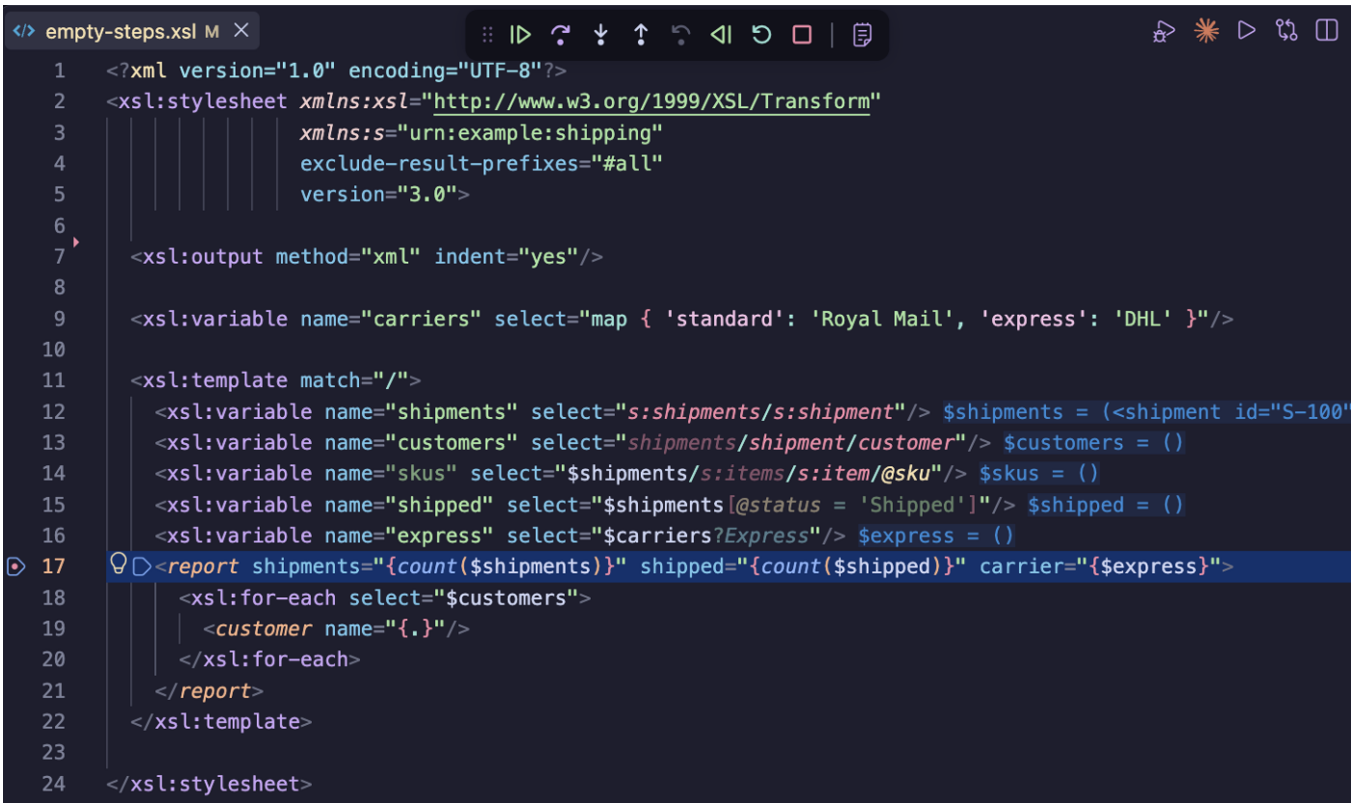
Inline values

While paused, the value of each variable declared earlier in the current template or function is shown at the end of the line where it's declared. This applies to `xsl:variable` and `xsl:param`, and in XPath, to `let`, `for`, `some` and `every` variables and inline function parameters.

The VS Code user setting `debug.inlineValues` turns inline values on or off. A change applies from the next debug session, not the one that's running.

Empty steps in paths

While paused, in the paths and lookups before the line paused at, the first step whose result is empty is faded - like an unused variable - so it's easy to see where a path stops matching. For `$doc/orders/order/item`, when `$doc` has no `orders` element, `orders` is faded. A step (after `/` or `//`), a predicate (`[...]`) and a map or array lookup (`?key`) can each be faded. Hovering over it says why it's empty:



Paused at line 17, the first empty step of each path is faded: `shipments`, which has no prefix for the shipping namespace; `s:items`; a predicate that no shipment matches; and the lookup `?Express`, as the map's key is `express`.

- for a step, how many nodes there were to select from, and any with the same name in another namespace, or the same name with different case - otherwise their names
- for a predicate, how many items it removed
- for a lookup, the map's keys, or the array's size

The paths checked are those in the template or function paused in (or the global variable, if paused while it's evaluated), on lines up to the one paused at. A relative path is checked when it's evaluated with the same context item as the instruction paused at - not within an `xsl:for-each` that's ended, for example, or a predicate. A path is evaluated as a Watch expression is, each step once, from the variable, root (`/`),

parenthesized expression or function call it starts with. A path that starts with a call of a stylesheet function calls it once more, which may write `xsl:message` output again; a path isn't checked beyond a stylesheet function called after its start. Checking stops after about 300ms at each pause, so a slow path doesn't slow down stepping.

Watch, hover and the Debug Console

While paused, type an XPath expression at the prompt at the bottom of the **Debug Console** and press `Enter` to evaluate it - for example, `line[@qty gt 1]/@product` on an `order` element, or `count(current-group())` in `xsl:for-each-group`. Expressions added to the **Watch** view are evaluated in the same way, and again each time execution pauses.

Expressions are evaluated in the context of the call stack frame selected in the **Call Stack** view, with:

- the variables in scope, and the context item
- the stylesheet's namespace declarations and functions, e.g. `f:line-total($line)`
- `current()`, `current-group()`, `current-grouping-key()` and `regex-group()`, which give their values where execution is paused

Results are shown, and can be expanded, as in the Variables view. Hovering over a variable reference such as `$total`, or the name in a variable declaration, shows its value.

Formatted values with xdm:trace()

If the stylesheet includes the xdm-view library - see Print Debugging - its `xdm:trace()` function can also be used while debugging, to see a value formatted and colored over several lines, as in a terminal. In the Debug Console, for example, `xdm:trace($order, 'order')` writes the formatted value, and gives the value as its result. A logpoint doesn't need it: its message formats values in the same way already.

Use `xdm:trace()` rather than `xdm:debug()` or `xdm:debug-color()` here: those give the formatted text as a string, which the Debug Console shows as a short, quoted value, without color.

Output while debugging

The output written so far - the principal result, and each `xsl:result-document` - can be shown beside the stylesheet while paused, and is updated at each pause. It's serialized by Saxon as the result is, but indented for readability, and shown in the language of its output method: XML, HTML, JSON or plain text. The result itself is written as the stylesheet specifies.

- **XSLT: Show Debug Output**, or its button on the debug toolbar, opens the output at any time - the output being written to, or, when none is and there's more than one, the one you choose.
- To open the output automatically, at the first pause where there is some, turn on the `XSLT.debug.openOutput` setting - it's off by default.
- **Show Output**, on the context menu of an output in the Variables view's **Output** section, opens that output.
- `xsl:message` output, and the result when there's no `resultPath`, are written to the Debug Console.

XML output can be traced back to the stylesheet. With the cursor on a line of the output, a hint at the end of the line - e.g. `→ report.xsl:4` - shows where that part of the output was written. To go there, use **Go to XSLT Source** on the output's context menu, `Cmd/Ctrl` -click, or `F12` - or `Option/Alt` -click, when `editor.multiCursorModifier` is `ctrlCmd`. The stylesheet is shown in another editor group from the output's. For HTML, JSON or text output, **Go to XSLT Source** reports that only XML output can be traced back to the stylesheet.

When the session ends, a notification offers to **Open** the result file - or, for HTML, to **Open in Browser**.

Saxon errors

Saxon's error messages in the Debug Console are linked to the location of the error. The errors are also listed in the **Problems** view, as for XSLT tasks. A file's errors are removed from the Problems view when the file is edited, as they may no longer apply.

When the transform fails, the notification at the end of the session gives Saxon's reason - for example, that the XML source file doesn't exist - or, for a dynamic error, its message, code and line. For a stylesheet with static errors, which can't be debugged, it gives the first of them.

Sample: stepping through XPath

The debugger-demo sample has a stylesheet, `debug-xpath.xsl`, and an XML file, `orders.xml`. Open `orders.xml`, so it becomes the XML context file, then `debug-xpath.xsl`, and start debugging. Things to try are listed in a comment at the top of the stylesheet, including:

- stepping through a multi-line `for ... return` expression, and into a function it calls
- a breakpoint inside an expression in `xsl:for-each-group`, with `count(current-group())` in the Watch view
- conditional breakpoints, hit counts and a logpoint
- pausing at a dynamic error, after changing a price in `orders.xml` to `n/a`

Limitations

- For debugging, the stylesheet is compiled without optimization, so all variables can be inspected. Saxon still evaluates constant expressions when it compiles - e.g. an `xsl:if` whose test uses only constant values - and an instruction removed that way can't be stopped at.
- Values can't be changed in the Variables view.
- Going back relies on the transform running the same way again; `current-dateTime()` is kept the same. A transform that reads a document that changes in the meantime, or calls an extension function with side effects, may not - then it runs to the end, with a notification. Breakpoint hit counts aren't wound back, and the output view keeps any output written after the point gone back to until it's written again.
- An expression that's stepped through is evaluated in full - except a test, e.g. an `xsl:if`'s, which is evaluated only as far as needed for true or false, as Saxon does. Where Saxon itself would evaluate only part of it - e.g. an `xsl:iterate`'s `select`, when `xsl:break` ends the loop, or a variable that's only partly used - stepping through it can call more functions, and so write more `xsl:message` output, than a normal run.

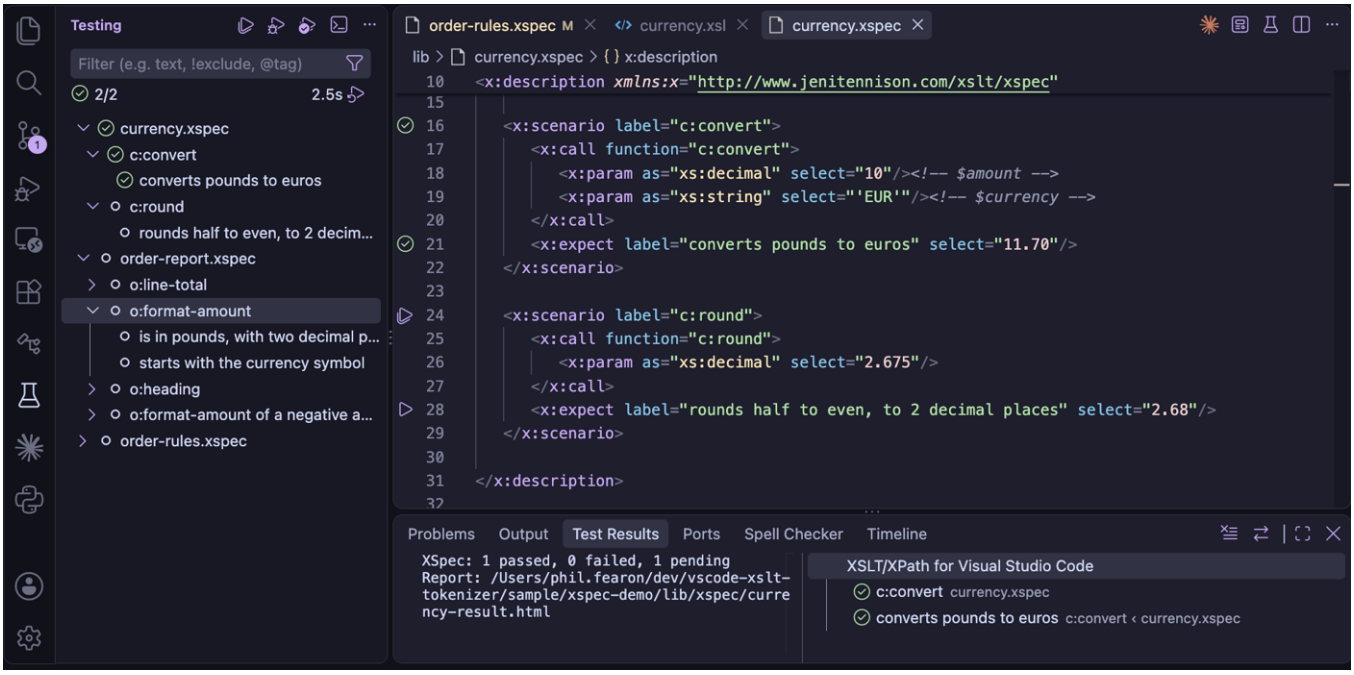
- With `"messageEscaping": "on"` in the `xslt` task, Saxon writes `xsl:message` output itself, so a message from a function called by an expression that's stepped through is written twice: once as the expression is stepped through, and again as the instruction runs.
- Only modules loaded from files can be stopped in.
- Saxon only evaluates `xsl:assert` with assertions enabled - `"enableAssertions": "on"` in the `xslt` task - so without that, its breakpoints are unverified.
- Saxon evaluates some instructions lazily - e.g. an `xsl:for-each` in an `xsl:map` runs when the map is built - so the call stack may not show them while their content is stopped in. The context item is still the right one.
- Accumulators: Saxon computes an accumulator's values for the whole document the first time one of them is needed - e.g. by `accumulator-before()` - so a breakpoint in an `xsl:accumulator-rule` stops for each node in turn at that point, not as each node is processed. If a breakpoint condition or a Watch expression is the first to need the values, they're computed while it's evaluated, and the rule's breakpoints aren't stopped at. Accumulator rules are reported to the debugger by Saxon 12.8 or later.
- Keys: Saxon builds a key's index for the whole document the first time `key()` is called for it - so a breakpoint on an `xsl:key` stops for each node indexed at that point. As for accumulators, if a breakpoint condition or Watch expression is the first to need the index, it's built while that's evaluated, and the key's breakpoints aren't stopped at.

Troubleshooting

When a debug session doesn't start, a notification gives the reason, with a **Show Output** button - for example, when Java isn't on the PATH, the Saxon jar doesn't exist, or the Java runtime is too old for the Saxon jar.

The **XSLT Debugger** log, in the Output panel, shows how each session was launched: the Java command line, the Saxon product, the Java runtime and the Saxon options used.

XSpec Tests



Tests can be run or debugged from Test Explorer, the XSpec file, or the XSLT component itself.

XSpec is a testing framework for XSLT and Schematron. An XSpec file (`.xspec`) describes scenarios: each calls a function or a named template, or applies template rules to a context, and its expectations say what the result should be:

```
<x:description xmlns:x="http://www.jenitennison.com/xslt/xspec"
               xmlns:f="urn:example:functions" stylesheet="prices.xsl">
    <x:scenario label="a price with VAT">
        <x:call function="f:with-vat">
            <x:param select="100"/>
        </x:call>
        <x:expect label="adds 20%" select="120"/>
    </x:scenario>
</x:description>
```

The XSLT/XPath extension checks XSpec files as you edit them - against XSpec's schema, and against the stylesheet they test - and runs and debugs their tests from VS Code's **Test Explorer**. XSpec 4.1.2 is included with the extension, so there's nothing else to install: tests are run with the Saxon jar of the `XSLT.tasks.saxonJar` setting, as Quick Run uses.

Creating a test

In a stylesheet, with the cursor in the start tag of an `xsl:function` , a named `xsl:template` or a template rule, the code action (⌘.) **Create XSpec scenario for …** adds a scenario testing it, to complete:

- **A function**: an `x:call` , with an `x:param` for each of its parameters, in order, with its type - and an `x:expect` with a `select` for the expected result
- **A named template**: an `x:call` , with an `x:param` for each parameter by name - not tunnel parameters - and an `x:expect` for the expected content

- **A template rule**: an `x:context` in its mode, with an element it matches, e.g. `<item/>` for `match="list/item[@id]"`

The scenario is inserted as a snippet: `Tab` moves through its label, the parameters' values and the expected result. It's added to the end of the XSpec file that tests the stylesheet, or a new one beside it, with the stylesheet's name - e.g. `prices.xspec` for `prices.xsl`. The namespaces its names use are declared on `x:description`.

In a new, empty XSpec file, typing `<` offers the root element, with the XSpec and XML Schema (`xs`) namespaces, for the stylesheets beside it - or in the folder above, for a file in an `xspec` folder. The one with the XSpec file's name, the usual convention, e.g. `order.xsl` for `order.xspec`, is first and selected:

- **x:description with pending scenarios for …** - one for each stylesheet: with a pending scenario for each of its functions and templates, as **XSpec: Create Test File** makes - the quickest start
- **x:description for a stylesheet**, with a choice of the stylesheets, or **for a Schematron** - empty. The code action in its start tag (`⌘.`), **Add pending scenarios for …**, adds the scenarios later.

For a package, it has `run-as="external"`. In its start tag, completion after `xmlns` offers the namespaces of the stylesheet under test, then common ones.

A test file for a whole stylesheet or package

XSpec: Create Test File - in the Command Palette, or the editor's context menu, for a stylesheet - or the code action in the start tag of its `xsl:stylesheet` or `xsl:package`, makes a skeleton of tests: a scenario, as above, for each function, named template and template rule. Each is pending - `pending="to do"` - so the file runs, with the scenarios skipped, until you give each its values and expected result and remove its `pending` attribute. Run the command again after adding declarations to the stylesheet: a scenario is added for each one that has no scenario yet.

In the XSpec file, for the functions, templates and template rules added to the stylesheet since:

- auto-completion after `<`, in `x:description` or an `x:scenario`, offers **x:scenario for …** for each one with no scenario yet - inserted as a snippet, with the namespaces it uses added to `x:description`
- the code action in the start tag of `x:description`, **Add pending scenarios for …**, adds a pending scenario for each of them, as the command does

For an `xsl:package`, the scenarios are for what a stylesheet using it can call or apply: its public functions and named templates, and its template rules in public modes. By default XSpec imports the stylesheet it tests into its test runner, which a package can't be, so a package is tested in one of two ways - the code action offers both, and the command asks which:

The XSpec file has	How XSpec tests the package
`run-as="external"`, with `stylesheet` the package's relative URI, e.g. `stylesheet="shapes.xsl"`	as a separate transformation: only its public components can be called, with `x:call` - not in the XSpec file's XPath expressions
`x:helper package-name="…"`, with `stylesheet` a stylesheet that does nothing - created beside the package, e.g. `shapes-xspec-harness.xsl` - as XSpec requires one	used by the test runner, with `xsl:use-package`, as by a stylesheet using it: its public functions

The XSpec file has	How XSpec tests the package
	can be called in XPath expressions too, e.g. in an `x:expect`'s `test`

Either way, calling a component that isn't public is reported. When tests are run, the packages they use - the package's own, and those of an `x:helper package-name="…"` - are passed to Saxon with `-lib`, found as for XSLT packages.

Within an `x:call`, auto-completion after `<` offers an `x:param` for each parameter not passed yet: by name for a template, and for a function, the next in order, with its type.

Editing XSpec files

A `.xspec` file is checked as you type, as a stylesheet is:

- **XSpec elements and attributes** are checked against XSpec's schema, including values such as `catch="yes"` - with the prefix the file uses for XSpec, from its `x:description`, e.g. `t:scenario` in a file using `t:`
- **XPath expressions** in `select`, `test` and `use-when` are checked, with the variables in scope: `x:variable`, a global or scenario `x:param`, and those XSpec provides - `$x:result`, `$x:context` and `$x:xspec-uri`
- **Files** - the stylesheet, Schematron, XQuery module or XProc step tested, each `x:helper` and `x:import`, and the `href` of `x:context`, `x:expect`, `x:param`, `x:variable`, `x:input` and `x:option` - are links, and a file that isn't found is reported. They're resolved as XSpec resolves them: relative to the XSpec file, or to an `xml:base`, or with the XML catalog of the `XSLT.resources.catalog` setting
- **Test data** - the content of `x:context`, `x:expect`, `x:param` and `x:variable` - may be any XML, including XSLT, and `x:text`. With `expand-text="yes"`, the value templates in it are checked

Auto-completion offers the elements, attributes and values XSpec allows where you are, and hover help on an XSpec element's name shows its description from the schema. The Outline view lists the scenarios and expectations by their labels.

The stylesheet under test

The stylesheet of `x:description` - and of each `x:helper` - is read, as an imported stylesheet is, so the functions, global variables, templates and modes it declares are known in the XSpec file. So are the global variables of the XSpec files it imports with `x:import`. Hover help shows a function's signature, and these are checked, with auto-completion for each:

In the XSpec file	Checked against the stylesheet
`x:call function="…"`	a function with that name - or a built-in function
`x:call template="…"`	a named template
`x:param name="…"` in an `x:call` of a template	a parameter of the template - unless it's a tunnel parameter
`x:context mode="…"`	a mode (a warning)

In the XSpec file	Checked against the stylesheet
`x:param` elements of `x:description`	the stylesheet's global parameters: one with `required="yes"` that isn't set is a warning, as each test would fail with XTDE0050
function calls and variable references in XPath	the stylesheet's functions and global variables

> *When the declarations can't be known, these aren't reported: for the XSpec file of a Schematron or an XQuery module, or when the stylesheet isn't found - or `x:description` has an `xml:base`. A function called with `call-as="variable"` is a variable's value, so it's not checked as a function.*

> *With `run-as="external"`, the stylesheet under test isn't imported into the test runner, so its functions and global variables can't be used in the XSpec file's XPath expressions - only called with `x:call`. A call of one in an expression is reported; those of an `x:helper` can be used.*

Navigating between tests and the stylesheet

From an XSpec file to the stylesheet it tests, **Go to Definition** (F12) and **Peek Definition** (⌥F12) open the declaration of:

- the function or template of `x:call function="…"` or `x:call template="…"`
- the global `xsl:param` an `x:param` of `x:description` sets
- a template parameter, from the `name` of an `x:param` in its `x:call`
- the mode of `x:context mode="…"`
- a function called, or a global variable used, in an XPath expression, e.g. in an `x:expect`'s `test`

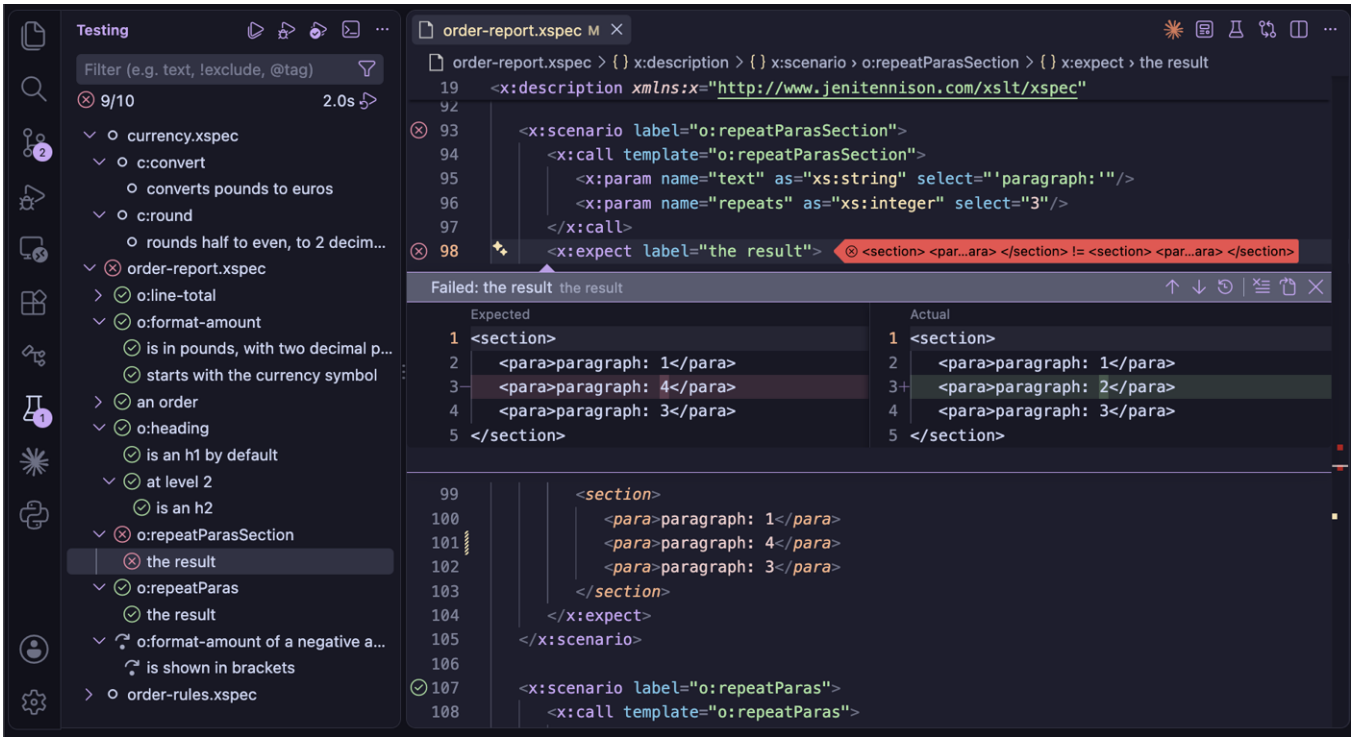
From the stylesheet back to its tests, **Find All References** (⌥⇧F12) on a function or named template also lists its uses in the XSpec files that test it - in `x:call`, and its calls in XPath expressions - so you can see which scenarios cover it. An XSpec file tests the stylesheet of its `x:description` or `x:helper`, and the modules that stylesheet imports or includes. A name is matched by its namespace, so an XSpec file may use another prefix for it, e.g. `g:double` for `f:double`.

Rename Symbol (F2) on a function or named template renames its uses in the XSpec files too, keeping the prefix each file uses.

Running tests

The **Test Explorer** (the beaker icon in the Activity Bar) lists the XSpec files in the workspace, each with its scenarios and expectations. The ▶ button beside an `x:scenario` or `x:expect` in the editor, or on an item in the Test Explorer, runs it:

- **A file** runs all its scenarios - also with the **XSpec: Run Tests** button in the editor's title bar
- **A scenario** runs it alone, with XSpec's `force-focus` - the others are pending - and **an expectation** runs its scenario



A test failure in Test Explorer and the test itself - click any failure label to see the diff.

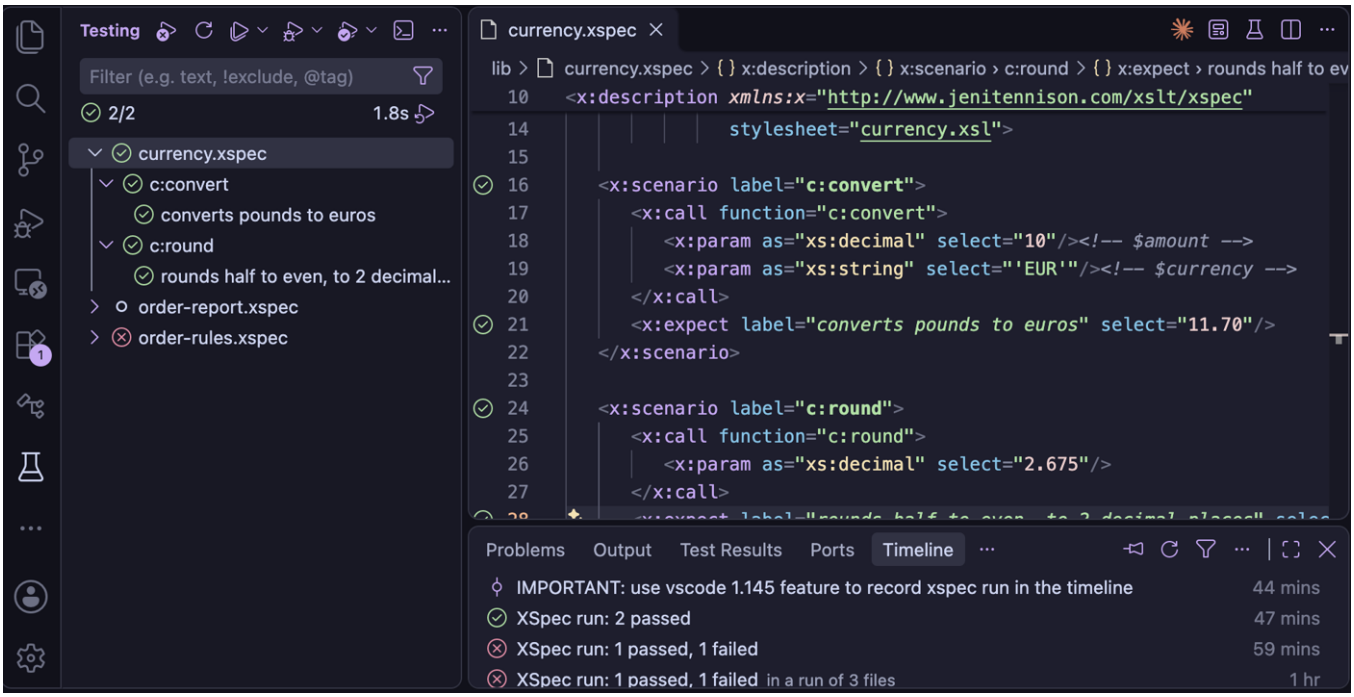
Each expectation is then marked passed, failed or pending (skipped), and each scenario by the expectations in it - the expectations are the tests, as XSpec counts them. For a failed expectation, the Test Explorer's peek view shows the expected and actual results side by side, as a diff - as XSpec compares them: without namespace declarations, with attributes in name order, without the whitespace between elements and with a map's entries in the expected map's order. What an expected `...` matches - an attribute's value, text, or an element's content - is shown as `...` in the actual result too, so only the differences that made it fail are shown. **XSpec: Open Report** opens XSpec's HTML report of the last run.

A test is run in XSpec's steps - each a Saxon transform - and nothing is hidden. The **XSpec** output channel shows each Saxon command, and Saxon's messages, including the errors in a test that doesn't compile. The files are where XSpec's own command line puts them, in an `xspec` folder beside the XSpec file:

File	What it is
`name-compiled.xsl`	the test runner: the XSpec file compiled to a stylesheet, which imports the stylesheet under test
`name-result.xml`	XSpec's result for each scenario and expectation
`name-result.html`	the HTML report
`name-diffs.json`	the expected and actual results of each failed expectation, for the diffs

They're kept after a run, e.g. to see why a test runner fails, or to take an actual result for an expected one. **XSpec: Delete Generated Files** deletes them in the workspace - all of them, or all but the reports - and an `xspec` folder left empty. Only files named as a run names them, in the `xspec` folder beside the XSpec file they're for, are deleted - not the XSpec files, even in a folder named `xspec`.

XSpec Timeline



The XSpec file's Timeline view, below the file editor, shows XSpec run events along with git events.

Once a test run for an XSpec file completes, the file's timeline is updated with the test run details. The Timeline view for the selected XSpec file shows these timeline events, along with other events for this file such as git commits. Hover over an XSpec event in the timeline to see a list of scenarios that failed.

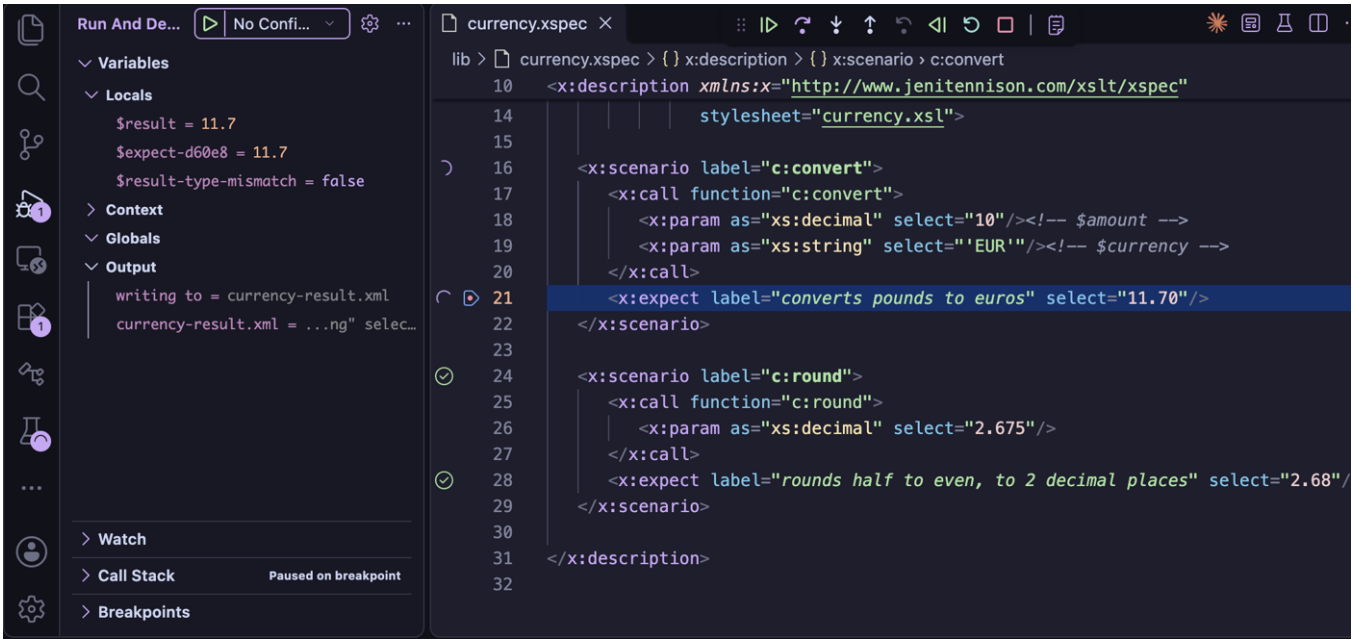
Testing a Schematron

For an XSpec file whose `x:description` has a `schematron` attribute, two steps come first, as in XSpec's command line: the Schematron is compiled to XSLT with SchXslt2 (included with XSpec), to `name-sch-preprocessed.xsl`, and the XSpec file is converted to test it. Schematron expectations such as `x:expect-assert` and `x:expect-not-report` appear in the Test Explorer with the labels XSpec gives them, e.g. *not assert a001*.

> *Only a Schematron with an XSLT query binding, e.g.* `queryBinding="xslt3"`*, is supported - not one for XQuery.*

Debugging a test

The **Debug** button beside a scenario - or the Test Explorer's **Debug Test** - runs the test runner with the XSLT debugger, and the outcomes are shown in the Test Explorer when it ends. As the runner imports the stylesheet under test, a breakpoint in the stylesheet pauses there, with the scenario's input in the Variables view - so a bug can be reproduced as a scenario, then debugged from it.



The code pauses at breakpoints within an XSpec test, allowing the XSpec context to be inspected.

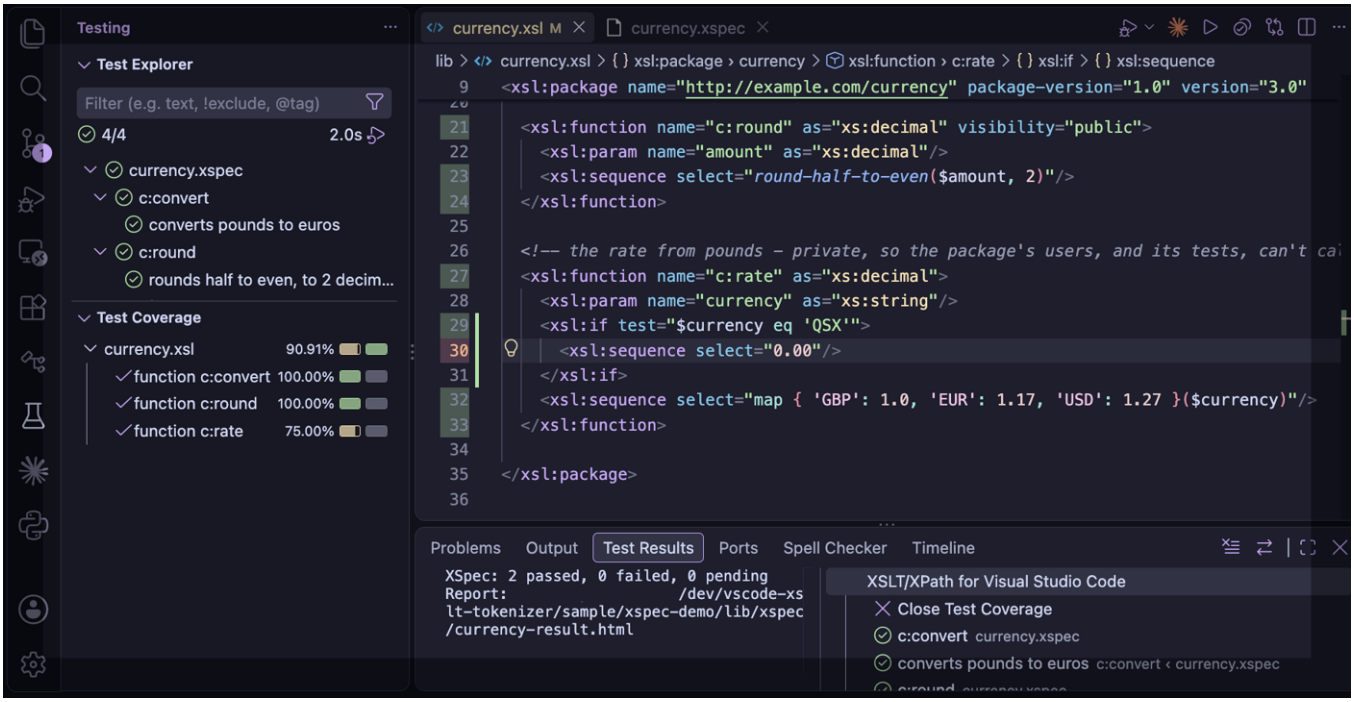
Breakpoints can also be set in the XSpec file itself, and the debugger pauses at the XSpec file's line - not in the generated test runner:

A breakpoint on	Pauses
`x:scenario`	as the scenario starts
`x:call` or `x:context`	where the stylesheet under test is called - **Step Into** goes into it
`x:expect`	where the expectation is tested, with `$x:result` in the Variables view

A breakpoint added elsewhere is moved to the line it belongs to, and the status bar says where: on an `x:param` or in test data, to its `x:call` or `x:context`; on a scenario's end tag, to the `x:scenario`. A breakpoint outside the scenarios, e.g. on a comment, is removed. A moved breakpoint keeps its condition, hit count or log message.

> *With `run-as="external"`, XSpec runs the stylesheet under test as a separate transformation - its breakpoints pause, as they do otherwise.*

Coverage



The Test Coverage view summarises covered components while margin highlights show the covered code

The Test Explorer's **Run Tests with Coverage** runs the tests and shows which parts of the stylesheet they ran - in the editor's gutter, and in the Test Coverage view: each instruction, and each function and template, run or not, with the number of times. It's measured as the XSLT debugger runs the tests, without pausing, so it works with any edition of Saxon. The modules the stylesheet imports or includes have their coverage too - not XSpec's own modules, or the test runner.

Settings

Setting	Used for
`XSLT.tasks.saxonJar`	the Saxon jar the tests are run with - Saxon-HE is enough
`XSLT.xspec.home`	another XSpec installation, e.g. a newer release - by default, the XSpec 4.1.2 included
`XSLT.resources.catalog`	an XML catalog, passed to Saxon, as for tasks
`XSLT.xspec.allowSyntaxExtensions`	Saxon's syntax extensions for the tests: `"auto"` (the default) - on only for tests that need them, unless Saxon-HE - `"on"` or `"off"`
`XSLT.xspec.legacyCompatibility`	tests written for XSpec before 1.0 run as that ran them - see below
`XSLT.xspec.emphasiseLabels`	scenario labels in bold and expectation labels in italic, so the scenarios read like headings - on by default
`XSLT.xspec.fadePending`	pending scenarios and expectations faded, other than the reason they're pending - on by default

> *By default, tests are run with Saxon's syntax extensions on (`--allowSyntaxExtensions:on`) only when they're needed: when the stylesheet tested - or a module it imports, includes or uses, or a helper - is XSLT 4.0, or uses item types or notes, e.g. record types with `version="3.0"` , with `run-as="external"` too. Not otherwise, as with them on, Saxon 13 runs a stylesheet before XSLT 4.0 with some of 4.0's rules - e.g. a union pattern's priority, which can change the template rule chosen (Saxon's warning SXWN9052). Never with Saxon-HE, which fails every transform with them on. The `XSLT.xspec.allowSyntaxExtensions` setting turns them on or off.*

The stylesheet's task

When the stylesheet tested has a Saxon task in `.vscode/tasks.json` - e.g. its Quick Run task - the tests are run and debugged with the task's Saxon settings: `classPathEntries` , e.g. for extension functions, `configFilename` , `catalogFilenames` (instead of the `XSLT.resources.catalog` setting), `features` , `dtd` , `enableAssertions` and `expandValues` . Not the settings of the transform, e.g. its source, parameters or initial template, as XSpec's test runner has its own - or the task's `saxonJar` : the tests use the `XSLT.tasks.saxonJar` setting.

With more than one task for the stylesheet, and different Saxon settings, you pick the task, once in each run of the tests - the one picked last is first in the list.

Tests written for older XSpec

XSpec 4 stops the run for some tests that XSpec before 1.0 (e.g. 0.5) ran. With the `XSLT.xspec.legacyCompatibility` setting, they're run as that did - without changing the XSpec files:

In the XSpec file	**XSpec 4**	**With the setting**
an `x:expect` whose `test` is boolean, with `select` , `as` , `href` or content	stops: *Boolean @test must not be accompanied by…*	the boolean decides
an `x:expect` whose `test` isn't boolean, with no expected value	stops: *Non-boolean @test must be accompanied by…*	passes if the `test` returns nothing - it's compared with the empty sequence
the `xs` prefix, e.g. `xs:integer(6)` , not declared	stops: XPST0081	declared for the test runner

Each `x:expect` run as XSpec before 1.0 ran it is listed in the **XSpec** output when it's run, e.g. *XSpec legacy compatibility: x:expect ('…') has a boolean @test and an expected value…*, so the tests can be updated for XSpec 4 over time. That includes a `test` that's boolean as written but returns nothing when it's run - e.g. `@colname eq 'colA'` for an element without the attribute - which passed before.

As you edit, a warning marks an `x:expect` whose `test` is clearly of the wrong kind for XSpec 4 - from the expression, whatever the setting - with quick fixes for the form XSpec 4 runs:

Warning	**Quick fixes**
a boolean `test` , e.g. `exists(…)` , a comparison or `and` , with `select` , `as` , `href` or content	remove `select` , `as` and `href` - the boolean decides, as before
a `test` that isn't boolean, e.g. a path such as `$x:result/p[@a]` , or `distinct-values(…)` , with no expected value	wrap it in `empty()` - it passes if the test returns nothing, as before - or in `exists()`

A `test` that may be either, e.g. `$x:result` or a call of a function of the stylesheet, isn't marked. The `xs` prefix used without a declaration is an error already.

> *XSpec files for XQuery modules and XProc steps are checked as you edit them, but their tests aren't run from the Test Explorer, which needs Saxon's XQuery processor or an XProc processor.*

Print Debugging

To set breakpoints and step through a stylesheet, see the Debugger.

While debugging, a logpoint can do much of what an `xsl:message` with `xdm:debug-color()` does, without changing the stylesheet or including the xdm-view library. Its message can include XPath values, formatted and colored in the same way, and `{$*}` writes all the local variables, with a banner. A logpoint has less flexibility, though:

- Messages are only written while debugging, with SaxonJ - not when the stylesheet runs as a task or with Quick Run, or with SaxonJS or SaxonC.
- There's less control over formatting: the banner always names the template, function or instruction, there's no `$level` indentation, and `{$*}` writes every local variable, not a chosen set.
- An expression in the message can't contain braces, as in a map constructor.
- Logpoints aren't saved with the code. VS Code keeps them for the workspace, so they can't be committed or shared with the stylesheet.

Adding **xsl:message** instructions is another way to debug XSLT - one that works with any Saxon processor, including SaxonJS and SaxonC, and shows how values change over a whole run. The output is written to the terminal when the stylesheet runs as a task or with Quick Run, and to the **Debug Console** when it's debugged, with ANSI colors kept in both. Diagnostics are added using the **xdm:debug()** function, from the xdm-viewer XSLT library, which renders one or more named in-scope variables to a formatted string for use in an xsl:message.

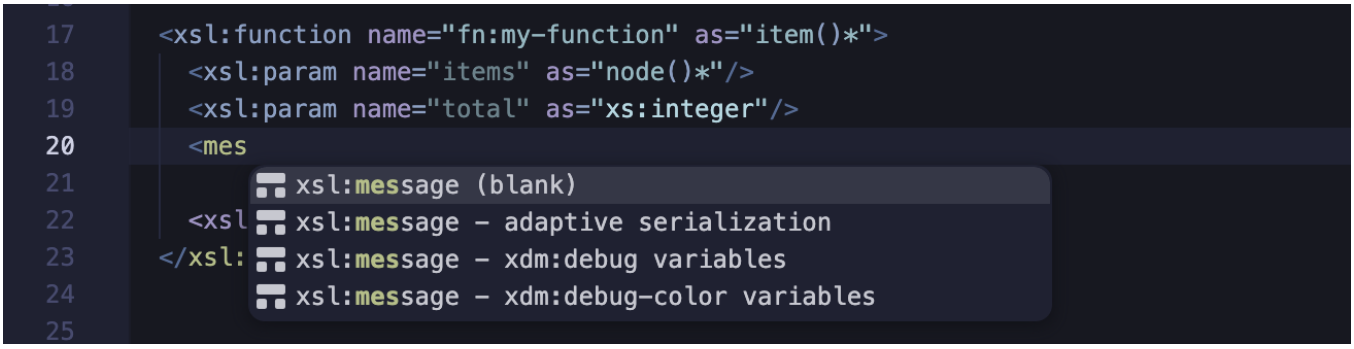
Diagnosing state of in-scope XSLT variables

There is special provision for conveniently reviewing the state of in-scope XSLT variables. This approach is outlined here.

1. Use Auto-complete for xsl:message

To assist with debugging, the XSLT editor has special auto-completion for the xsl:message instruction. Four completion items are offered when in-scope variables are found - otherwise, only **xsl:message (blank)**:

- **xsl:message (blank)** - inserts a simple placeholder, for entering your own xsl:message content
- **xsl:message - adaptive serialization** - serializes the last declared variable using `serialize()` with the `adaptive` method
- **xsl:message - xdm:debug variables** - lists all in-scope variables in a single **xdm:debug()** call
- **xsl:message - xdm:debug-color variables** - as above, with ANSI-colored output

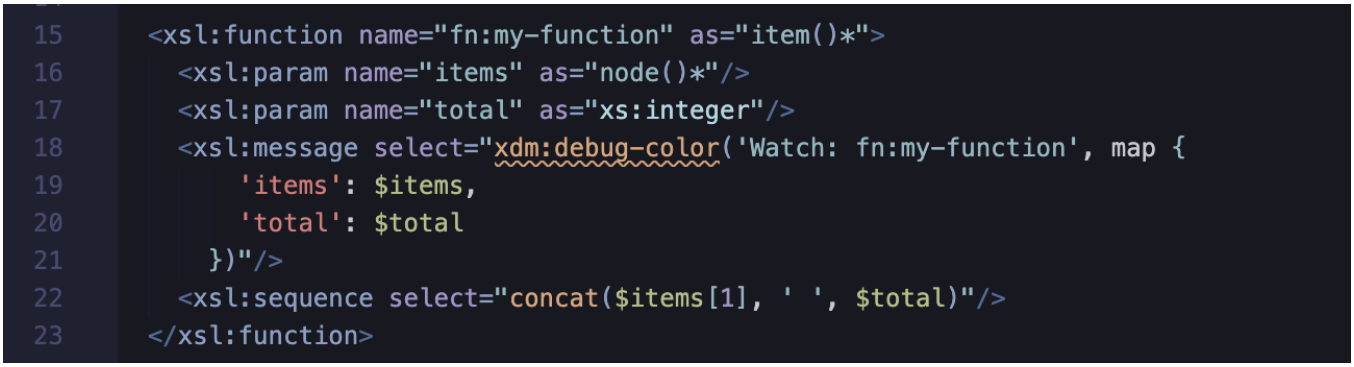


The auto-complete list for xsl:message when variables or parameters are in context.

2. Review Inserted xsl:message

When **xsl:message - xdm:debug variables** (or **xsl:message - xdm:debug-color variables**) is selected from the auto-complete list, an xsl:message snippet is inserted at the cursor with all in-scope variables collected into a single XPath map, passed as the second argument to **xdm:debug()**. A title is inserted as the first argument: `Watch:` followed by the name of the element that the xsl:message is in - e.g. an xsl:function or named template - or the mode of an xsl:template. If that element has neither, the title is `Watch Variables`.

```
<xsl:message select="xdm:debug('Watch: my-function', map {
    'items': $items,
    'total': $total
})"/>
```



The generated `xsl:message` inserted with unresolved call to `xdm:debug-color()`

3. Quick Fix: Include the xdm-view Library

As can be seen above, the `xdm:debug()` function is not initially resolvable - it will be marked as a 'problem' until its defining library is included. Put the cursor on the problem function and open **Quick Fix** with `Cmd/Ctrl+.` - or choose **Quick Fix...** when hovering over it.

Two Quick Fix options are offered:

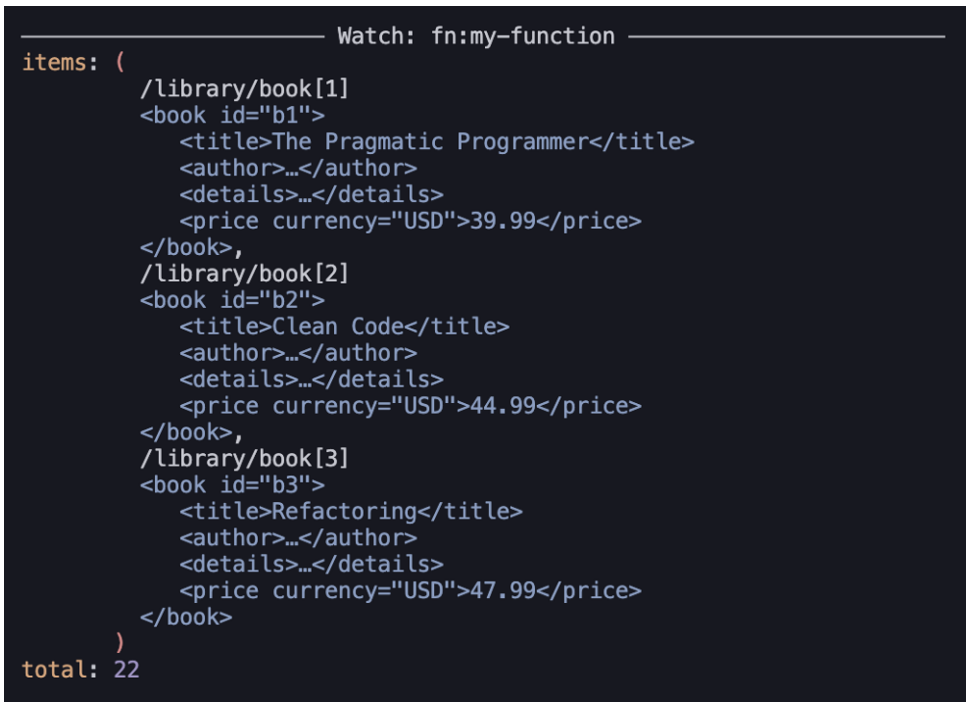
- **include XSLT module for xdm:debug and xdm:trace** (the preferred option) - inserts the **xdm** namespace declaration (`http://deltaxignia.com/ns/xdm-persistence`) and an **xsl:include** instruction referencing a workspace-local copy of **xdm-view.xsl** if one is found, otherwise the copy of the **xdm-viewer** library bundled with this extension.
- **copy xdm-view library into workspace** - copies the bundled **xdm-viewer** library (and its **xdm-persistence** dependency) into an `xslt-resources/xdm-view` folder in your workspace, then applies the same include as above. Use this if you want the library files available locally, for example to inspect or version them alongside your own stylesheets.



Quick Fix menu to resolve the call to `xdm:debug-color()`

4. Run XSLT and Review State in Terminal

You can now run the XSLT as normal, but you will now see the xsl:message output in the terminal window, with each in-scope variable shown under a title banner. Node values are shown with their location path and a pruned, truncated rendering, keeping the output readable for large trees.



`xsl:message` *output* `xdm:debug-color()`.

> *Colorizing for* ***xdm:debug-color()*** *should work from VS Code with all SaxonJ, SaxonC and SaxonJS versions. Because later SaxonJ versions don't support use of a built-in TextEmitter command-line option, an extension class is used instead of* `net.sf.saxon.Transform` *when invoking a transform via an XSLT task. This prevents escaping of ANSI codes used for coloring the output.*
> *For this same reason, in SaxonC, the XML output from SaxonC is decoded prior to output to the terminal.*

5. Benefits of xdm:debug() over fn:serialize()

We recommend the `xdm:debug-color()` function as it supplements the built-in `serialize` function with features that are especially important when you're interested in several variable values or the values have complex types.

The auto-complete fills in the arguments, so you rarely need to change them. The function signatures are:

```
xdm:debug($title as xs:string, $labels as map(xs:string, item()*)) as xs:string
xdm:debug($title as xs:string, $labels as map(xs:string, item()*), $level as xs:integer) as
xs:string
xdm:debug-color($title as xs:string, $labels as map(xs:string, item()*)) as xs:string
xdm:debug-color($title as xs:string, $labels as map(xs:string, item()*), $level as xs:integer)
as xs:string
```

- The output begins with a title to help locate the xsl:message in the code
- The remaining content is structured in two columns: variable names in the first, values in the second
- Nested maps, arrays and sequences are formatted for readability
- Bracket-pair coloring helps identify the start/end of maps, sequences or arrays
- A set of named variables or parameters can be serialized with a single function
- Lexical XPath locations are shown above each node, and above each XPath 4.0 JNode, e.g. `/lines/*[2]/sku` - see JSON and JNodes
- Node content is truncated when required to keep message output concise
- Syntax highlighting helps identify types and map keys etc.
- Values are annotated with their type where this is non-obvious
- The optional `$level` argument adds indentation proportional to this value - helpful for recursive calls

Tracing a value within an expression: xdm:trace()

To see a value part-way through an XPath expression, without taking the expression apart, wrap it in **xdm:trace()**. Like XPath's built-in `trace()`, it returns the value unchanged, so the expression gives the same result, and writes it with `xsl:message`, after a label. Where `trace()`'s output is up to the processor - Saxon writes one line per item, with a map on one line - `xdm:trace()` writes the value as `xdm:debug()` does: formatted over several lines where it has structure, and in color.

```
<xsl:variable name="total" select="sum(xdm:trace($order/line ! (@qty * @price), 'line
totals'))"/>
<xsl:variable name="customer" select="xdm:trace($customers?($id), 'customer')"/>
```

```
line totals: (7.5, 12, 17.5)
customer: {
            'name': 'Ada',
            'langs': ['xslt', 'xpath']
          }
```

To trace an existing expression, select it and choose **Wrap with xdm:trace()** (or **Wrap with trace()**) from the refactoring code actions, e.g. with ⌘. . The call's label is the expression itself, selected so you can type another. When you've finished debugging, put the cursor on a call's name - `xdm:trace`, `trace` or `fn:trace` - and choose **Remove xdm:trace() - keep the expression** to leave just the expression traced. It's kept in brackets only where it needs them to keep its meaning, e.g. `(1 + 2) * 2` from `xdm:trace(1 + 2, 'x') * 2`.

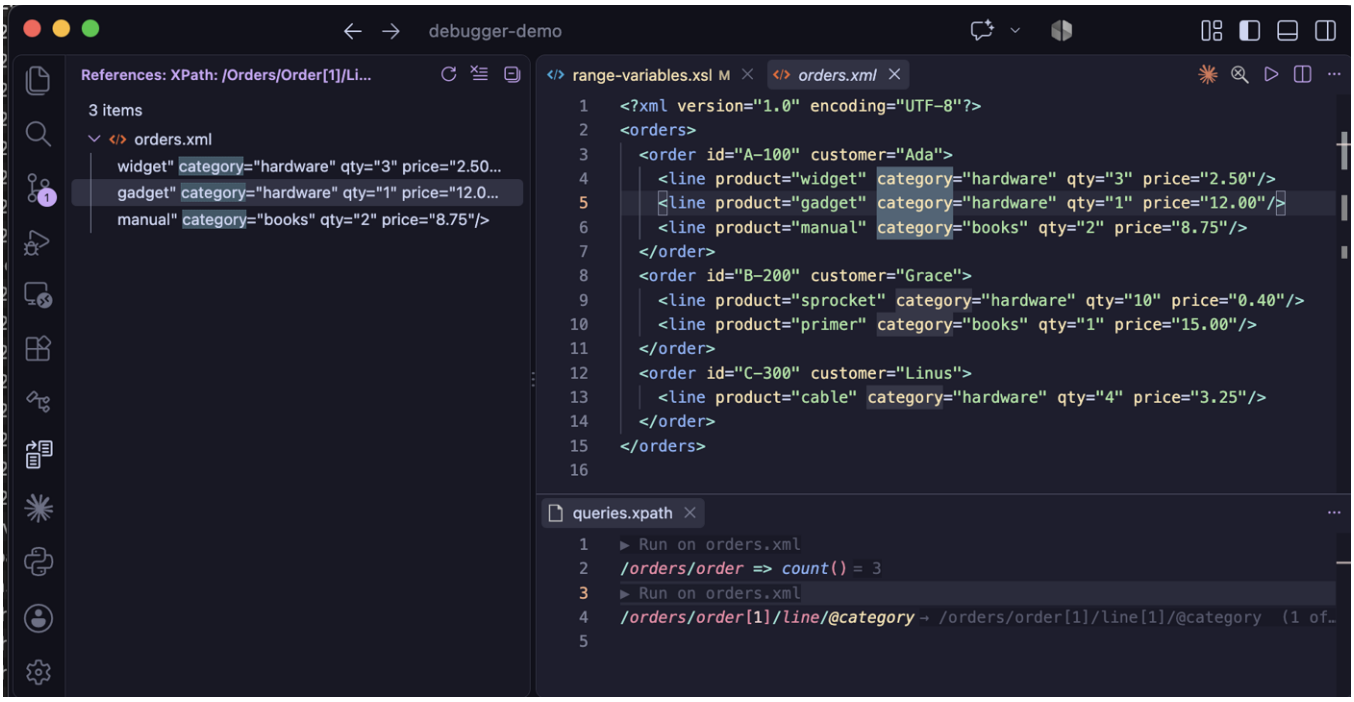
At the start of an XPath expression, auto-completion offers **xdm:trace(..., label)**. It's from the same xdm-viewer library as `xdm:debug()`, so it's marked as a problem until the library is included, with the same Quick Fix. The function signatures are:

```
xdm:trace($value as item()*) as item()*
xdm:trace($value as item()*, $label as xs:string?) as item()*
xdm:trace($value as item()*, $label as xs:string?, $useColor as xs:boolean) as item()*
```

- Color is on by default. To turn it off for every call - e.g. when messages are written to a log file - set the `NO_COLOR` environment variable, to any value. To turn it off for one call, pass `false()` as the third argument.
- Node values are shown with their location path and a pruned rendering, as with `xdm:debug()`.
- `xdm:trace()` reads the whole value before returning it, to format it. `trace()` may instead report the items of a lazily evaluated value one at a time as they're used - so, for example, `head(xdm:trace($items))` reads all of `$items`.
- In the debugger, the output is written to the Debug Console, in color.

Find Nodes with XPath

To find nodes in an XML file with XPath, write expressions in a query file, an XPath editor below the XML file, and run them one at a time. The nodes an expression finds are listed as VS Code lists references, to step through in the XML file. Other results, such as a count or a list of strings, are shown after the expression.



Enter XPaths in the query window to navigate the matching nodes.

> ***Requirements:*** *SaxonJ - any edition, including Saxon-HE - set with the* `XSLT.tasks.saxonJar` *setting, and a Java runtime on the PATH. Expressions are XPath 4.0 or XPath 3.1 - see XPath version. XPath 4.0 needs Saxon-PE or Saxon-EE.*

Opening the query file

In the editor of an XML file, click the search button in the editor's title bar, or run **XML: Find Nodes with XPath...** - from the Command Palette, the **XML Selection** submenu of the editor's context menu, or the **XML: Select Element...** quick pick (⇧⌘2 / Ctrl+Shift+2).

The query file, `queries.xpath`, opens in an editor group below the XML file, a quarter of its height. A new query file starts with an expression for the XML file's root element, such as `/orders`. There's one query file for each workspace, kept by VS Code, so your expressions are there next time.

> *With no workspace folder open, there's one query file shared by all the VS Code windows without a folder.*

Writing expressions

The query file has several expressions, separated by blank lines. An expression can take several lines, with no blank line within it, and can have comments, `(: like this :)`.

```
/orders

//order[@status = 'late']/@id

count(//item)

//order
  [item = 'ink']
  /@id
```

As in any XPath file, expressions are highlighted and checked as you type, and auto-completion includes the names of the elements and attributes in the XML file. Each expression is checked on its own.

XPath version

While the query file is in the editor, the status bar shows the XPath version of its expressions: **XPath 4.0** or **XPath 3.1**. Click it to switch between them, or run **XML: Switch XPath Version of Queries (3.1 / 4.0)**. The version is kept for the workspace.

Expressions are checked, auto-completed and run as that version:

- **XPath 4.0** - with XPath 4.0's syntax and functions, e.g. `{ 'a': 1 }` or `characters('abc')`, as Saxon-PE and Saxon-EE run them
- **XPath 3.1** - as in XSLT 3.0: XPath 4.0 syntax and functions are reported as problems and aren't offered by auto-completion, and Saxon runs the expression without its syntax extensions - so an expression that works here works in an XSLT 3.0 stylesheet

The version starts as XPath 4.0 - or XPath 3.1 if the `XSLT.tasks.saxonJar` setting is a Saxon-HE jar. Other XPath files, such as XPath notebooks, are always XPath 4.0.

> *Saxon-HE runs XPath 3.1 only. If XPath 4.0 is chosen with Saxon-HE, the status bar item shows a warning, and the **XPath Query** output panel says the expression was run as XPath 3.1.*

Namespaces

The namespaces declared on the XML file's root element can be used without declaring them:

- a name without a prefix matches an element in the root element's default namespace - so for `<orders xmlns="urn:orders">`, `//order` finds its `order` elements
- a prefix declared on the root element, e.g. `xmlns:x="urn:extra"`, can be used as it is: `//x:note`

For another namespace - one first declared further down, or with a prefix of your choosing - start the expression with a `declare namespace` line:

```
declare namespace svg = "http://www.w3.org/2000/svg";
//svg:rect
```

A declaration applies only to its own expression.

Inserting the location of a node

To start from a node in the XML file, put the cursor on it - on an element or an attribute - and press `⌥G` / `Alt+G` (**XML: Add XPath Location to Queries**). Its location, such as `/orders/order[2]/@id`, is added to the end of the query file, with the cursor on it, ready to run or change. If it's already in the query file, the cursor moves to it. With the cursor on no node, the query file is just opened.

In the query file, at the start of an expression, the same location is offered by auto-completion - type `/`, or press `^Space` / `Ctrl+Space`. It's the first suggestion, marked *cursor in orders.xml*. The XML file must be shown in an editor.

Running an expression

Above each expression, on the blank line before it, is ▶ **Run on orders.xml**, naming the XML file it runs on. Click it to run the expression, or press `⌘Enter` / `Ctrl+Enter` to run the expression at the cursor. The text in the XML file's editor is used, including any changes not yet saved.

> *With no blank line before an expression, e.g. on the first line, ▶ **Run** is a CodeLens above it - or, with CodeLens turned off (`editor.codeLens`), at the start of the expression's line.*

If Saxon reports an error, such as a syntax error, it's shown in the Problems view, at the place in the expression where Saxon found it.

Results

After each expression run, its result is shown in brief, at the end of its last line:

- `→ /orders/order[1]/@id (1 of 3)` - the location of the first node found, and how many there are
- `= 3`, or `= ('pen', 'ink', 'pad')` - values, written as XPath literals and cut short if they're long
- `= () nothing found` - an empty sequence

Click it to see the whole result in the **XPath Query** output panel. The panel is cleared each time an expression runs, so it shows only the latest result - or Saxon's error. The result is written as `xdm:debug()` writes values, with each sequence item, map entry and array member on a line of its own:

```
(
  {
    'name': 'Ada',
    'tags': (
      'a',
      'b'
    )
  },
  /orders/order[1]/@id
  @id="A1"
)
```

A node is shown as its location above a copy of it, with long text cut short. At most 100 items are shown in the panel - to see part of a large result, change the expression to select it.

When the result has nodes, the XML file's editor shows the first, with the nodes listed beside it, as VS Code lists references. Click one to go to it, or press F4 / ⇧F4 to go to the next or previous node. The name of an element or attribute is selected, and for a text node, its text.

> *At most 5000 items of a result are shown.*

Switching between XML files

Expressions run on the XML file you were last in. Click into another XML file's editor, and ▶ **Run** names that file: run the same expressions on it, as you would search it with Find. The checks and auto-completion in the query file follow the switch, for the new file's names and namespaces.

The results shown after the expressions are those for the file they'd run on. Switch back to an earlier file to see its results again.

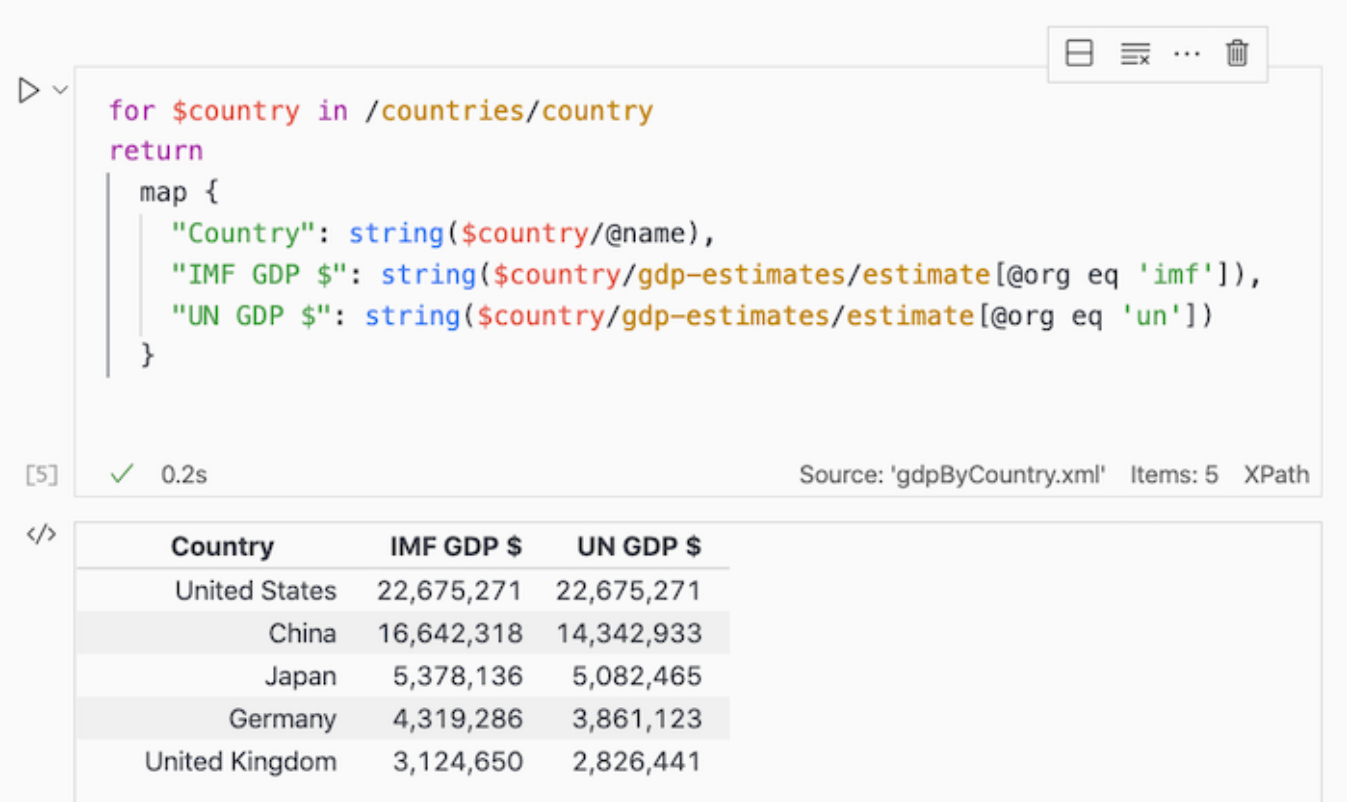
This is separate from the XML context file shown in the status bar for Quick Run and tasks: switching between XML files doesn't change it.

Limitations

- The first run starts Java, so it takes about a second. Later runs reuse it and are quicker, until no expression has run for 5 minutes.
- In the output panel, a result with a function item, e.g. `fn:count#1`, is listed with each item on one line, as Saxon writes it.
- Results shown after the expressions aren't kept when VS Code is closed.

XPath Notebooks

This section describes DeltaXignia's XPath Notebook extension for Visual Studio Code. Here we cover the fundamentals of using XPath 3.1 in a Visual Studio Code Notebook.



Why use an XPath Notebook?

XPath Notebooks are used for a wide range of purposes, including:

- Quick analysis of XML or JSON
- Learning and experimenting with XPath
- Tutorials on XPath
- Tutorials on a specific XML or JSON format

Quick Start

For this guide, we're assuming you've already installed the XPath Notebook extension for VS Code.

1. Install Node.js

This tool uses a builtin SaxonJS XSLT processor from Saxonica. This processor requires a recent installation of Node.js. Available for download from https://nodejs.org/en/.

2. Select an XML 'source' file

Open any XML (or JSON) source file in VS Code to set this as the context item for XPath expressions in an XPath Notebook. The most recent XML/JSON file opened is used.

3. Create and save a new XPath Notebook

From Visual Studio Code, open the Command Palette (⇧⌘P) and execute the **New XPath Notebook** command.

A new XPath Notebook with a **Markdown** cell and a **Code** cell is shown. It is untitled at first, with an `.xbook` file extension. Save this notebook in the usual way (⌘S), keeping the `.xbook` extension.

4. Edit and execute a new code cell

To add a new XPath code cell, press the **+ Code** button at the top of the notebook. In the new code cell, enter the `/*` XPath expression, then press `^Enter` to execute the cell.

The result cell is shown immediately below the Code cell.

AI Assistants

AI coding assistants in Visual Studio Code - such as GitHub Copilot or Claude Code - can make good use of the XSLT/XPath extension. Most of what they use needs nothing special from it: the extension's problems, documentation notes and debugging output are all in forms that assistants read and write well. For more, the extension has an MCP server, so an assistant can ask it directly - for example, for the problems of a stylesheet that isn't open, or where a function is declared.

Fixing problems

The problems reported by the extension's linter are standard Visual Studio Code diagnostics, shown in the Problems view. They identify the exact symbol at issue - an unresolved variable, a function called with the wrong number of arguments, a record field that doesn't exist - so an assistant that reads the diagnostics can explain and fix them precisely, and check its own changes the same way. For example, GitHub Copilot can include the problems with `#problems` in a chat prompt, or fix a problem from its Quick Fix menu, and Claude Code reads them through its Visual Studio Code integration.

The problems are kept up to date for every stylesheet open in an editor - not only the active one. An assistant usually changes files while your cursor is in another, so a stylesheet it changes is checked again, whether it's shown beside the active editor or in a tab that isn't shown, and the assistant sees the problems its change caused. When it changes several stylesheets at once, each is checked with its own declarations, XSLT version and imports, so no problems are reported that aren't there.

The MCP server (experimental)

The extension can run a Model Context Protocol (MCP) server, with tools an assistant can call to ask the extension about your stylesheets. The tools use the same code as the editor - the linter, Go to Definition, Find All References and the XSLT Imports view - so their answers match what you see, and they work for files that aren't open. The server is off by default: to turn it on, set `XSLT.mcp.enabled` to `true` - see Settings.

Tool	What it gives the assistant
`lint_file`	The problems in an XSLT module or XSpec file - as the Problems view shows them, checked with its imports, includes and packages - whether or not it's open. Optionally, only those of a severity or worse.
`lint_files`	The problems in each XSLT and XSpec file in a folder: a summary, then only the files with problems. By default, warnings and errors - not hints, such as unused variables.
`module_imports`	The tree of imports, includes and packages from a module's top-level stylesheet down, with the module marked - as in the XSLT Imports view.
`find_declaration`	Where a function, named template, global variable or parameter, mode, key, attribute set, accumulator, item type or character map is declared, as seen from a module - with its parameters and types, and its documentation: its documentation note, or a comment just before it. When more than one module declares it, the one used, with the highest import precedence, is first.
`find_references`	The references to one of these, as Find All References finds them - in the modules that use it and the XSpec files that test it - with each reference's line. The XSpec uses are listed last, as the tests, each with the scenario it's in.
`describe_type`	

Tool	What it gives the assistant
	An XSLT 4.0 named item type: a record type's fields, with their types and `@field` documentation, or an enumeration type's values.
`xpath_function`	The signature and description of a built-in XPath or XSLT function, for XPath 3.1 or 4.0 - for a function declared in a stylesheet, the assistant uses `find_declaration`.

The tools only read files: they don't change them, or run stylesheets or tests.

GitHub Copilot

With the server on, Copilot finds it by itself: its tools are listed with the other MCP tools in Copilot Chat's agent mode.

Claude Code

Run **XSLT: Set Up MCP Server for Claude Code** from the Command Palette - it offers to turn the server on, if it isn't, and asks whether to set it up for all your projects or only this one. It runs Claude Code's `claude mcp add` command for you. A Claude Code session started after that can use the tools, whether it runs in Visual Studio Code or in a terminal of its own; in a session already running, run `/mcp` to connect.

If the `claude` command isn't found - for example, it isn't on the path Visual Studio Code uses - the `claude mcp add` command is copied instead: run it in a terminal - for one project, in that project's folder.

Security

The server only accepts connections from your own computer (`127.0.0.1`), and each request needs a token that the extension makes once and keeps in Visual Studio Code's secret storage - it doesn't expire. Claude Code keeps the token in its own settings; a copied `claude mcp add` command includes it, so keep the command private. To make a new token - for example, if the old one has been shared - run **XSLT: Reset MCP Server Token**. Copilot uses the new token straight away, and Claude Code is set up again with it - for all projects, or for a project in the workspace - where it was set up with the old one.

The server listens at the port of `XSLT.mcp.port` (6745 by default). With more than one Visual Studio Code window open, the first one opened has the server. Its tools work best for the files of that window's workspace: a relative path is taken from its first folder, and a module's top-level stylesheet is found from the workspace's modules. The **XSLT MCP Server** output channel shows whether the server started, or why it didn't - for example, because another program uses the port.

Documentation notes

XSLT 4.0 documentation notes - `xsl:note format="xdoc-md"` - use Markdown with `@param` and `@return` tags, in the style of Javadoc: a format that assistants already know. An assistant can read a function's note to understand it, and can write notes too - for example, when asked to document the functions in a stylesheet. As the extension checks the `@param` tags against the parameters, a missing or misnamed parameter in a note is reported straight away.

Debugging output

The output of xdm:debug() in `xsl:message` is plain, labelled text: each variable's value in XPath syntax, with its type where it isn't obvious, and each node with its path. Copied from the terminal into a chat, it gives an assistant the actual values to reason about, rather than a description of the problem.

Running XSLT

Quick Run creates ordinary tasks in the workspace's `tasks.json` file, and each task shows the Saxon command line it runs in the terminal. An assistant that can run commands can run a transform in the same way.

The Agents window

Visual Studio Code's Agents window, for running agent sessions and reviewing their changes, only runs the extensions on a list chosen by Microsoft, and the XSLT/XPath extension isn't on it yet. Without the extension, XSLT files in the Agents window have plain XML highlighting, with none of the extension's features - and if Red Hat's XML extension is installed, it may report errors against its XSLT 3.0 schema. To run the extension in the Agents window, add this to your user settings - see Settings:

```
"extensions.supportAgentsWindow": { "deltaxml.xslt-xpath": true }
```

Current limitations

The MCP server's tools only read files: an assistant can't yet ask the extension to run a stylesheet, an XSpec test or an XPath expression, though it can run a Quick Run task's Saxon command itself. Tools for these may be added in a future release.

In the Problems view, problems are only shown for files open in an editor, as by most language extensions: a stylesheet an assistant changes that isn't open has none there until it's opened. With the MCP server on, an assistant can check such a file with `lint_file`.

XSLT 3.0

XSLT 3.0, published by W3C on 8 June 2017, is the current W3C Recommendation for XSLT, alongside **XPath 3.1**. This extension supports XSLT 3.0, and also XSLT 4.0 as implemented in Saxon 13.

Learning XSLT 3.0

The online W3C documentation for XSLT 3.0, along with its XPath 3.1 sub-language, comprises 3 main parts:

- XSL Transformations (XSLT) Version 3.0
- XML Path Language (XPath) 3.1
- XPath and XQuery Functions and Operators 3.1

> *Note: The XSLT and XQuery Serialization 3.1 specification is published separately as this is shared with XQuery.*

While the size of each part of the XSLT specifications may appear daunting at first, each part is well-structured, with a Table of Contents panel that makes it easy to navigate. Many code samples are included with the specification.

Offline Reading Material

There are no published books for XSLT 3.0, but the XSLT 2.0 and XPath 2.0 Programmer's Reference by Michael Kay is frequently recommended. XSLT 3.0 is a significant enhancement to XSLT 2.0 but it builds on the fundamental principles and language in XSLT 2.0.

Video Tutorials

A set of video tutorials by Dimitre Novatchev is available:

- XSLT 2.0 and 1.0 Foundations.
- The Evolution of XPath: What's New in XPath 3.0.

Supporting XSLT Processors

This extension integrates with the Saxon XSLT processor by Saxonica. Stylesheets are run with XSLT tasks or Quick Run, using the command-line interface of SaxonJ (Java), SaxonJS (JavaScript) or SaxonC (C/C++). Saxon is probably the most popular/ubiquitous XSLT 3.0 processor. It publishes specification conformance data and supports many platforms. A free open source version, Saxon-HE, is available along with paid-for 'Professional' and 'Enterprise' variants. These are summarised in a feature matrix.

Saxon PE and EE 12.8 or later, run with syntax extensions, also accept some XSLT 4.0 features in an XSLT 3.0 stylesheet: named item types - with record and enumeration types - and `xsl:note`. To edit such stylesheets with the same support as for XSLT 4.0, enable the `XSLT.validation.xslt30ItemTypesAndNotes` setting - see XSLT 4.0 Item Types and Notes in XSLT 3.0. For the full set of XSLT 4.0 features implemented in Saxon 13, see XSLT 4.0.

Altova's RaptorXML server supports XSLT 3.0 - no integration with Visual Studio Code is available as Altova offer their own paid-for XML/XSLT development solution. No XSLT standards conformance data is published for this processor.

XSLT 4.0

XSLT 4.0 currently comprises a set of draft proposals maintained by the W3C XSLT 4.0 Community group.

DeltaXignia's XSLT/XPath language extension provides full language support for the XSLT 4.0 and XPath 4.0 features implemented in Saxon 13, as documented here: type-aware auto-completion, checks, hover help and refactoring, as well as syntax highlighting.

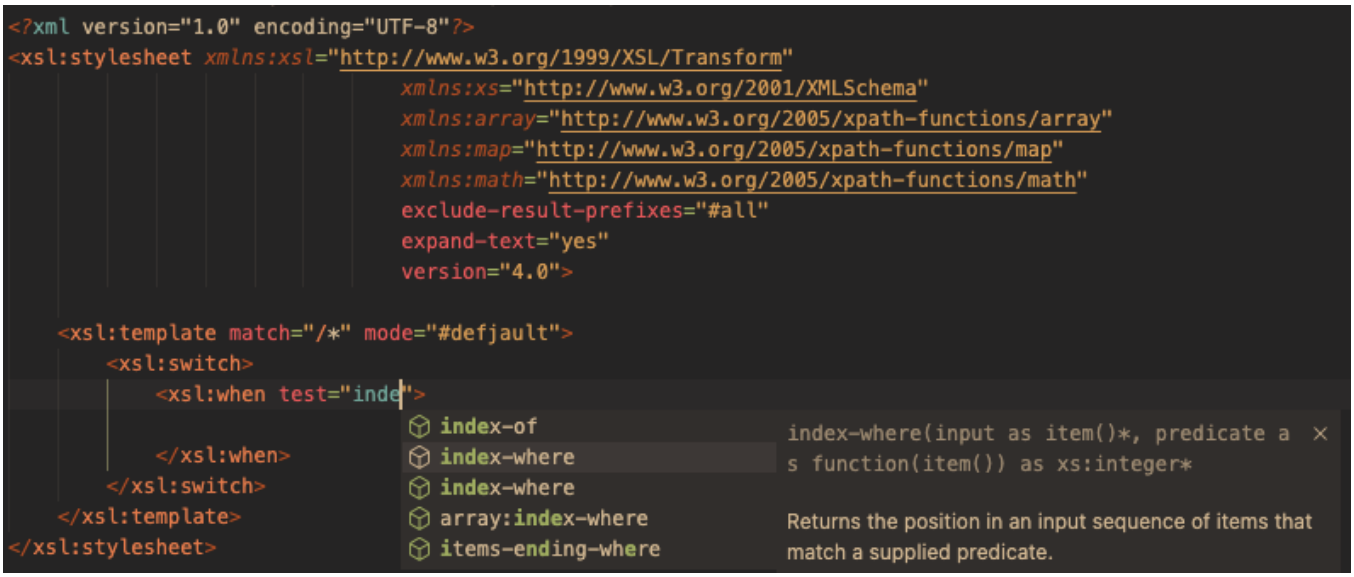
Note that, given the independent release cycles for Saxon and the XSLT 4.0 draft, there will be differences between the implementation and the draft specification. Some parts of the specification are likely to change significantly before it is finalized.

We will endeavour to keep this extension up to date with the latest Saxon XSLT 4.0 implementation. Earlier versions of XSLT 4.0 will not be supported for auto-completion and linting.

Editing XSLT 4.0

The **version** attribute on the stylesheet root element controls the XSLT version supported by the linter and for auto-completion. When the version attribute on the stylesheet root element is set to **4.0**, auto-completion for XSLT instructions and attributes and XPath functions will be included. In an XSLT 3.0 stylesheet, the linter reports XPath 4.0 syntax as an error.

For Saxon PE or EE 12.8 or later, run with syntax extensions, XSLT 4.0 item types - with record and enumeration types - and `xsl:note` can also be enabled in XSLT 3.0 stylesheets: see XSLT 4.0 Item Types and Notes in XSLT 3.0.



XPath files, with the `.xpath` extension, are always treated as XPath 4.0.

XSLT 4.0 support

The syntax highlighter, linter and auto-completion support:

- **New XPath 4.0 syntax** - the pipeline `->` and mapping arrow `=!>` operators, the method call operator `=?>`, e.g. `$shape =?> area()` - with the fields of a record type suggested after it - the `×` and `÷`

operators, braced `if` expressions, map constructors without the `map` keyword, string templates, and the new `-or-self` axes

- **Literals and bindings** - hexadecimal and binary numeric literals, and `_` digit separators, e.g. `0xFF_FF`, `0b101` and `1_000_000`; QName literals, e.g. `#xs:integer`; positional variables, e.g. `for $x at $i in $items return ...`; `for key` and `value` map bindings, e.g. `for key $k value $v in $map return ...`; and typed `for`, `let` and `some` bindings, e.g. `let $n as xs:integer := 3`
- **New item types** - choice types, such as `(xs:date | xs:time)`, and the `enum`, `record`, `fn` and `jnode` types
- **Named item types** - declared with `xsl:item-type`, with checks on the names used, go to definition and find references
- **Function syntax** - `fn` and focus functions, keyword arguments such as `ex:area(scale := 2)`, optional parameters (`xsl:param required="no"`) and partial function application
- **Functions in no namespace** - an `xsl:function` name without a prefix, e.g. `name="double"`, called as `double(2)` - a Saxon 13 extension, not available in Saxon-HE. A function with the name and arity of a built-in function, e.g. `count`, replaces it in calls without a prefix, so is reported as a warning.
- **JNodes** - path expressions on trees of maps and arrays, with `jtree()`, `jkey()`, `jvalue()` and `get(...)` steps
- **New instructions** - snippets for `xsl:switch`, `xsl:array` and `xsl:select`, template rules declared within an `xsl:mode` (enclosed modes), and `xsl:record`, with its attributes completed and checked against the record type - see Building records

Features in the XSLT 4.0 draft that Saxon 13 doesn't implement - such as extensible record types - are reported by the linter.

Some of the more significant XSLT 4.0 features have their own pages:

- Records and Enums - auto-completion, checks and refactoring for record and enumeration types, including `xsl:switch` on an enumeration
- JSON and JNodes - path expressions on JSON, with auto-completion, checks and hover help for JNodes with a record type
- Code Documentation - documenting functions and templates with `xsl:note`, for hover help and signature help

Running XSLT 4.0

XSLT 4.0 stylesheets run with Quick Run, or with an XSLT task. Saxon needs the `--allowSyntaxExtensions:on` configuration property for XPath 4.0 syntax. This is controlled by the **allowSyntaxExtensions40** task property:

- **auto** (the default) - on for a stylesheet that needs it: XSLT 4.0, or one that uses item types or notes, in it or a module it imports, includes or uses - unless the Saxon jar file, or SaxonC path, is for Saxon-HE. Not for other stylesheets, as with it, Saxon 13 runs a stylesheet before XSLT 4.0 with some of 4.0's rules, e.g. a union pattern's priority
- **on** - always on
- **off** - never on

> *The free Saxon-HE version **does not** support the language extensions for XSLT 4.0. You will need to reference a Saxon-PE or Saxon-EE Jar file (with license file) in the **saxonJar** property of the task definition, or the equivalent SaxonC edition.*

Other Editors for XSLT 4.0

The Oxygen XML editor also provides language support for XSLT 4.0, with its own bundled Saxon implementation.

While Oxygen does need to be purchased, it has the advantage of including Saxon-EE built-in, so you will not need to purchase Saxon separately. Oxygen XML provides an integrated solution for XML developers and content authors, and hosts an array of other features.

Records and Enums

XPath 4.0 adds **record** types, for maps with a known set of fields, and **enumeration** types, for strings with a known set of values. XSLT 4.0 adds the `xsl:item-type` declaration to give these types names.

When a variable, parameter or function result is declared with one of these types, the XSLT/XPath extension uses the declared type for auto-completion, linting, hover help and navigation - for example, to list the fields when building a map, or the values of an enumeration inside a string literal.

> *The extension only uses **declared** types - from `as` attributes, `xsl:item-type` declarations and function return types. It doesn't infer types from expressions: that is left to Saxon when the stylesheet is compiled.*

These features are enabled when the stylesheet has **version="4.0"**. See the XSLT 4.0 page for the XSLT 4.0 features supported.

For Saxon PE or EE 12.8 or later, run with syntax extensions, they can also be enabled for XSLT 3.0 stylesheets, with the setting `XSLT.validation.xslt30ItemTypesAndNotes` - see XSLT 4.0 Item Types and Notes in XSLT 3.0. That page also describes converting Saxon's earlier `saxon:type-alias` and `tuple(...)` syntax.

Declaring types

A record type lists its fields, each with an optional type. A field name followed by `?` is optional. An enumeration type lists its values as string literals:

```
<xsl:item-type name="address" as="record(city as xs:string, postcode? as xs:string)"/>
<xsl:item-type name="person" as="record(name as xs:string, age as xs:integer, address as
address)"/>
<xsl:item-type name="colour" as="enum('red', 'green', 'blue')"/>
```

A type can also be written inline in an `as` attribute, for example `as="record(x as xs:double, y as xs:double)"`.

In an attribute value, quotes within a type can be written as references, as Saxon allows - for example `as="enum("it's", 'is')"` or a field name quoted with `'` - and the extension reads them as the quotes they stand for.

Auto-completion for types

- Within an `as` attribute, and after `instance of`, `castable as` and the other type operators, the names of declared item types are offered with the built-in types.
- The `record(...)` and `enum(...)` suggestions insert a snippet with placeholders for the fields or values.

Checks on declarations

- An item type name that isn't declared is reported, as are duplicate declarations and a type that refers to itself (directly or through other named types).
- A duplicate field name in a record type is an error, as in Saxon. A duplicate value in an enumeration type is a warning, with a **Remove duplicate enum value** quick fix.

- Extensible record types, such as `record(a, *)`, have been dropped from XPath 4.0 and are reported as an error.

Go to Definition (F12) and Find All References (⌥⇧F12) work for item type names.

Building records

A record is built with a map constructor, or with the `xsl:map` or `xsl:record` instruction. Wherever the record type is known, the extension helps you complete the map and checks it against the type.

Map constructors

In the `select` attribute of a declaration with a record type, type `{` (or `map {`) and accept the suggestion to insert a complete map constructor, with an entry for each required field - when the record has optional fields, a second suggestion includes them too:

```
<xsl:variable name="p" as="person" select="{
  'name': __TODO.name,
  'age': __TODO.age,
  'address': {
    'city': __TODO.city
  }
}"/>
```

Each `__TODO.field` placeholder is reported as a warning until it's replaced with a value. A double-click selects the whole placeholder, ready to type over.

After typing `,` at the end of an entry, the suggestions list the fields that aren't in the map yet, including optional fields.

The xsl:map instruction

Where a record is built with `xsl:map`, for example in the content of an `xsl:variable` with a record type, typing `<` offers an `xsl:map` suggestion with an `xsl:map-entry` for each required field (or, as a second suggestion, for all fields). Inside an existing `xsl:map`, the `xsl:map-entry` suggestions are named after the fields not yet added, for example `xsl:map-entry 'age'`, and the `key` attribute offers the field names.

The xsl:record instruction

The XSLT 4.0 `xsl:record` instruction builds a record from its attributes: each attribute in no namespace is a field, with an XPath expression as its value. Its record type is the type in its own `xsl:as` attribute, or else the declared type of its value, as for `xsl:map`:

```
<xsl:variable name="p" as="person">
  <xsl:record name="'Ann'" age="42" address="{ 'city': 'Oxford' }"/>
</xsl:variable>
```

- **Auto-completion** - typing `<` offers an `xsl:record` suggestion with an attribute for each required field, e.g. `xsl:record person: required fields`, and a second suggestion with all fields. Within the start tag, the fields not yet added are offered as attributes, required fields first, and within the content, typing `<` offers an `xsl:map-entry` for each field not yet added. A field whose name isn't an NCName, e.g. `'first name'`, is added as an `xsl:map-entry` child.
- **Checks** - as for a map: an attribute that isn't a field is a warning, a literal value of the wrong type is an error, and a missing required field is an error, with the **Add missing record fields** quick fix. A field can

also be added by an `xsl:map-entry` child - and when other content may add fields, missing fields aren't reported.

- **Renaming** - renaming a field renames its `xsl:record` attributes, and Find All References lists them.

Checks on records

- **Missing fields** - a required field that's missing is an error, with an **Add missing record fields** quick fix that adds an entry, with a placeholder, for each one.
- **Unknown fields** - an entry for a field that isn't in the record type is a warning. For a map constructor key, or an `xsl:record` attribute, there's a **Change to '*name*'** quick fix for each field not yet in the map, closest name first - e.g. **Change to 'name'** for `'nam'`.
- **Field values** - a literal value of the wrong type, such as a string for an `xs:integer` field, is an error.
- **Duplicate keys** - two entries with the same literal key are an error, in any map constructor or `xsl:map`, whether or not it has a record type - except in an `xsl:map` with a `duplicates` attribute, which handles them, where they're a warning.

Where the record type comes from

Maps are checked, and completed, where their type is declared:

- the `select` or content of an `xsl:variable`, `xsl:param` or `xsl:with-param` with an `as` attribute - for an `xsl:with-param` without one, the type of the called template's parameter is used
- the result of an `xsl:function` with an `as` attribute - its last `xsl:sequence`
- `xsl:sequence` and `xsl:select` with an `as` attribute
- the arguments of calls to user-defined functions, including keyword arguments such as `ex:move(point := { ... })` and the left-hand operand of the arrow operators
- a typed `let` binding, such as `let $p as person := { ... }`

Using records

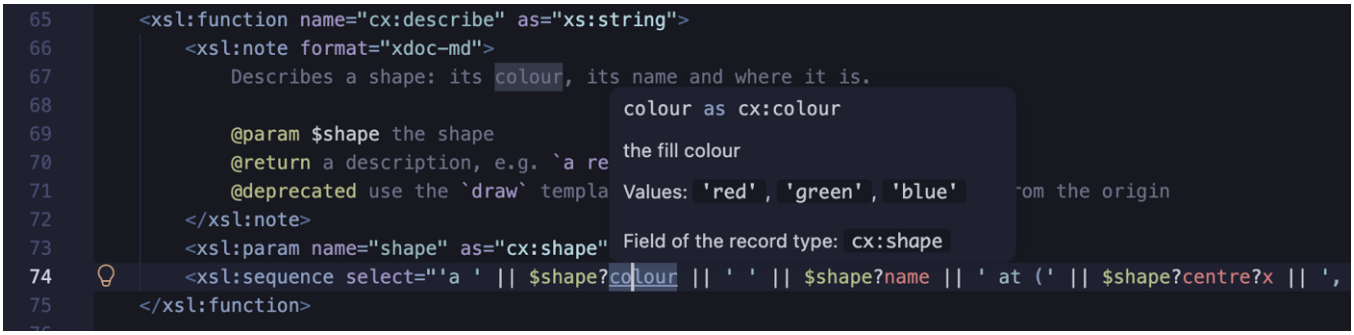
Lookups

After the `?` lookup operator on a value with a record type, the suggestions list the record's fields. This works for variables, calls to functions with a declared record result, and chains of lookups such as `$p?address?city`. A lookup of a field that isn't in the record type is a warning.

Hover and navigation

Hovering over a field name - in a lookup, a map constructor key or the `key` of an `xsl:map-entry` - shows the field's declaration. Go to Definition on a field name goes to the field in the record type.

Hovering over the name of a named item type - where it's used, or on its declaration - shows its declaration, with its values for an enumeration type. When the `xsl:item-type` has a documentation note, the note is shown too, and a field's `@field` text is shown when hovering over the field, and with the field in the auto-completion suggestions.



Hover over a record field in a lookup, e.g. $shape?colour, showing the field's declaration and its @field text from the xsl:item-type's documentation note.

Renaming a field

Rename Symbol (F2) on a field name - in the record type, a lookup, a map constructor key, the `key` of an `xsl:map-entry`, an `xsl:record` attribute, or an `@field` tag of a documentation note - renames the field, and all its references. Find All References (⌥⇧F12) lists them.

- Only the field of that record type is renamed - not a field with the same name in another record type.
- The references are found in the stylesheet, the modules it includes or imports, and - from the workspace's modules - each stylesheet that imports or includes it, with the modules they use. So a rename in a module of types reaches the stylesheets that use them.
- A quoted field name keeps its quotes, e.g. `'unit name'`. A new name must be an NCName, unless the field's name isn't one.

Path steps on records (JNodes)

XPath 4.0 path expressions can navigate maps and arrays as trees of **JNodes**, e.g. `jtree($p)/address/city`. Field names are suggested, and checked, for steps after `jtree($p)/`, and after a variable declared as a JNode for a record type, such as `jnode(*, person)`. See JSON and JNodes.

Refactoring: Extract record type

The **Extract record type** refactoring creates a record type from an existing map - for example, when updating code written for XSLT 3.0. It's offered when the stylesheet has **version="4.0"**, or with the `XSLT.validation.xslt30ItemTypesAndNotes` setting:

1. Select a map constructor or `xsl:map` whose keys are string literals - or simply place the cursor on the start tag of the `xsl:variable`, `xsl:param`, `xsl:with-param`, `xsl:function`, `xsl:sequence` or `xsl:select` that it's the value of.
2. Open the refactorings with the 💡 light-bulb, or ⌘ . , and choose **Extract record type**.
3. An `xsl:item-type` declaration is added for the record type, and the declaration's `as` attribute is set to it - replacing a generic type such as `map(*)`. Field types are taken from literal values, and a nested map becomes a nested record type.
4. The new type is named `record-type`, and Rename Symbol is started so you can give it a meaningful name.

If an existing `xsl:item-type` already has the same field names, a **Use record type '*name*'** refactoring is also offered, in the **Rewrite** section of the list. It sets the `as` attribute to the existing type without adding a declaration.

Extract record type is also offered for a `record(...)` type written inline in an `as` attribute - with the cursor on it, or a selection within it - except the whole type of an `xsl:item-type`. The innermost record type is replaced by the name of a new `xsl:item-type`, named after the field whose type it is, if any, and Rename Symbol is started:

```
<xsl:item-type name="person" as="record(name as xs:string, address as record(city as
xs:string))"/>
```

becomes:

```
<xsl:item-type name="address" as="record(city as xs:string)"/>
<xsl:item-type name="person" as="record(name as xs:string, address as address)"/>
```



Before refactor: The two 💡 refactorings for a candidate record-type variable.



After refactor: a prompt to rename the record-type now it's declared and set on the variable.

Enumeration types

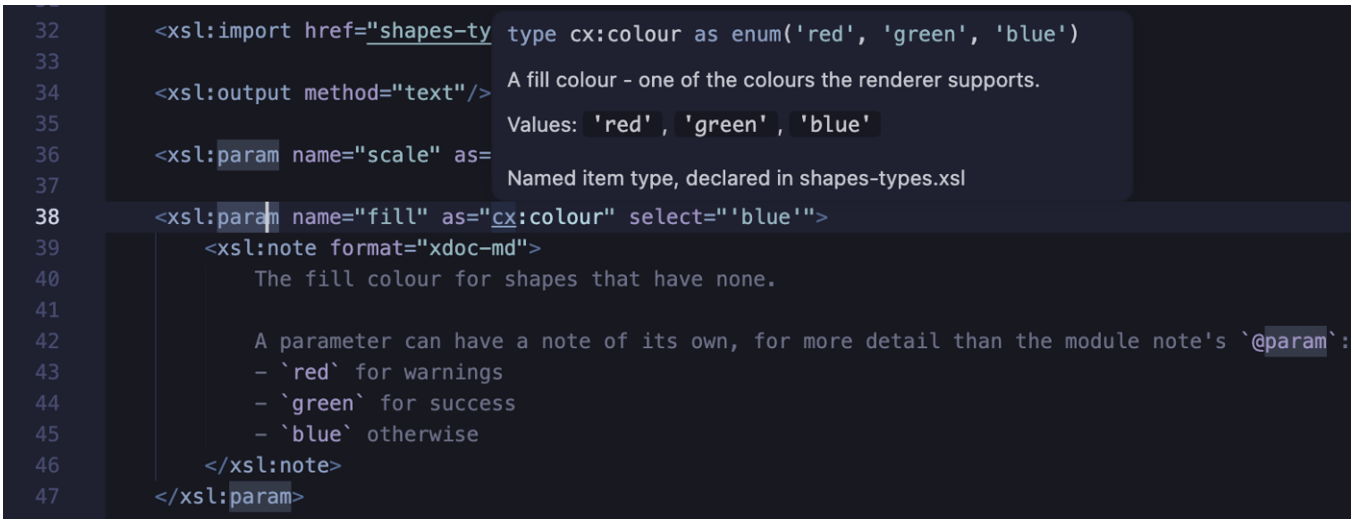
Auto-completion for values

Where an enumeration type is expected, typing a quote character offers the enumeration's values. This applies to:

- the `select` of an `xsl:variable`, `xsl:param` or `xsl:with-param` declared with the type (or `xs:boolean`, for `true()` and `false()`)
- arguments of calls to user-defined functions whose parameter has the type, and typed `let` bindings
- record fields with the type, in map constructors
- the `test` of an `xsl:when` in an `xsl:switch` - see below

A string literal that isn't one of the values is reported as an error.

Hovering over the name of an enumeration type - or a global variable or parameter, a parameter of a function or template, or a record field, with the type - lists its values - on one line for a few values, otherwise as a list. For a type declared as another, or as a choice of enumeration types, the values are those it resolves to.



Hover over the name of an enumeration type, showing its values

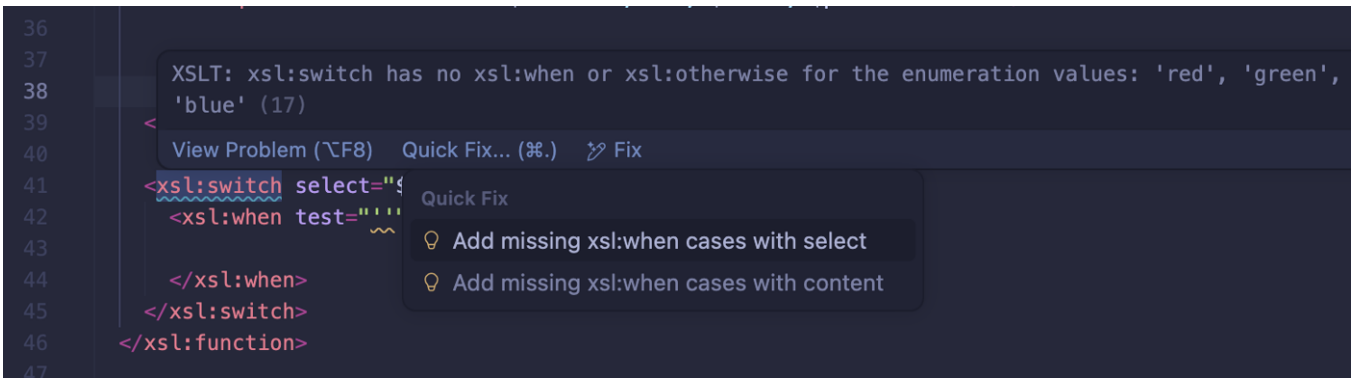
Renaming a value

Rename Symbol (F2) on an enumeration value - in its `enum(...)` declaration, or a string literal for it - renames the value and the string literals that refer to it: variable values, function arguments (including keyword arguments), record field values in map constructors, typed `let` bindings and `xsl:switch` cases. Find All References (⌥⇧F12) lists them.

- Only the values of that type are renamed - not the same value in another enumeration type. An inline `enum(...)` type is identified by its values.
- The value is renamed in imported modules too.
- A new name that's already a value of the type is rejected. A quote in the new value must be doubled, e.g. `don''t`.

xsl:switch on an enumeration

The XSLT 4.0 `xsl:switch` instruction selects an `xsl:when` by comparing a value with each `test`. When the `select` has an enumeration type - a variable, function call or record field lookup with a declared type - the extension works with its values.



Quick Fix options shown for `xsl:switch` on an enum type.

- **Missing values** - when an enumeration value has no `xsl:when`, and there's no `xsl:otherwise`, the `xsl:switch` is marked with the information message *xsl:switch has no xsl:when or xsl:otherwise for the enumeration values: ...*. Its two quick fixes add an `xsl:when` for each missing value:
  - **Add missing xsl:when cases with select** - e.g. `<xsl:when test="'green'" select=""/>`
  - **Add missing xsl:when cases with content** - e.g. `<xsl:when test="'green'">...</xsl:when>`
- **Values that never match** - a `test` value that isn't one of the enumeration's values, or that an earlier `xsl:when` already tests, is a warning, as the `xsl:when` never matches it.
- **Auto-completion** - inside an `xsl:switch`, the completion list offers an `xsl:when` for each value not yet tested.

JSON and JNodes

XPath 4.0 path expressions can navigate JSON - maps and arrays - as they do XML. The `jtree()` function wraps a map or array in a tree of **JNodes**, with a JNode for each map entry and array member, so a path such as `$order/customer/name` works on JSON data, with predicates, the descendant and parent axes, and the other features of path expressions.

When a JNode is declared with a record type, for example `jnode(*, order)`, the XSLT/XPath extension knows which fields each step can reach. It checks each step, offers the field names as auto-completion suggestions, and shows each field's type and documentation on hover.

JNodes need **version="4.0"** and Saxon-PE or Saxon-EE 13, run with syntax extensions enabled (Quick Run enables these for a `version="4.0"` stylesheet). Saxon-HE 13 doesn't support XPath 4.0 syntax. In an XSLT 3.0 stylesheet, the `jnode(...)` item type and the `get(...)` node test are reported as needing XPath 4.0.

> *A sample project, xslt-json-schema-records, shows the features on this page in use. It converts a JSON Schema to XSLT 4.0 record types, reading the schema as JNodes, then reads JSON order data typed with the generated record types.*

Navigating JSON with JNodes

A JNode tree is made with `jtree()`, from a map or array - for example the result of `json-doc()` or `parse-json()`:

```
<xsl:variable name="root" select="jtree(json-doc('order.schema.json'))"/>
<xsl:variable name="defs" as="jnode()*" select="$root/get('$defs')/*"/>
```

- **Name steps** select map entries by key, e.g. `$order/customer/name`. `*` selects every entry of a map, or every member of an array, e.g. `$order/lines/*`.
- **get(...)** is the node test for a key that isn't an NCName, such as `get('$ref')` or `get('first name')`, or for an array member by position, such as `get(1)`. The extension reports `get('a')` used on its own, as a function call, as a call to an unknown function.
- **Axes** work as they do for XML: `$schema//get('$ref')` finds every `$ref` at any depth, and `$line/../../currency` goes up from an order line, through the `lines` array, to the order.
- **jkey()** gives a JNode's key - the map key, or the position of an array member - and **jvalue()** gives the value it wraps. Where a value is atomized, such as in an arithmetic expression or a text value template, a JNode gives its value, so `$line/quantity * $line/unit-price` works without `jvalue()`.
- A JNode is matched with the `jnode()` item type, e.g. `$tree/type instance of jnode(type)`. `jnode()` is offered in auto-completion for `as` attributes in XSLT 4.0.

> *Saxon 13 doesn't support type node tests such as `~record(...)` or `~xs:string` in a path step, so the extension reports these as errors.*

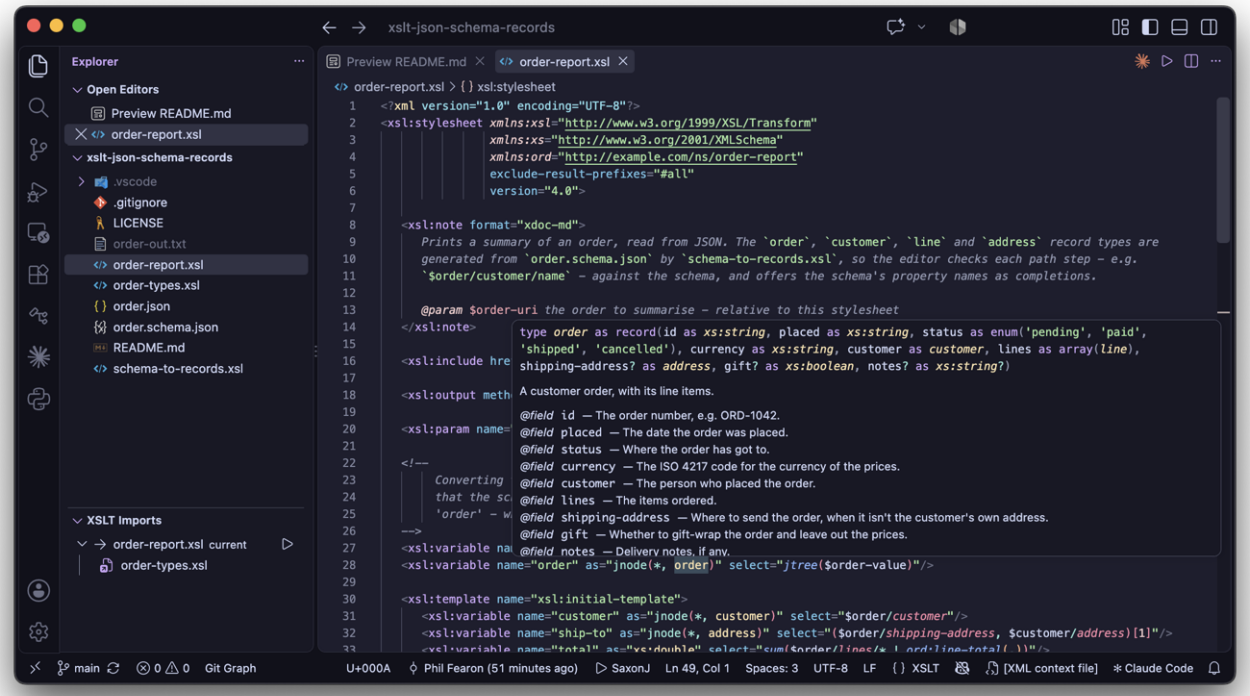
JNodes with a record type

A variable, parameter or function result declared as `jnode(*, T)`, where *T* is a record type, is a JNode whose value is a *T*. A named item type can be used too, e.g. `<xsl:item-type name="order-node" as="jnode(*, order)"/>`. Child steps from the JNode are then auto-completed and checked against the record's fields, at any depth:

```
<xsl:variable name="order-value" as="order" select="json-doc($order-uri)"/>
<xsl:variable name="order" as="jnode(*, order)" select="jtree($order-value)"/>
...
<xsl:function name="ord:line-total" as="xs:double">
  <xsl:param name="line" as="jnode(*, line)"/>
  <xsl:sequence select="$line/quantity * $line/unit-price"/>
</xsl:function>
```

- **Auto-completion** - after a `/`, the suggestions list the record's fields. Typing `$order/customer/` lists the fields of the `customer` record type.
- **Checks** - a step that isn't a field is a warning, e.g. *Child step 'nam' - this is not a field of the record type: customer* for `$order/customer/nam`. A `?` lookup on the JNode, e.g. `$order?id`, is checked in the same way.
- **Hover** - hovering over a field name in a step, such as `unit-price` in `$line/unit-price`, shows the field's type. Its `@field` text is shown too, from the `xsl:item-type`'s documentation note. Hovering over the type name in `jnode(*, order)` shows the record type, with its fields and documentation.

Child steps after `jtree($p)/`, where `$p` has a record type, are auto-completed and checked in the same way.



Hovering on `order` in `jnode(*, order)` shows the record type, imported from a module of types generated from a JSON Schema: its fields, their types, and each field's description from the schema.

Maps need jtree() before '/'

A `/` step needs a JNode on its left, not a map. Saxon 13 reports XPTY0019 for a step such as `$order-value/id`, where `$order-value` is a map with a record type, so the extension reports this as an error. The message suggests converting the map first, e.g. `jtree($order-value)/id`. Alternatives are a lookup, `$order-value?id`, or a variable declared as a `jnode(*, order)`, as above. A step after a lookup, such as `$p?address/city`, is reported in the same way.

Debugging: xdm:debug() shows JNode paths

The xdm:debug() function shows a JNode's location in its tree, as an XPath path from the tree's root, above the value it wraps - in the same way it shows the location of an XML node. For an order line, this is a path such as `/lines/*[2]`, so you can see which part of the JSON each value came from:

```
<xsl:message select="xdm:debug('ord:line', map {
  'line': $line,
  'sku': $line/sku,
  'bundle': $line/bundle-items
})"/>
```

The `xsl:message` output, for the second order line:

```
──────────────────────────────── ord:line ────────────────────────────────
line:   /lines/*[2]
        {
          'sku': 'KIT-7',
          'description': 'Electronics starter kit',
          'quantity': 1,
          'unit-price': 64,
          'bundle-items': [
            {'sku': 'TL-1', 'description': 'Soldering iron', 'quantity': 1, 'unit-price': 0},
            {'sku': 'TL-2', 'description': 'Multimeter', 'quantity': 1, 'unit-price': 0}
          ]
        }
sku:    /lines/*[2]/sku
        'KIT-7'
bundle: /lines/*[2]/bundle-items
        [
          {'sku': 'TL-1', 'description': 'Soldering iron', 'quantity': 1, 'unit-price': 0},
          {'sku': 'TL-2', 'description': 'Multimeter', 'quantity': 1, 'unit-price': 0}
        ]
```

The **xdm:debug variables** auto-completion for `xsl:message` inserts a call like this one, with every in-scope variable. See Print Debugging.

Things to know about JNodes and record types in Saxon 13

- **A `jnode(*, T)` type matches only an *instance* of *T*.** A map with keys that *T* doesn't declare isn't an instance, so `jtree(json-doc('order.json'))` doesn't match `jnode(*, order)` when the data has extra properties. Converting the value to the record type first - as for `$order-value` above - drops the extra keys, at every depth, and fails if a required field is missing or has the wrong type.
- **No recursive item types.** An `xsl:item-type` can't refer to itself, directly or through other types (Saxon reports SXTA0001). A field that would refer back, such as a bundle of order lines within an order line, has to be `map(*)` or `array(map(*))` instead, so the steps after it aren't checked.

- **JSON numbers are `xs:double`.** `json-doc()` and `parse-json()` give every JSON number as an `xs:double`, so a record field for a number must be `xs:double` - `xs:integer` would reject the data. A JSON `null` is an empty sequence, so a nullable field needs an occurrence indicator, e.g. `xs:string?`.
- **No `??` operator.** Saxon 13 doesn't support the deep lookup operator, so use the descendant axis on JNodes instead, e.g. `$schema//get('$ref')`.

Sample: JSON Schema to record types

The xslt-json-schema-records sample works in two stages:

1. `schema-to-records.xsl` reads a JSON Schema, `order.schema.json`, as JNodes, and writes `order-types.xsl`: an `xsl:item-type` for the schema and for each of its `$defs`. Each property's `description` becomes an `@field` in the type's documentation note, so it's shown on hover.
2. `order-report.xsl` includes `order-types.xsl`, and prints a summary of `order.json`, using `jnode(*, order)`, `jnode(*, line)` and other typed JNodes.

Run each stylesheet with Quick Run, choosing 'None' as the XML context file, so it starts from `xsl:initial-template`. Then, in `order-report.xsl`, try:

- typing `$order/customer/` to see the customer's fields as suggestions
- changing `$order/customer/name` to `$order/customer/nam`, to see the warning
- hovering on `unit-price` in `$line/unit-price`, or on `line` in `jnode(*, line)`, to see the documentation taken from the schema
- changing `$order/id` to `$order-value/id`, to see the error that suggests `jtree()`
- adding a property to `customer` in `order.schema.json`, running `schema-to-records.xsl` again, and seeing it offered after `$order/customer/`

The sample's README describes how each JSON Schema keyword maps to an XPath type.

Code Documentation

XSLT 4.0 adds the `xsl:note` element, for documentation within a stylesheet. The XSLT/XPath extension supports notes written in a simple format, **xdoc-md**: Markdown text followed by tags such as `@param` and `@return`, in the style of Javadoc and xqDoc.

A note in this format, as the first child of an `xsl:function`, a named `xsl:template` or an `xsl:item-type`, is shown in hover help wherever the declaration is used - including from other stylesheet modules - and on the declaration's own name. For functions and templates, it's also shown in signature help. For a record type, declared with `xsl:item-type`, the note can also describe each of the record's fields.

A module note, as the first child of the `xsl:stylesheet`, describes the module and its global parameters and variables - and a global parameter or variable can have a note of its own, for more detail.

The XSLT 4.0 specification allows `xsl:note` anywhere, with any attributes and content, and leaves the format of the documentation open. The `format="xdoc-md"` attribute is the extension's convention for the notes it shows: XSLT processors ignore it, with the rest of the note.

These features are enabled when the stylesheet has **version="4.0"** - and for XSLT 3.0, see Notes in XSLT 3.0.

Writing a documentation note

Set the `format` attribute to `xdoc-md`. The text before the first tag is a description, in Markdown. Each tag starts a new line, and continues on the following lines up to the next tag:

```
23  <xsl:function name="ex:area" as="xs:double">
24    <xsl:note format="xdoc-md">
25      Returns the area of a rectangle, **scaled** by an optional factor.
26
27      @param $width the width, in `cm`
28      @param $height the height, in `cm`
29      @param $scale the scale factor
30      @return the area, in square centimetres
31    </xsl:note>
32
33    <xsl:param name="width" as="xs:double"/>
34    <xsl:param name="height" as="xs:double"/>
35    <xsl:param name="scale" as="xs:double" required="no" select="1"/>
36    <xsl:sequence select="$width * $height * $scale"/>
37  </xsl:function>
```

The xdoc-md tags within xsl:note are highlighted along with any markdown used

Tags

- `@param $name` - a parameter of a function or template - or, in a module note, a global parameter
- `@return` - the result of a function or template
- `@field name` - a field of a record type, for an `xsl:item-type`
- `@variable $name` - a global variable, in a module note
- `@see` - a related function, template, item type, variable or URI - see References to other declarations
- `@since` - the version it was added in

- `@deprecated` - why it should no longer be used, and what to use instead
- `@error` - an error a function or template may raise
- `@author` and `@version` - for a module note

`@field` isn't an xqDoc tag: the extension adds it, in the same style, for documenting records. A field name that isn't an NCName is quoted, as in the record type - for example `@field 'unit name' e.g. cm`.

```
31    <xsl:item-type name="cx:shape" as="record(name as xs:string, centre as cx:point, colour as cx:colour, size? as xs:double)">
32        <xsl:note format="xdoc-md">
33            A shape to draw.
34
35            @field name the shape's name, e.g. `circle`
36            @field centre the centre of the shape
37            @field colour the fill colour
38            @field size the size, in points - `1` if it's absent
39            @since 1.0
40        </xsl:note>
41    </xsl:item-type>
42
43</xsl:stylesheet>
```

A note on an `xsl:item-type`, with an `@field` tag for each field of its record type.

Markdown

The description and tag text can use Markdown: `**bold**`, `*italics*`, `` `code` ``, `[links](https://example.com)` and `#` headings.

As the note is XML, the characters `<` and `&` must be escaped as normal XML, for example `<`. Alternatively, put text containing markup in a CDATA section: its content is shown exactly as written.

Adding a note

With the cursor on the start tag of an `xsl:function`, `xsl:template`, `xsl:item-type`, the `xsl:stylesheet`, or a global `xsl:param` or `xsl:variable`, that has no `xsl:note`, choose **Add documentation note** from the refactorings (💡 or ⌘ .). A note is inserted as the first child, with:

- an `@param` for each parameter, and an `@return` for a function (or a template with an `as` attribute)
- for an `xsl:item-type` with a record type, an `@field` for each field
- for a module note, an `@param` or `@variable` for each global parameter or variable that has no note of its own

An empty element, e.g. `<xsl:item-type .../>`, gets an end tag for the note.

Press `Tab` to move between the description placeholders.

Notes can also be added with auto-completion, after `<`:

- **xsl:note** inserts a note on one line. It's offered within any XSLT element, and within literal result elements - XSLT 4.0 allows `xsl:note` anywhere.
- **xsl:note xdoc-md** inserts a documentation note. It's offered where one is used - within a function, named template, item type, global parameter or variable, or at the top level for the module note - and when there isn't one yet. There, it's listed first, and preselected.

Within the start tag of an `xsl:note`, `format` is offered as an attribute, and `xdoc-md` as its value.

Within a note, typing `@` offers the tags that apply: `@see`, `@since` and `@deprecated` for every note, with `@field` for an `xsl:item-type`; `@param`, `@return` and `@error` for a function or template; and `@param`, `@variable`, `@author` and `@version` for a module note. The note of a global `xsl:param` or `xsl:variable` is offered only `@see`, `@since` and `@deprecated`. After `@param`, the parameter names without an `@param` yet are offered - and likewise after `@field` and `@variable`.

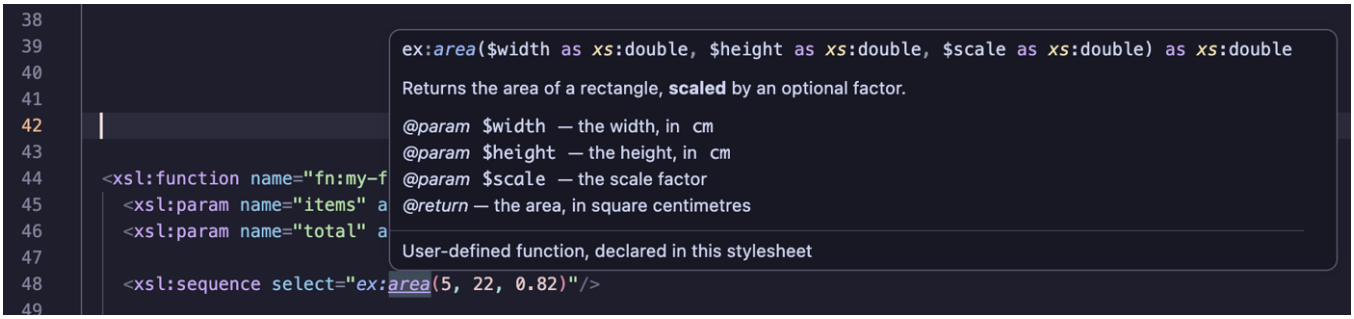


The code action 'Add documentation note' is offered on selecting a function element.

Where notes are shown

- **Hover help** - on a call to the function, or on the `name` of an `xsl:call-template`, the note's description and tags are shown with the signature. On the `name` of an `xsl:with-param`, the parameter's `@param` text is shown.
- **Hover help on item types** - on a named item type where it's used, e.g. in an `as` attribute or after `instance of`, the note is shown with the type's declaration.
- **Hover help on declarations** - on the `name` of an `xsl:function`, named `xsl:template` or `xsl:item-type`, the note is shown just as for a use of it. On the `name` of an `xsl:param` of a function or template, the parameter's `@param` text is shown.
- **Record fields** - the `@field` text is shown in hover help on a field, e.g. `r` in `$c?r`, and with the field suggestions of auto-completion: for a lookup, a child step on a JNode, a map constructor key and an `xsl:map-entry`. For an item type declared as another, e.g. `as="cx:point"`, the other type's `@field` text is used.
- **Variable references** - on a reference to a global parameter or variable, e.g. `$scale`, or the `name` of its declaration, its type and documentation are shown - see Module notes. On a reference to a parameter of a function or template, its `@param` text is shown.
- **Signature help** - while typing the arguments of a function call, or the `xsl:with-param` children of an `xsl:call-template`, the parameter hints show the `@param` text for the current parameter.
- **Auto-completion** - the `xsl:with-param` suggestions within an `xsl:call-template` show the `@param` text for each parameter.
- **Packages** - on the `name` of an `xsl:use-package`, the package's module note is shown with the components the stylesheet can use - and a declaration in `xsl:override` without a note of its own has the note of the package component it overrides, see XSLT Packages.

Notes on functions, templates and item types in included and imported stylesheets, and in the packages a stylesheet uses, are shown too.



To see `xdoc-md` documentation on the function (or named template) hover over the referring code.

References to other declarations

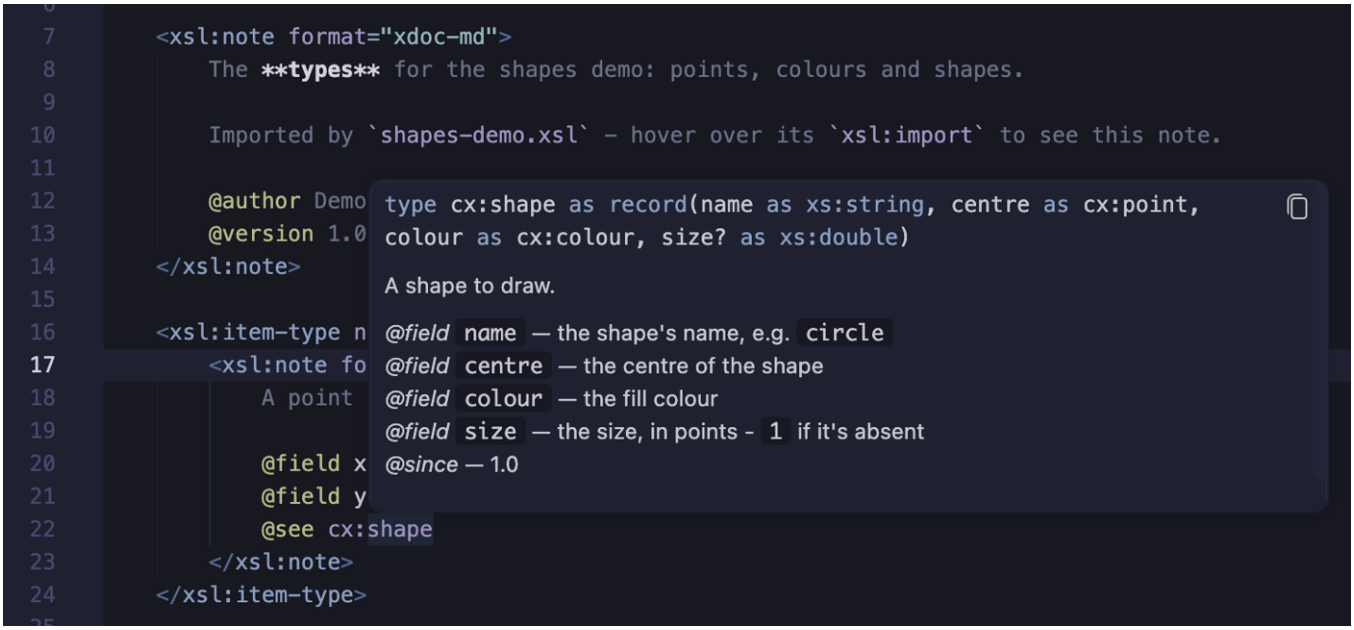
A note can refer to a function, template, item type or variable declared elsewhere - after `@see`, or as a Markdown code span anywhere in the note - using the XPath syntax for it:

- `my:area#2` - the function with that arity
- `my:area()` or `my:area` - a function of any arity - or for `my:area`, else a named item type or template
- `$scale` - a parameter of the function or template the note documents, or else a global parameter or variable
- `template my:draw` - a named template

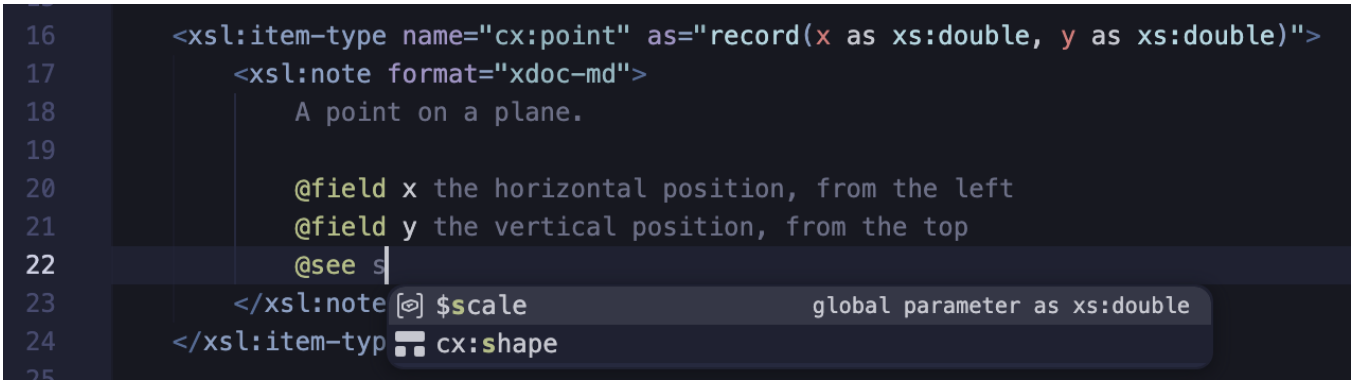
For example, `@see my:area#2`, or `Uses `$scale` and `template my:draw`.` In a code span, a name without a prefix is only a reference as `$name`, `name#2`, `name()` or `template name`, as code spans are used for other text too.

- **Navigation** - **Go to Definition** and hover help work on a reference just as on a use of the declaration, including declarations in imported and included modules.
- **Find references and rename** - the references in notes are found, and renamed, with the declaration - e.g. renaming `my:area` changes `@see my:area#2` too.
- **Auto-completion** - after `@see`, and within a code span, the declarations are suggested, written so that they refer to them: parameters and variables, e.g. `$scale`, functions by arity, e.g. `my:area#2`, then item types and templates.

In the note of a declaration in `xsl:override`, a reference to the component it overrides is to the package's declaration - so `@see shp:area#1` in the note of the overriding `shp:area` goes to the package's `shp:area`, not to itself. The declarations that the override quick fixes and refactoring add have a note with that `@see`.



Hover on a reference in a note to review the referenced item's description.



Auto-completion hints are triggered when adding a reference in a note.

Module notes

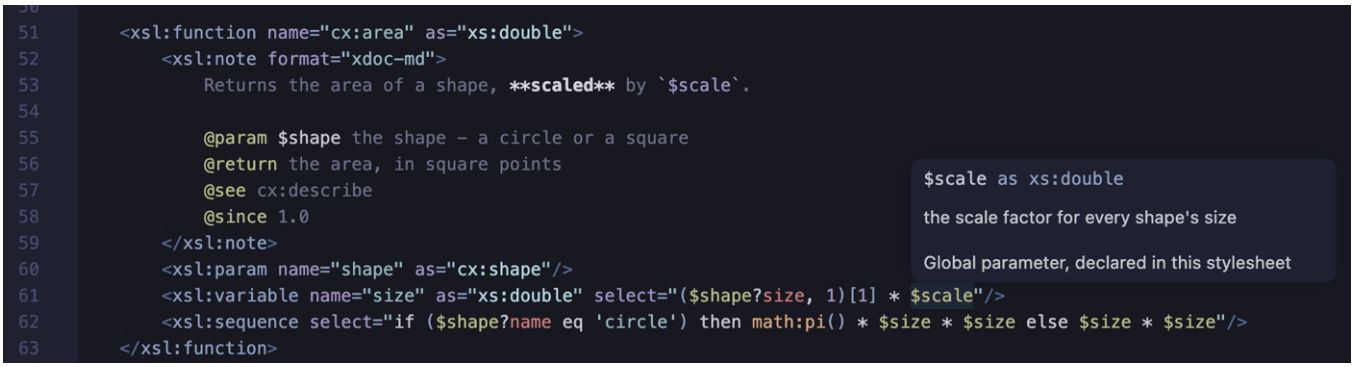
A documentation note as the first child of the `xsl:stylesheet` (or `xsl:transform` or `xsl:package`) describes the module. Its tags can describe the global parameters and variables briefly, so they don't need notes of their own:

```
<xsl:stylesheet ... version="4.0">
  <xsl:note format="xdoc-md">
    Draws shapes.

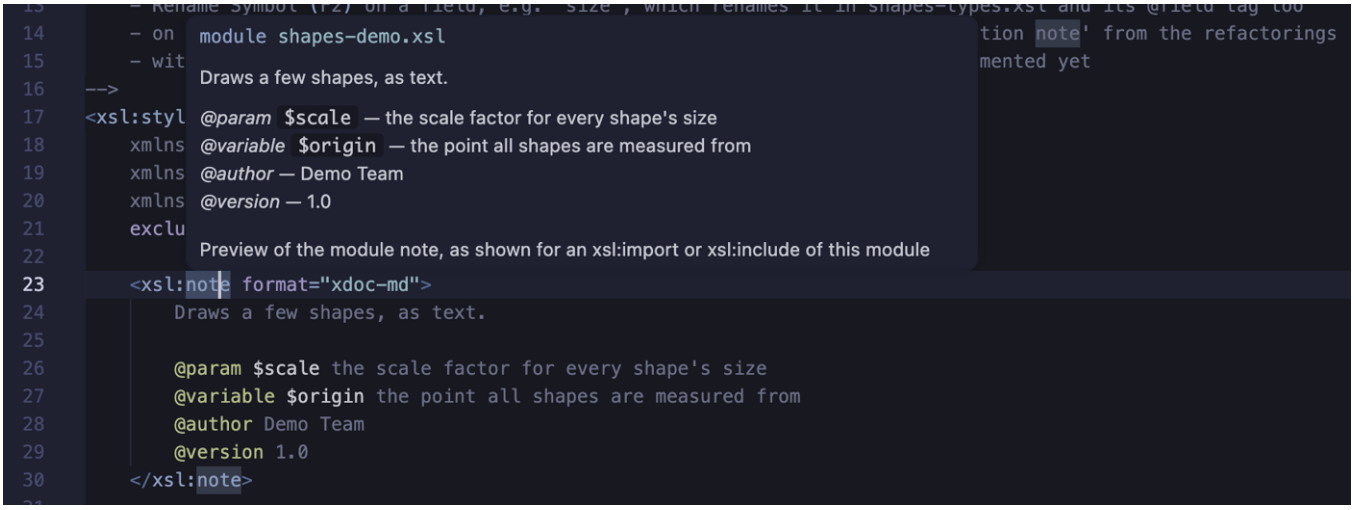
    @param $scale the scale factor
    @variable $origin the point all shapes start from
    @author Ann
    @version 1.2
  </xsl:note>
  <xsl:param name="scale" as="xs:double" select="1"/>
  <xsl:param name="colour" as="cx:colour" select="'red'">
    <xsl:note format="xdoc-md">The fill colour, used for **every** shape.</xsl:note>
  </xsl:param>
```

For more detail, a global `xsl:param` or `xsl:variable` can have a documentation note of its own, as its first child. It's shown in place of the module note's `@param` or `@variable` text - so a module note can give a brief summary, and the parameter's own note the details.

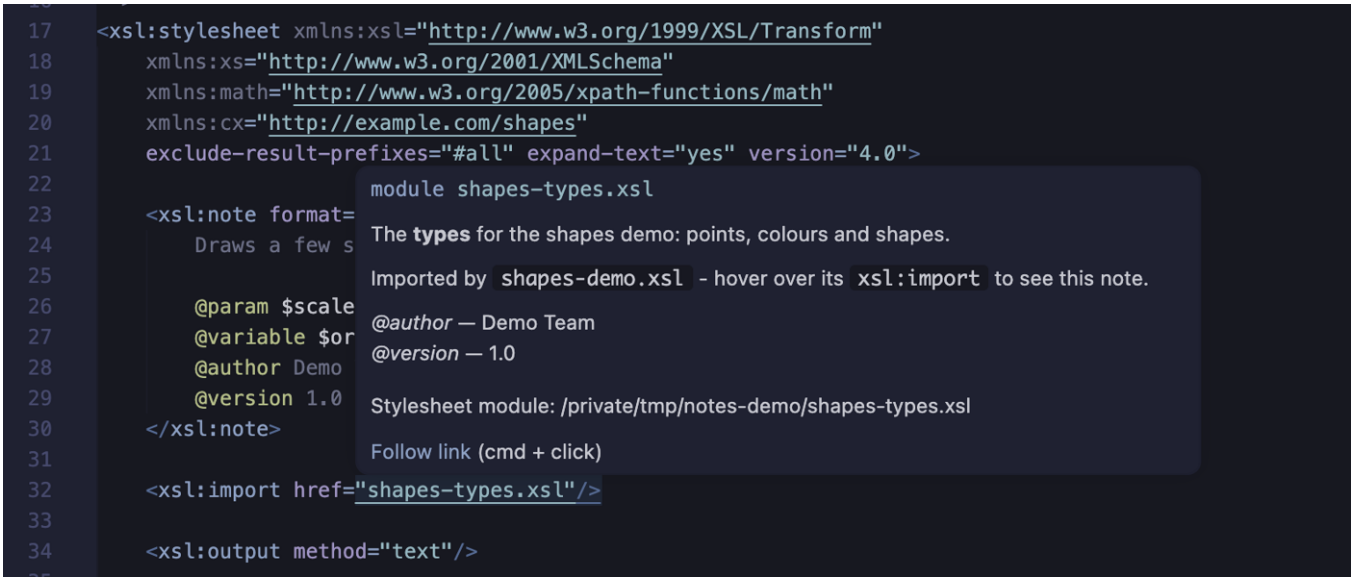
- **Hover help on variables** - on a reference to a global parameter or variable, or the `name` of its declaration: its type, its documentation, and for an enumeration type, its values.
- **Hover help on imports** - on the `href` of an `xsl:import` or `xsl:include`: the module's note, and its path - and on the `name` of an `xsl:use-package`, the note of the `xsl:package`, see XSLT Packages.
- **Preview** - on the start tag of the module note: the note as it's shown for an import of the module.
- **Rename** - renaming a global parameter or variable renames its `@param` or `@variable` tag too.



The hover on a reference to a global parameter, $scale, showing its @param text from the module note.



The hover on the start tag of a module note, previewing the note.



The hover on the href of an xsl:import, showing the imported module's note.

Notes in XSLT 3.0

Unless you're using a compatible Saxon version with syntax extensions enabled (see note below), an XSLT 3.0 processor reports `XTSE0010` for `xsl:note`, as an unknown XSLT element - the workaround is just to exclude this with `use-when="false()"`, which works with any processor. In a stylesheet with `version="3.0"`:

- an `xsl:note` without a `use-when` attribute is a warning, with the quick fix **Exclude with use-when="false()"**
- the `xsl:note` completions and **Add documentation note** insert the note with `use-when="false()"`
- documentation notes are shown, highlighted and checked, as in XSLT 4.0

> *Saxon PE and EE 12.8 or later, run with syntax extensions, accept `xsl:note` in an XSLT 3.0 stylesheet. With the setting `XSLT.validation.xslt30ItemTypesAndNotes`, the linter recognises notes so they need no `use-when`, and XSLT 4.0 `xsl:item-type` declarations are fully supported too - see XSLT 4.0 Item Types and Notes in XSLT 3.0.*

Highlighting Details

The content of a note is highlighted as documentation: tags, parameter and field names, code, bold and italic text, headings and links each have their own semantic token type. The token types have fallbacks, so they're highlighted in most color themes. To change their colors, add rules for the token types `xdocText`, `xdocTag`, `xdocParam`, `xdocCode`, `xdocBold`, `xdocItalic`, `xdocHeading`, `xdocLink` and `xdocCdata` to the `editor.semanticTokenColorCustomizations` setting.

Notes as XML

The extension includes an Invisible XML grammar for its documentation notes, xdoc-md.ixml. With the XPath 4.0 `invisible-xml()` function, a stylesheet can use it to turn its notes into XML - for example, to generate documentation for a library of functions.

Tag Checks

The `@param` tags are checked against the `xsl:param` elements of the function or template, and the `@field` tags against the fields of the item type's record type:

- an `@param` for a name that isn't a parameter, or an `@field` for a name that isn't a field, is a warning
- a second `@param` for the same parameter, or `@field` for the same field, is a warning
- when a note has `@param` tags, parameters without one are reported as information, with an **Add missing @param** quick fix - and likewise fields without an `@field`, with an **Add missing @field** quick fix
- an `@field` in the note of a function or template, or of an item type that isn't a record type, is a warning
- an `@param`, `@return` or `@error` in the note of an `xsl:item-type` is a warning
- an `@see` reference that refers to no declaration, e.g. `@see my:aera#2`, is a warning - but not other text, e.g. a URL, or a built-in function or type, e.g. `fn:sum#1`

In a module note, the `@param` and `@variable` tags are checked against the global parameters and variables in the same way: when the note has `@variable` tags, global variables without one are reported as information, with an **Add missing @variable** quick fix - as for `@param`, with **Add missing @param**. A global parameter or variable with a note of its own isn't reported as missing from the module note. `@return` and `@error` in a module note, and `@param`, `@variable`, `@return` and `@error` in the note of a global parameter or variable, are warnings.